



Fabric Lakehouse in a Day

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Introduction

What is Microsoft Fabric?

Microsoft Fabric provides a one-stop shop for all the analytical needs of every enterprise. It covers the complete spectrum of services including data movement, data lake, data engineering, data integration and data science, real-time analytics, and business intelligence. With Microsoft Fabric, there is no need to stitch together different services from multiple vendors. Instead, enterprises enjoy an end-to-end, highly integrated, single comprehensive product that is easy to understand, onboard, create, and operate. Combined with Microsoft Purview, which is included by default in every tenant, the entire data estate can be centrally governed, ensuring compliance with all data policies. There is no other product on the market that offers the breadth, depth, and level of integration that Microsoft Fabric offers.

Learn more about [Microsoft Fabric](#)

Purpose of This Workshop

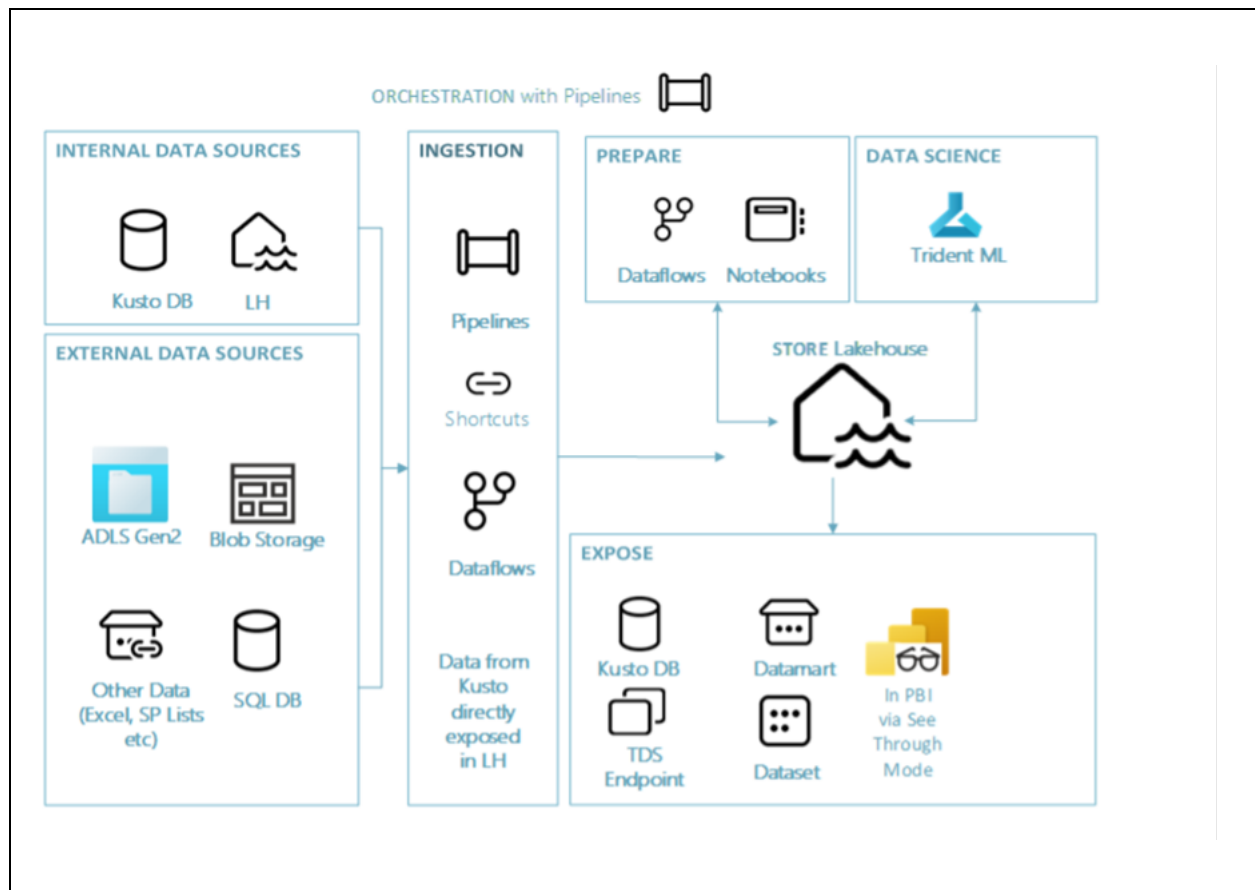
The aim of this workshop is to guide individuals through a Data Engineering end-to-end scenario using Microsoft Fabric. While some concepts may be familiar to data and analytics professionals, applying them in a new environment can be challenging.

In this workshop, participants will learn how to build a Fabric Lakehouse using Delta Lake. They will gain hands-on experience with data engineering tasks such as data ingestion, data transformation, and data modeling. By the end of the workshop, participants will have a solid understanding of how to build a data pipeline using available Data Engineering options in Microsoft Fabric.

OneLake and Lakehouse

Every Fabric tenant will always have exactly one OneLake, which is a unified logical SaaS data lake for the whole organization with data organized into domains. It prevents data silos and provides full and open access through industry-standard APIs and formats to any application, preventing platform lock-in. Within Fabric, new workspaces are created similar to how they are created with Azure Synapse or Power BI. All these workspaces are just folders within OneLake. Anything stored within a workspace gets stored in the workspace's corresponding folder in OneLake. All data is ultimately under the control of a tenant administrator.

End-to-End Architecture



Ingesting Data into a Lakehouse

Local file upload: Uploading data directly from a local machine through the Lakehouse explorer

Notebook Code: Using available spark libraries to connect to a data source directly and then loading data to dataframe and saving it in a lakehouse

Copy Tool in Pipelines: Connect to different data sources and land the data either in original format or convert it to a delta table

Transformations: Notebooks/Spark jobs/Dataflows can be used for data preparation, cleansing and transformations

Storage: Lakehouse, or Lakehouse(s) with medallion structure (Bronze/Silver/Gold)

Exposure: In Power BI via Direct Lake or SQL Endpoint

Orchestration: Pipelines triggering Notebooks

Hands-On Scenario Walkthrough

This comprehensive workshop has been meticulously crafted to guide you through a step-by-step journey, starting from the initial data acquisition all the way to data consumption. Our primary focus is on fostering a deep understanding of Microsoft Fabric.

The initial stage entails setting up your Fabric environment. You can either create a Fabric trial account or utilize your existing Power BI account, as explained in the subsequent section. Following this, you'll establish a dedicated Fabric workspace, where all your essential components such as lakehouses, warehouses, and reports will find their home.

Moving forward, you'll embark on a journey that adheres to the Medallion architecture by creating three distinct lakehouses – bronze, silver, and gold. These lakehouses will serve as the stages where your data undergoes transformations and enhancements. Starting with your bronze lakehouse, you'll ingest taxi data from a public storage account, perform necessary transformations, and store the refined data as a delta table in the silver lakehouse.

Within the silver lakehouse, you'll take your taxi data enrichment to the next level by incorporating holiday-related information. Once again, the enriched data will be saved as a delta table. Subsequently, you'll progress to the gold lakehouse, where you'll further enrich your dataset with weather-related data, execute essential aggregations, and archive your final dataframe as a delta table.

Finally, you'll conclude this immersive journey by harnessing the data you've prepared to create a compelling Power BI report, leveraging the capabilities of Direct Lake.

Datasets

During this workshop we will connect to a public Azure Storage account to load data. The containers inside the storage account contain all the data we will need but for your reference the datasets we are using are:

- [NYC Taxi Dataset](#)
- [Holiday Dataset](#)
- [Weather Dataset](#)

Getting Started with Fabric

Microsoft Fabric is currently in public preview and is temporarily provided free of charge when you sign up for the Microsoft Fabric (Preview) trial. Your use of the Microsoft Fabric (Preview) trial includes access to the Fabric product experiences and the resources to create and host Fabric items. The Fabric (Preview) trial lasts for a period of 60 days. The Microsoft Fabric (Preview) trial experience is subject to certain capacity limits.

Existing Power BI users

If you are an existing Power BI user, you can skip to [Start the Fabric \(Preview\) trial](#).

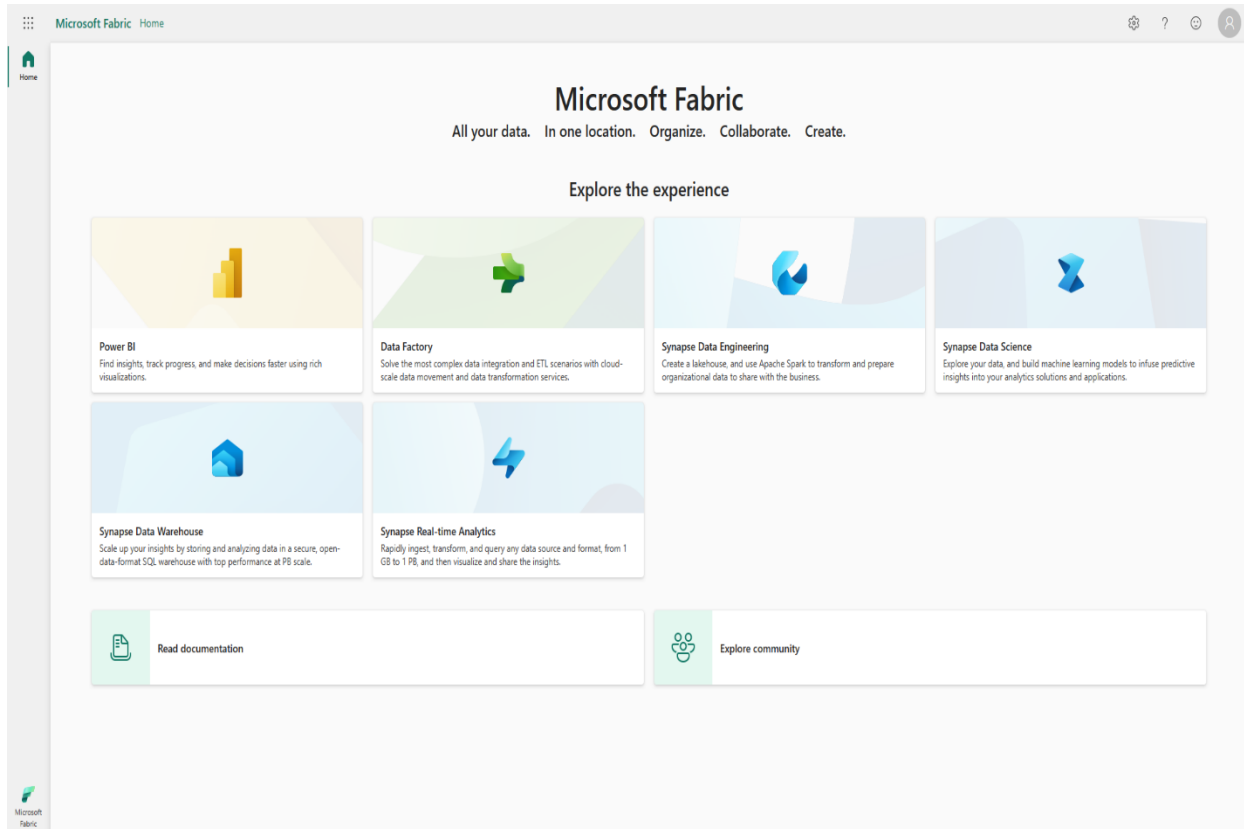
Users who are new to Power BI

For public preview, the Fabric (Preview) trial requires a Power BI license. Navigate to <https://app.fabric.microsoft.com> to sign up for a Power BI free license. Once you have a Power BI license, you can start the Fabric (Preview) trial.

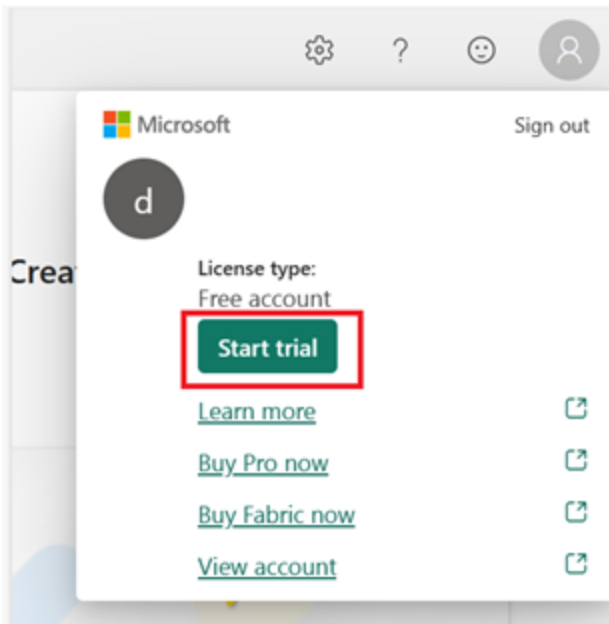
Start the Fabric (Preview) trial

Follow these steps to start your Fabric (Preview) trial.

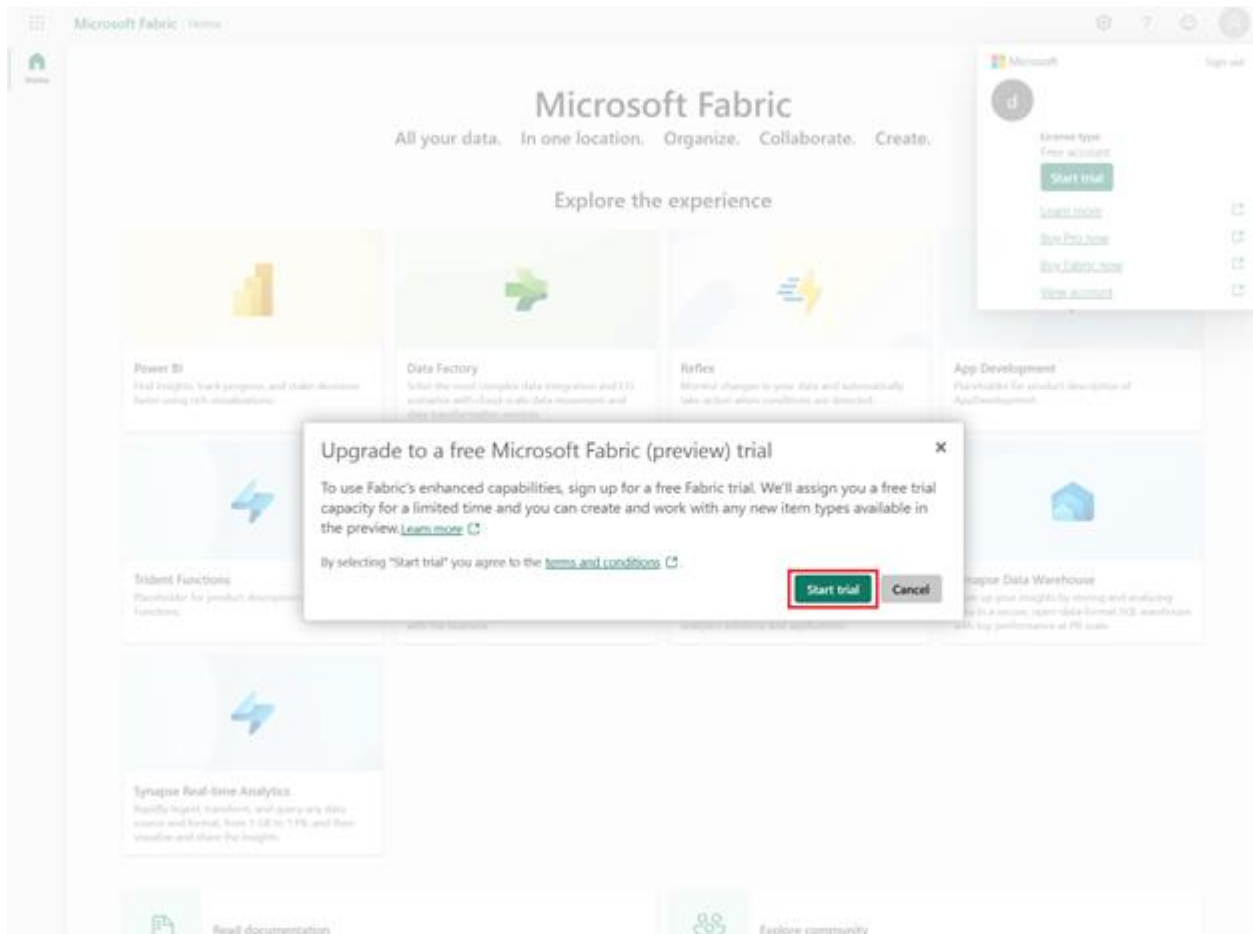
Open the [Fabric homepage](#) and select the Account manager.



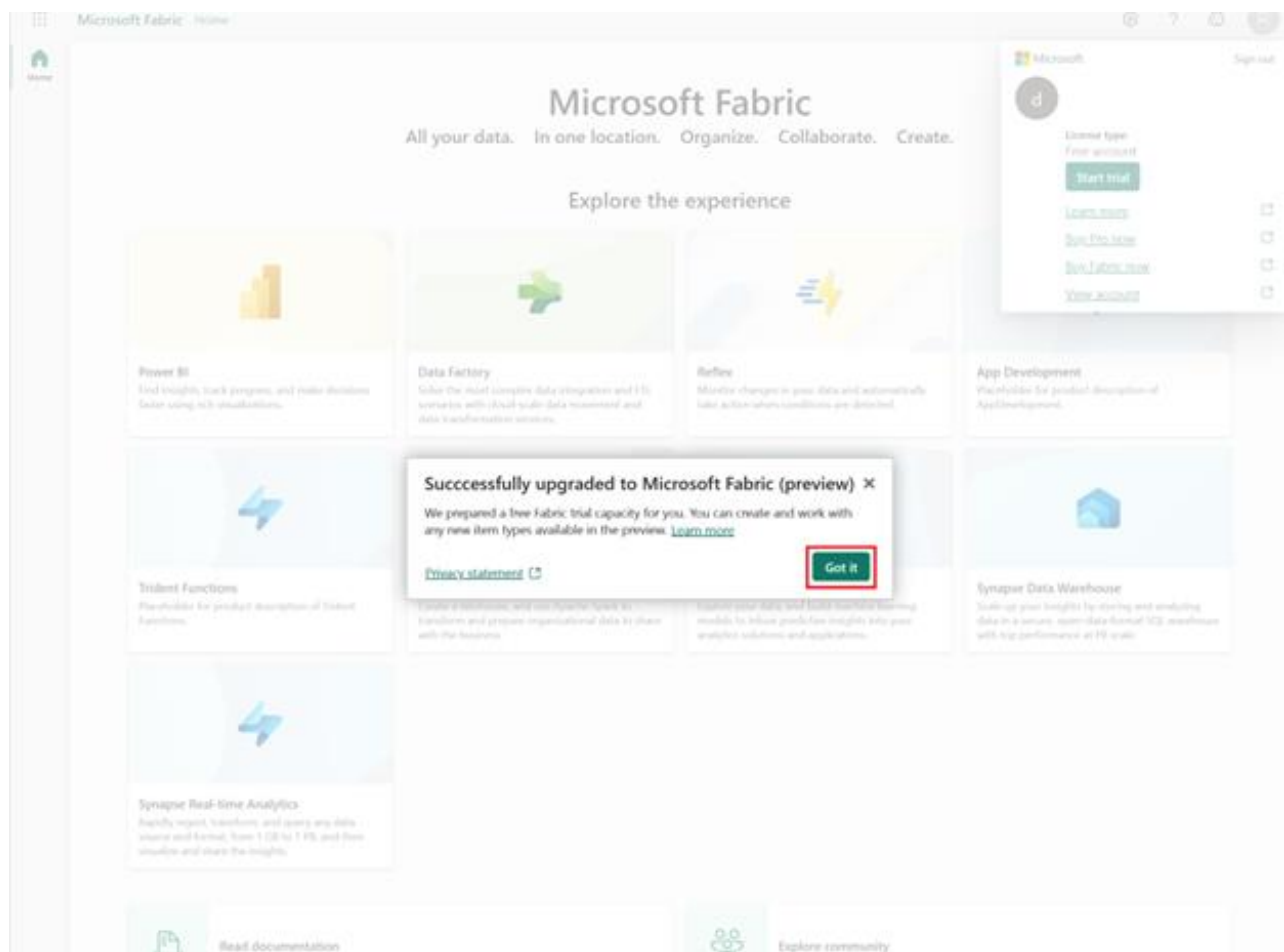
In the Account manager, select Start trial. If you don't see the Start trial button, trials may be disabled for your tenant. To enable trials for your tenant, see [Administer user access to a Fabric \(Preview\) trial](#).



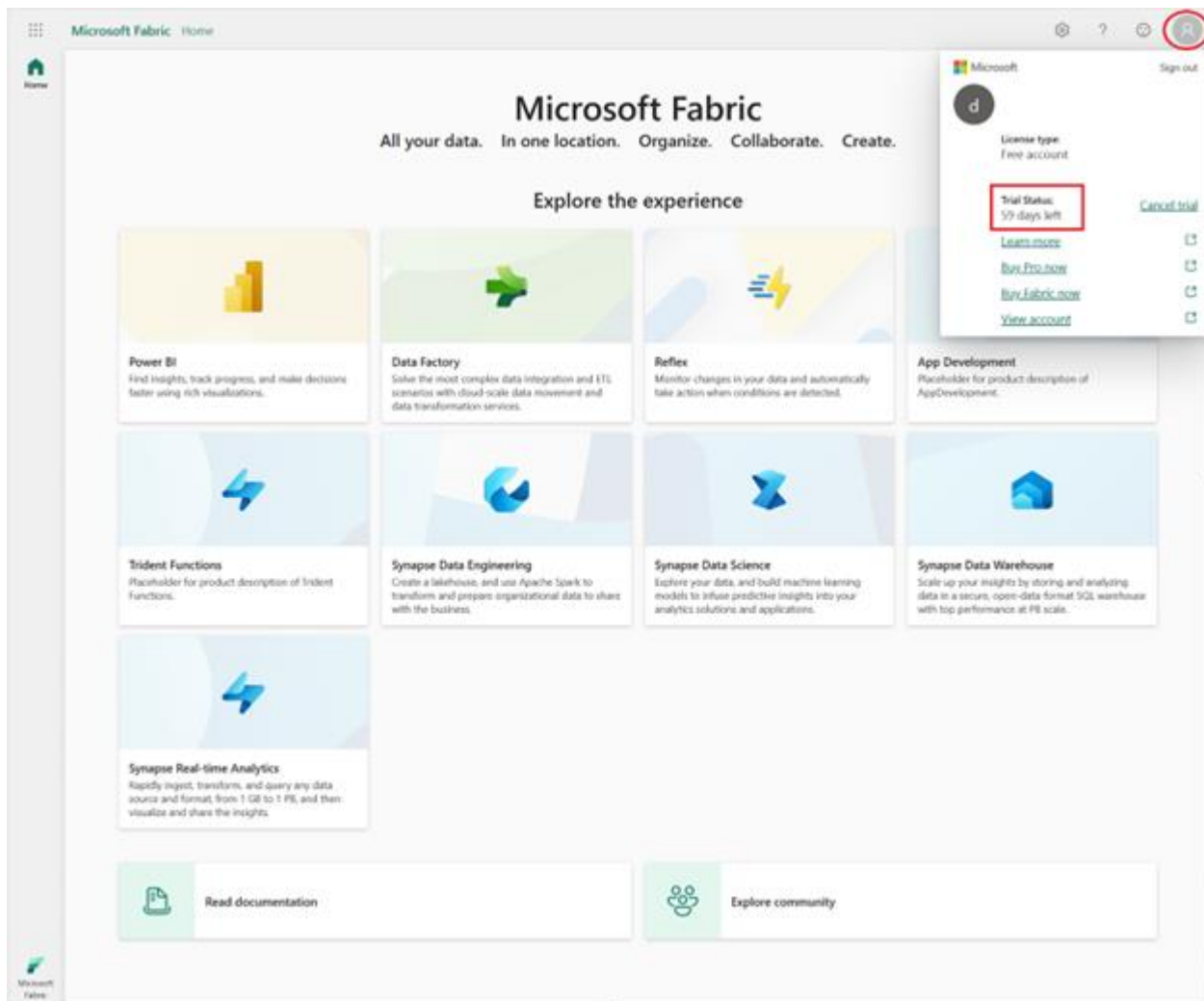
If prompted, agree to the terms and then select Start trial.



Once your trial capacity is ready, you receive a confirmation message. Select Got it to begin working in Fabric.



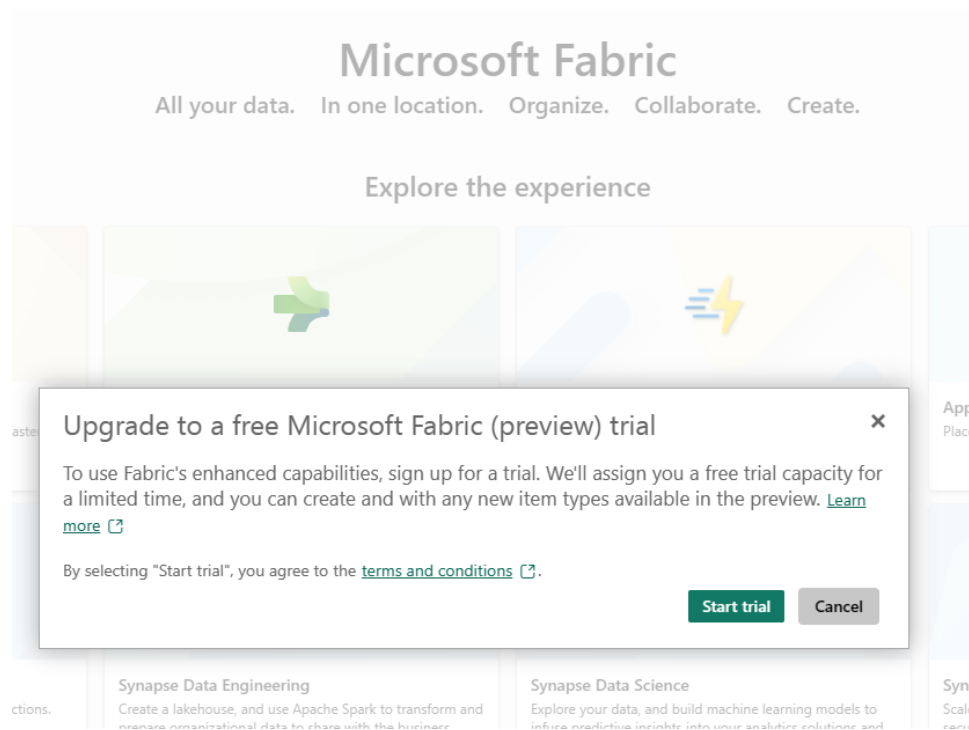
Open your Account manager again. Notice that you now have a heading for Trial status. Your Account manager keeps track of the number of days remaining in your trial. You also see the countdown in your Fabric menu bar when you work in a product experience.



Congratulations! You now have a Fabric (Preview) trial that includes a Power BI individual trial (if you didn't already have a Power BI paid license) and a Fabric (Preview) trial capacity.

Other ways to start a Microsoft Fabric (Preview) trial

If your Fabric administrator has [enabled the preview of Microsoft Fabric for the tenant](#) but you do not have access to a capacity that has Fabric enabled, you have another option for enabling a Fabric (Preview) trial. When you try to create a Fabric item in a workspace that you own (such as My Workspace) and that workspace doesn't support Fabric items, you're prompted to start a Fabric (Preview) trial. If you agree, your Fabric (Preview) trial starts and your workspace is upgraded to a trial capacity workspace.



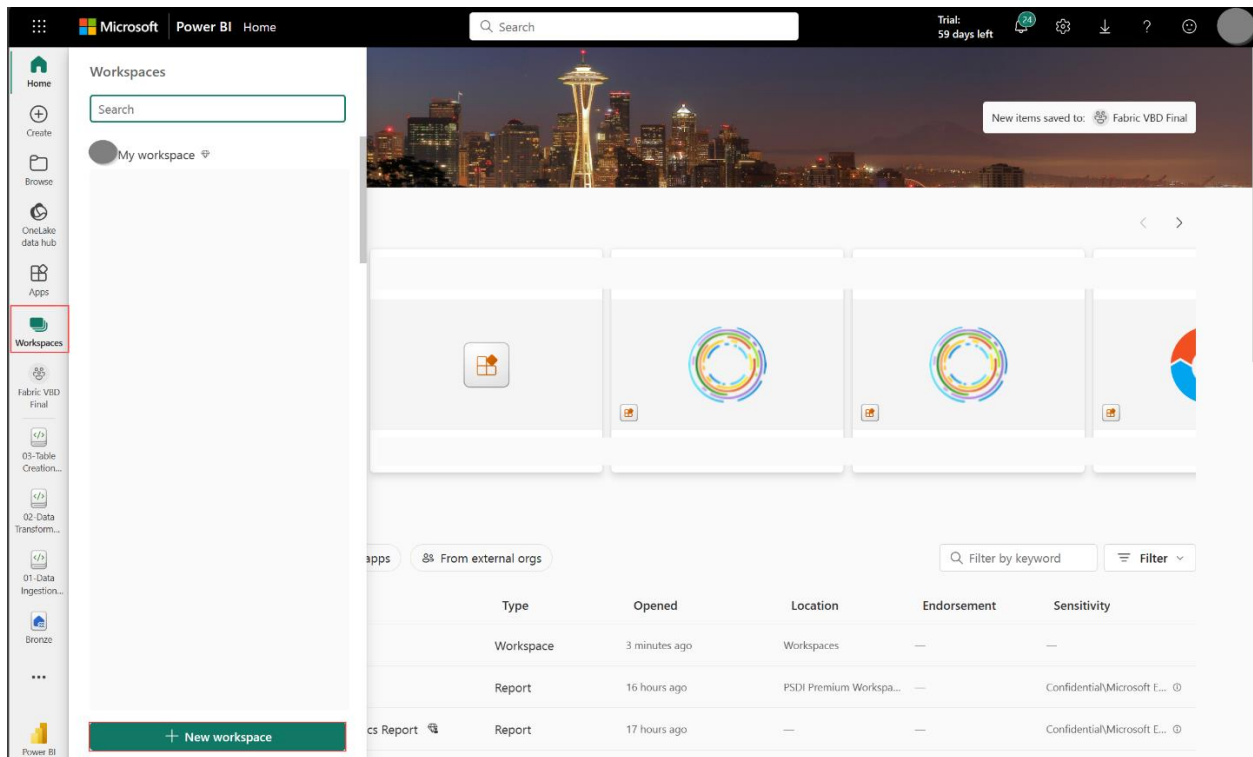
Data Engineering in a Lakehouse

Create a Workspace

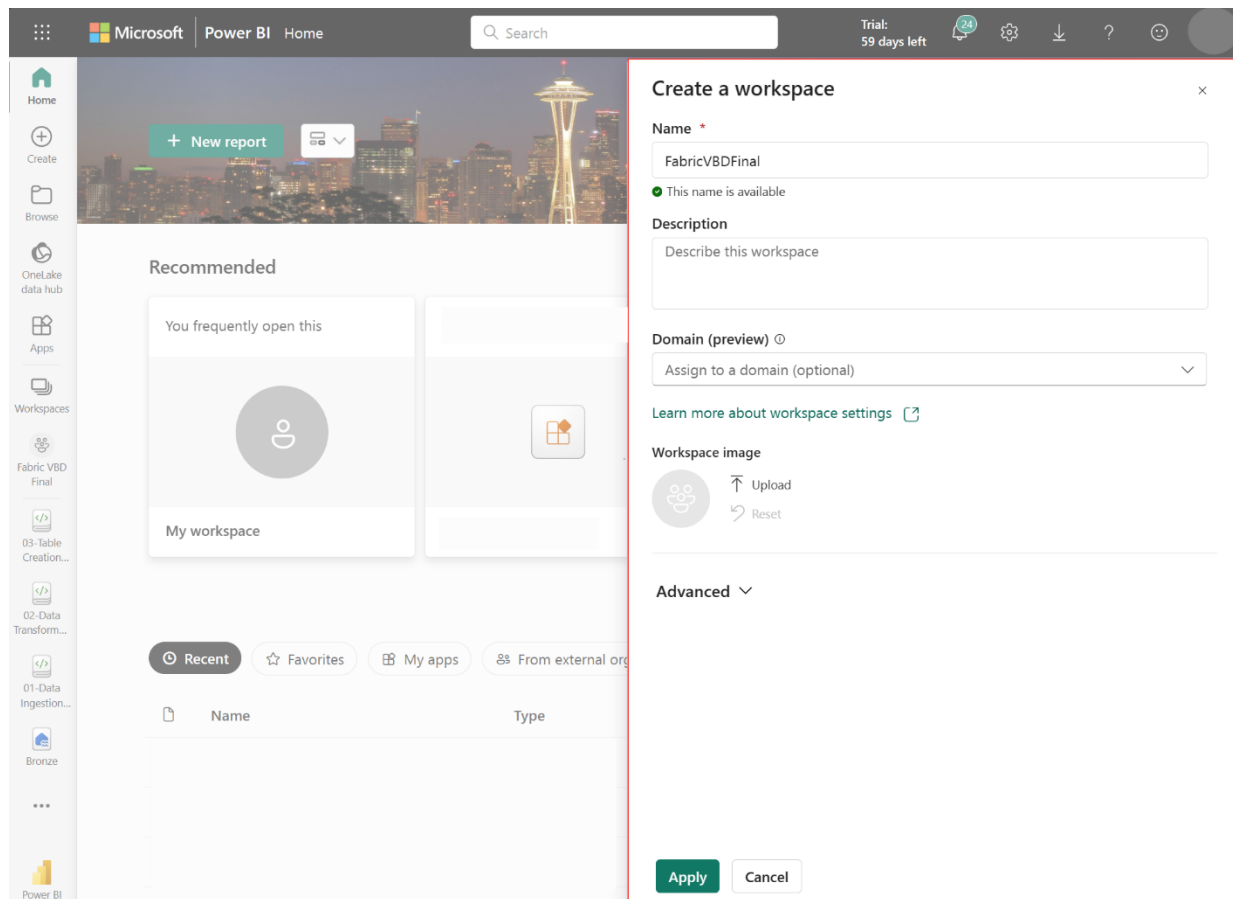
In this section we will create workspaces in Microsoft Fabric. In workspaces, you create collections of items such as lakehouses, warehouses, and reports. For more background, see the [Workspaces](#) article.

To create a workspace:

1. Select Workspaces > New workspace.



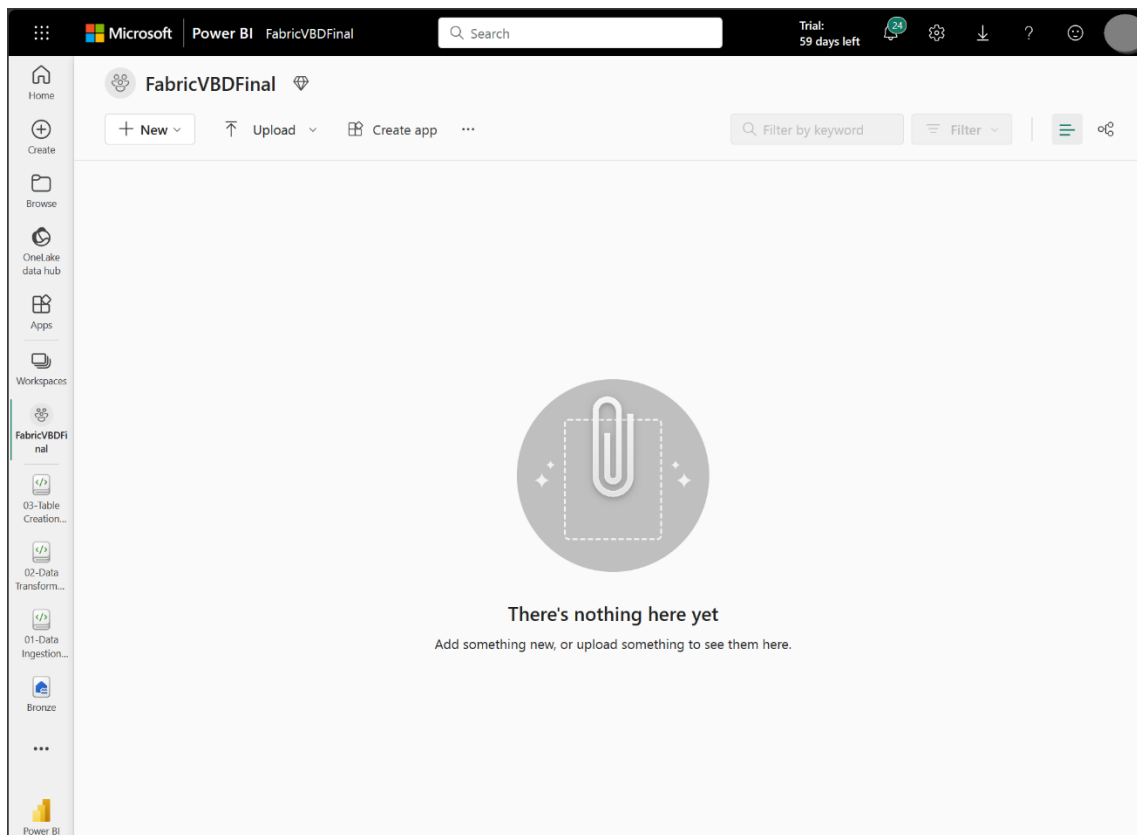
2. The Create a workspace pane opens.



3. Give the workspace a unique name (mandatory).
4. Provide a description of the workspace (optional).
5. Assign the workspace to a domain (optional).

If you are a domain contributor for the workspace, you can associate the workspace to a domain, or you can change an existing association. For information about domains, see [Domains in Fabric](#)

When done, either continue to the advanced settings, or select Apply. After clicking on Apply, you will land in your newly created workspace:

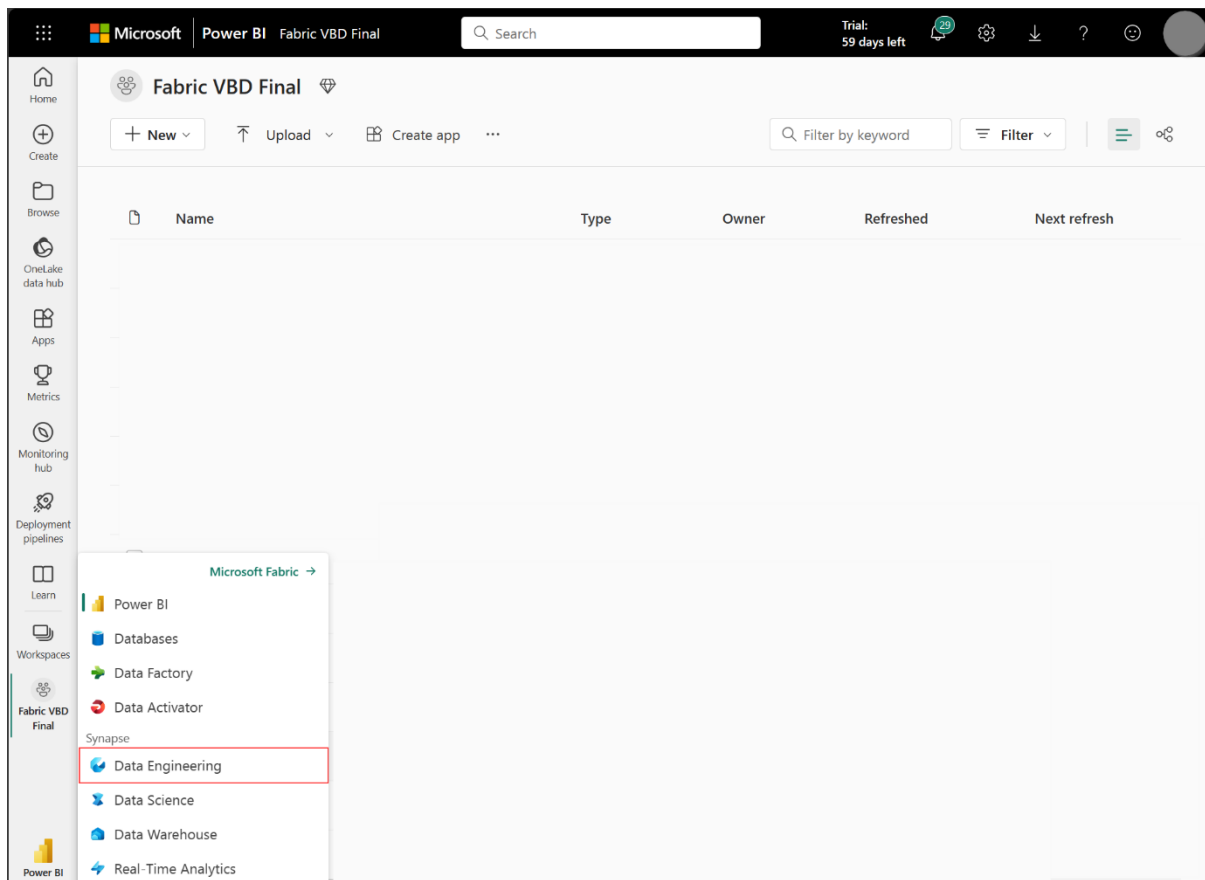


Learn more about [Workspaces](#)

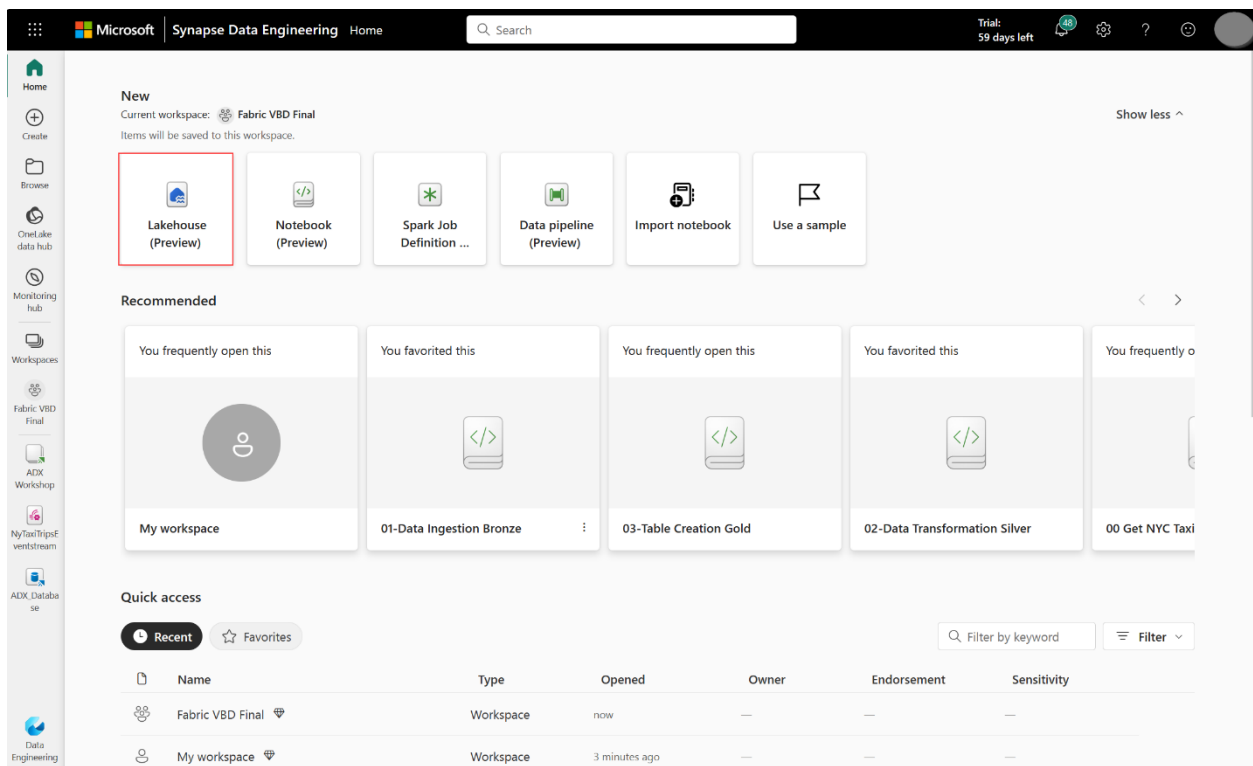
Create a Lakehouse

Now that you have a workspace, it's time to switch to the Data engineering experience in the portal and create a data lakehouse for your data files.

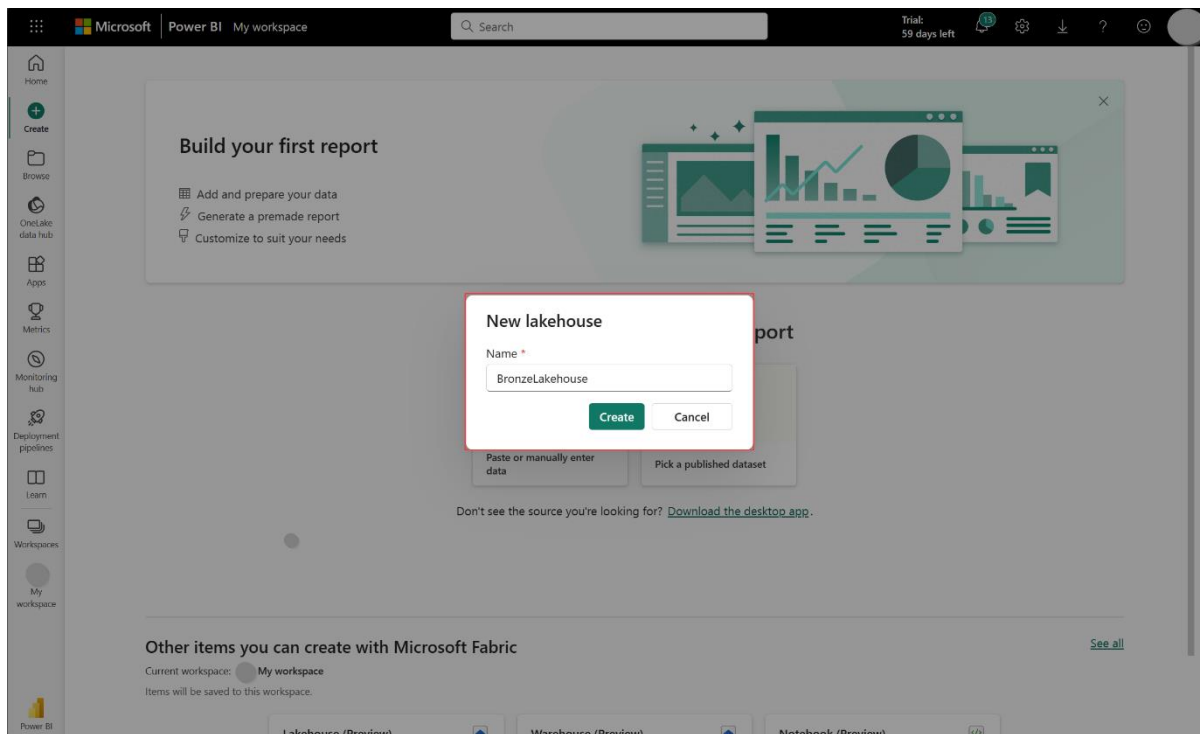
At the bottom left of the Power BI portal, switch to the Data Engineering experience. If you do not see the data engineering experience, contact your Fabric administrator with a request to [enable Fabric](#).



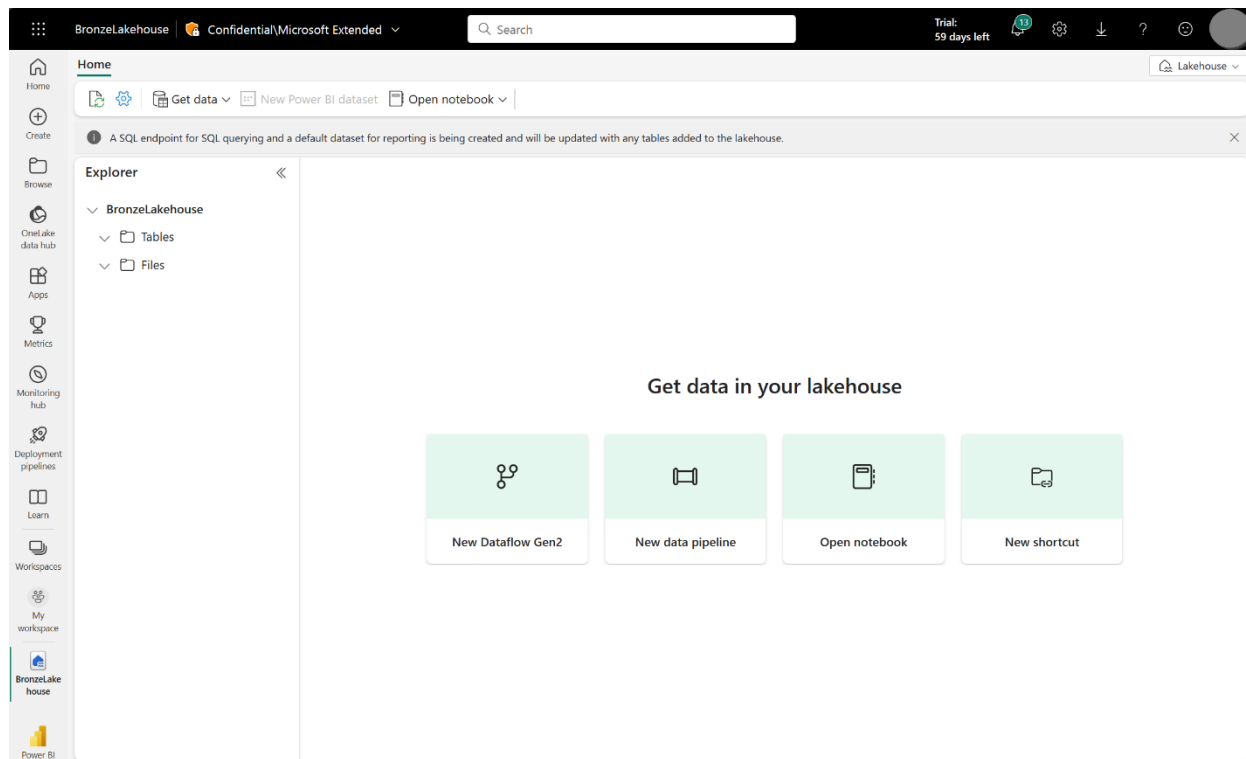
The data engineering home page includes tiles to create commonly used data engineering assets. In the Data engineering home page, click on Lakehouse:



Create one lakehouse with the name BronzeLakehouse:



After a minute or so, a new lakehouse will be created:



For this workshop, you will need to create 3 lakehouses in total inside your workspace to follow the Medallion architecture. Repeat the steps above to create the remaining two lakehouses.

1. BronzeLakehouse
2. SilverLakehouse
3. GoldLakehouse

After creating the 3 lakehouses, go back to your workspace:

Name	Type	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
BronzeLakehouse	Dataset (default)		9/28/23, 6:48:29 PM	N/A	—	Confidential/Microsoft...	
BronzeLakehouse	SQL endpoint		—	N/A	—	Confidential/Microsoft...	
BronzeLakehouse	Lakehouse		—	—	—	Confidential/Microsoft...	
GoldLakehouse	Dataset (default)		9/28/23, 6:48:46 PM	N/A	—	Confidential/Microsoft...	
GoldLakehouse	SQL endpoint		—	N/A	—	Confidential/Microsoft...	
GoldLakehouse	Lakehouse		—	—	—	Confidential/Microsoft...	
SilverLakehouse	Dataset (default)		9/28/23, 6:48:32 PM	N/A	—	Confidential/Microsoft...	
SilverLakehouse	SQL endpoint		—	N/A	—	Confidential/Microsoft...	
SilverLakehouse	Lakehouse		—	—	—	Confidential/Microsoft...	

You will see that 3 items with the same name but different type are created for each lakehouse:

1. [Lakehouse](#)
2. [Dataset\(default\)](#)
3. [SQL Endpoint](#)

Workspace Spark Compute

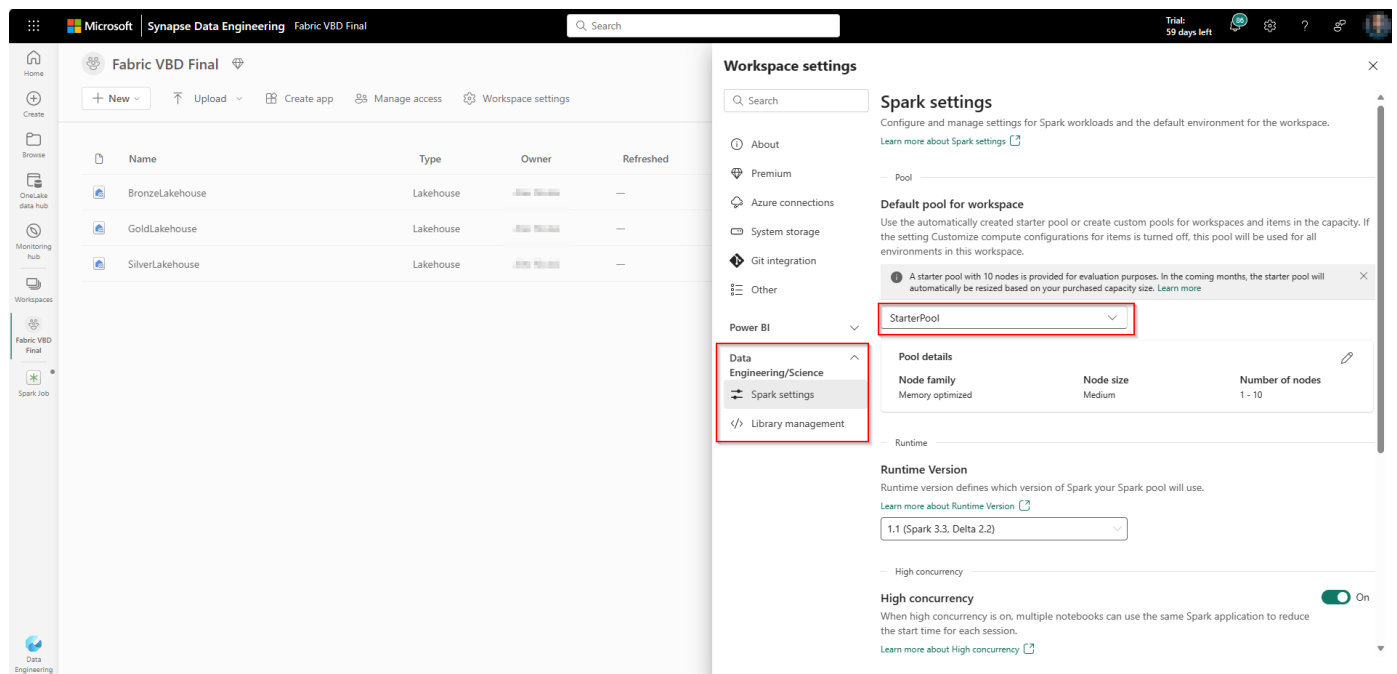
Microsoft Fabric Data Engineering and Data Science experiences operate on a fully managed Spark compute platform. This platform is designed to deliver unparalleled speed and efficiency. With starter pools, you can expect rapid Spark session initialization, typically within 5 to 10 seconds, and no need for manual setup. Furthermore, you also get the flexibility to customize Spark pools according to the specific data engineering and data science requirements. The platform enables an optimized and tailored analytics experience.

With Microsoft Fabric trial you get Starter pools which are a fast and easy way to use Spark on the Microsoft Fabric platform within seconds. You can use Spark sessions right away, instead of waiting for Spark to set up the nodes for you, which helps you do more with data and get insights quicker.

To check Spark pool settings for trial, click on the Workspace Settings within your trial workspace and then navigate to Spark settings under Data Engineering/Data Science section

The screenshot shows the Microsoft Fabric VBD Final workspace interface. The top navigation bar includes a search bar and a 'Workspace settings' link highlighted with a red box. Below the navigation bar, a table lists three lakehouses: BronzeLakehouse, GoldLakehouse, and SilverLakehouse, all owned by Alex Niculai. The left sidebar shows the 'Workspace' section selected, and the bottom right corner displays a Windows taskbar with various application icons.

Name	Type	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
BronzeLakehouse	Lakehouse	Alex Niculai	—	—	—	Confidential/Microsoft...	
GoldLakehouse	Lakehouse	Alex Niculai	—	—	—	Confidential/Microsoft...	
SilverLakehouse	Lakehouse	Alex Niculai	—	—	—	Confidential/Microsoft...	



Further Reading

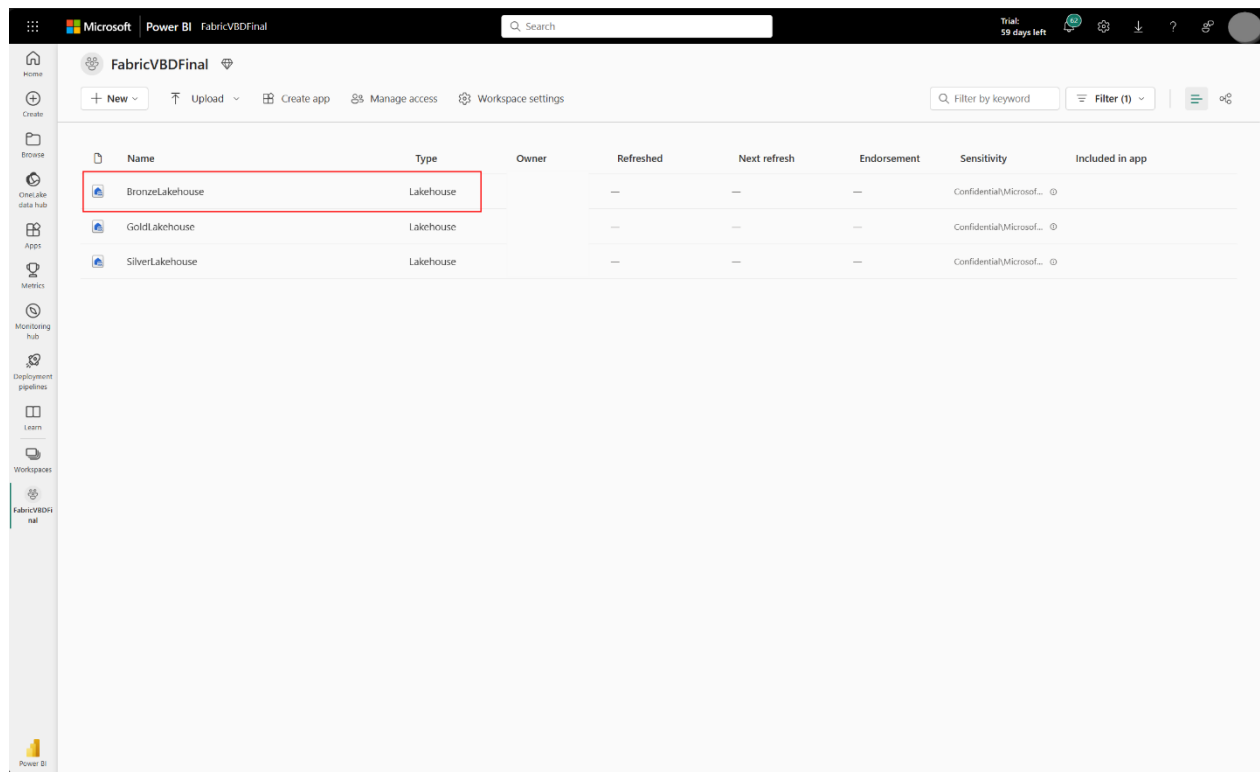
[Spark compute for Data Engineering and Data Science - Microsoft Fabric | Microsoft Learn](#)

Saving the ABFS Path of a Lakehouse

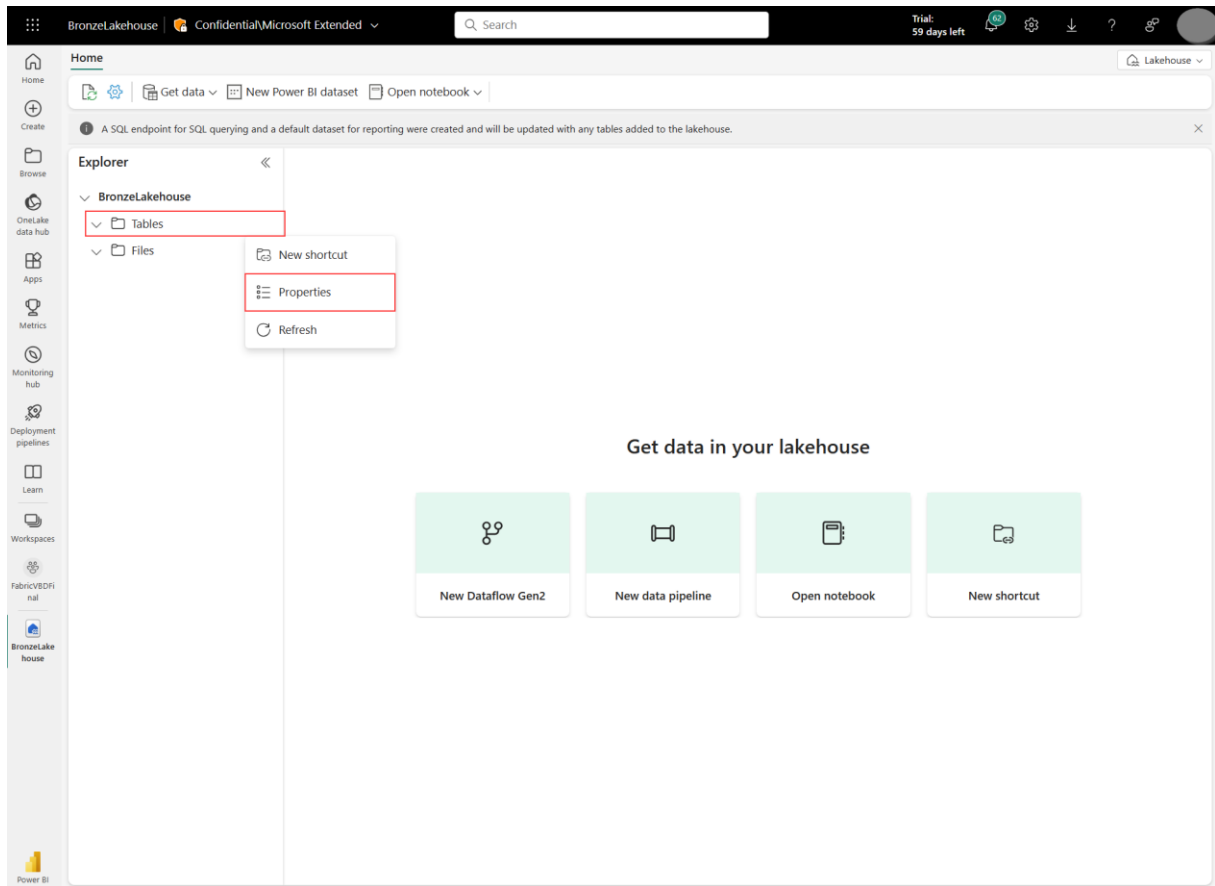
The ABFS (Azure Blob File System) path is needed when saving a DataFrame or any other data artifact to a different Lakehouse or storage location, such as the Bronze layer in your data processing pipeline. You will need these paths later in the notebook execution part.

To retrieve the ABFS path, please follow these steps:

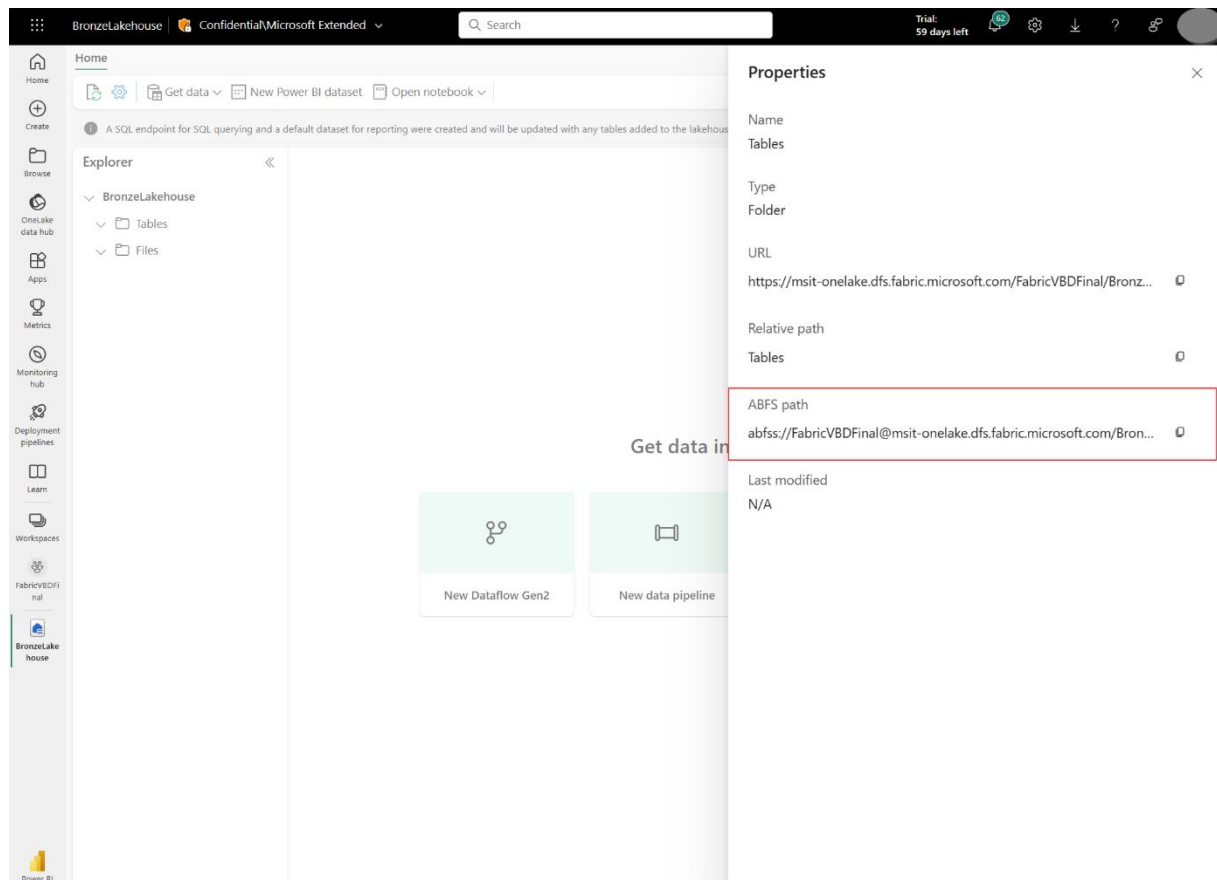
1. Go to the Workspace and click on the Bronze Lakehouse:



2. Navigate to the Tables section in the Explorer pane and click on Properties:



3. Copy the ABFS path provided.



Save this ABFSS path in your notes. Follow the same steps to save the ABFS path of all for the remaining two lakehouses.

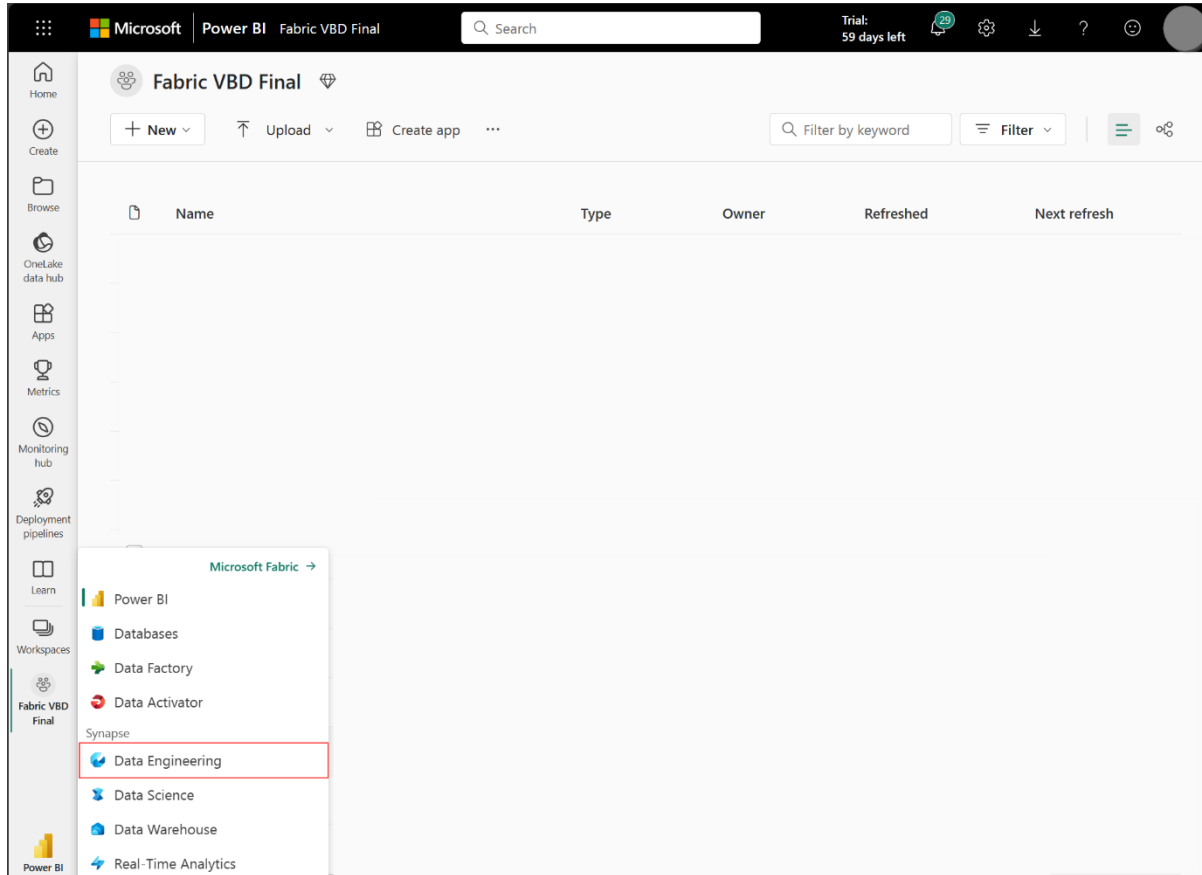
Later, when executing the notebooks, you will use these paths in specific notebook cells to ensure that your dataframe is saved in the appropriate location and can be easily accessed for any future data transformations.

After creating the Lakehouses, it's time to start exploring the Data Engineering scenario we mentioned in the previous section. To do so, the first step is to upload some notebooks in your workspace.

Uploading your Notebooks

For this part of the workshop, you will need to import 4 notebooks into your workspace. These notebooks contain code and comments that will guide you through each step. To be able to import the notebooks:

1. Download the notebooks from this [Path](#)
2. From the workload switch located at the bottom left of the screen, select Data Engineering. This will take you to the “Data Engineering” home page.



3. Select Import notebook from the new section at the top of the landing page.

Microsoft

Synapse Data Engineering

Home

Search

Trial: 59 days left

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

New

Current workspace: Fabric VBD Final

Items will be saved to this workspace.

Lakehouse (Preview)

Notebook (Preview)

Spark Job Definition ...

Data pipeline (Preview)

Import notebook

Use a sample

Recommended

You frequently open this

My workspace

You favorited this

01-Data Ingestion Bronze

You frequently open this

03-Table Creation Gold

You favorited this

02-Data Transformation Si

Quick access

Recent

Favorites

Filter by keyword

Filter

Name	Type	Opened	Owner
Fabric VBD Final	Workspace	2 minutes ago	—
01-Data Ingestion Bronze	Notebook	12 minutes ago	

Select Upload from the Import status pane that opens on the right side of the screen.

Microsoft

Synapse Data Engineering

Home

Q

Search

Trial:

59 days left

29

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

New

Current workspace: Fabric VBD Final

Items will be saved to this workspace.

Lakehouse (Preview)

Notebook (Preview)

Spark Job Definition ...

Data pipeline (Preview)

Import notebook

Recommended

You frequently open this

My workspace

You favorited this

01-Data Ingestion Bronze

You frequently open this

03-Table Creation Gold

Quick access

Recent

Favorites

	Name	Type	Opened
	Fabric VBD Final	Workspace	2 minutes ago
	01-Data Ingestion Bronze	Notebook	12 minutes ago

Import status

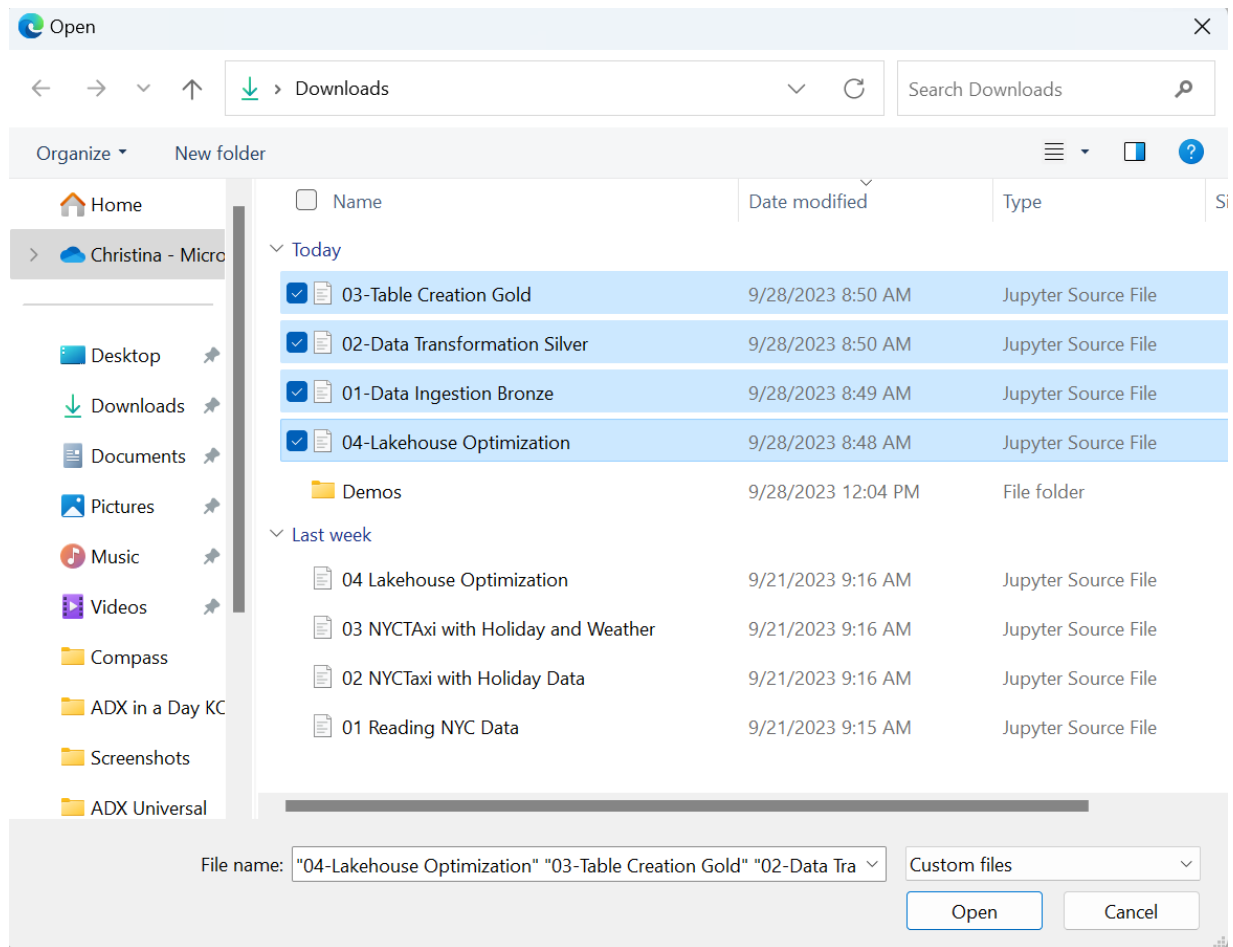
ⓘ

Upload and import notebook from local computer, you can select multiple notebooks and import them in a batch.

↑

Upload

Select all four the notebooks that were downloaded in step 1 of this section and click Open:



After clicking Open, notebooks will start importing and you will get this message on the top right of your screen. Click on "Go to workspace" and you will be able to see your imported notebooks:

Microsoft

Synapse Data Engineering

Home

Q Search

Trial: 59 days left

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

FabricVBDFinal

Data Engineering

New

Current workspace: FabricVBDFinal

Items will be saved to this workspace.

Import notebook source code files from your local machine and create Notebook items in workspace.

Lakehouse (Preview)

Notebook (Preview)

Spark Job Definition ...

Data pipeline (Preview)

Import notebook

Use a sample

Recommended

You frequently open this

My workspace

You favorited this

01-Data Ingestion Bronze

You frequently open this

03-Table Creation Gold

You favorited this

02-Data Transformation Silver

You frequently o

00 Get NYC Taxi

Quick access

Recent

Favorites

Q Filter by keyword

Filter

	Name	Type	Opened	Owner	Endorsement	Sensitivity
	FabricVBDFinal	Workspace	now	—	—	—
	My workspace	Workspace	a minute ago	—	—	—

You can also use the Filter functionality in the top right to filter only the Notebooks:

Microsoft | Synapse Data Engineering | Fabric VBD Final

Search

Trial: 59 days left

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

Fabric VBD Final

+ New | Upload | Create app

Filter by keyword | Filter (1)

refresh

Name	Type	Owner
01-Data Ingestion Bronze	Notebook	
02-Data Transformation Silver	Notebook	
03-Table Creation Gold	Notebook	
04-Lakehouse Optimization	Notebook	

Reset all filters

Type (1)

☒ Notebook

☐ Dataset (default)

☐ SQL endpoint

☐ Lakehouse

☐ Spark Job Definition

Product

Owner

Microsoft | Synapse Data Engineering | Fabric VBD Final

Search

Trial: 59 days left

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

Fabric VBD Final

+ New | Upload | Create app

Filter by keyword

Filter (1)

	Name	Type	Owner	Refreshed	Next refresh
				—	—
	01-Data Ingestion Bronze	Notebook		—	—
	02-Data Transformation Silver	Notebook		—	—
	03-Table Creation Gold	Notebook		—	—
	04-Lakehouse Optimization	Notebook		—	—
				—	—
				—	—

Executing the Notebooks

To navigate to the **01-Data Ingestion Bronze** notebook, first click on the workspace on the left pane and now search for your workspace name and open that workspace.

Microsoft

Synapse Data Engineering

Fabric VBD Final

Search

Trial: 59 days left

29

?

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

Workspaces

fabric

Pinned

Fabric VBD Final

FabricSession18September

All

Fabric Lab

Fabric_Session_Live

FabricVBDFinal

Deployment pipelines

New workspace

te app

Filter by keyword

Filter

Type	Owner	Refreshed	Next refresh
Notebook		—	—
Notebook		—	—
Notebook		—	—
Notebook		—	—
Notebook		—	—
Dataset (default)		9/19/23, 12:42:59 PM	N/A
SQL endpoint		—	N/A
Lakehouse		—	—
Dataset (default)		9/19/23, 10:35:10 PM	N/A
SQL endpoint		—	N/A
Lakehouse		—	—
Dataset (default)		9/19/23, 12:57:19 PM	N/A
SQL endpoint		9/21/23, 7:47:53 PM	N/A

From the list of existing notebooks, select the **01 – Data ingestion Bronze**

Microsoft

Synapse Data Engineering

Fabric VBD Final

Search

Trial: 59 days left

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

Fabric VBD Final

New

Upload

Create app

Filter by keyword

Filter (1)

	Name	Type	Owner	Refreshed	Next refresh
				—	—
	01-Data Ingestion Bronze	Notebook		—	—
	02-Data Transformation Silver	Notebook		—	—
	03-Table Creation Gold	Notebook		—	—
	04-Lakehouse Optimization	Notebook		—	—
				—	—
				—	—

After clicking on the notebook, you will land on this page. Please choose “Add Lakehouse” on the left:

01 Data Ingestion... ConfidentialMicrosoft Extended · Saved

Search

Trial: 59 days left

Home Edit Run Data View

Run all Stop session Language PySpark (Python) Open in VS Code

Lakehouses Resources

Lakehouses

Add lakehouse

Add

Enrich NYC Yellow Taxi Data with Holiday and Weather

We are going to enrich NYC Yellow Taxi Data as a process of multiple notebooks This notebook provides examples of how to load and clean the NYC Yellow Taxi Data

On the other Notebooks the NYC Yellow Taxi Data will be enriched with Holiday and Weather with focusing on :

read Open Datasets manipulate the data to prepare for further analysis, including column projection, filtering, grouping and joins etc. create a Spark table to be used in other notebooks for modeling training create a Lakehouse table and build PBI reports on top of it

Importing the Data types we need for this Notebook

```
1 from pyspark.sql.types import StructType, StructField, IntegerType, DoubleType, StringType, TimestampType
2 import pyspark.sql.functions as f
```

Data Ingestion New York Yellow Taxi Data Set

We are going to load the New York Yellow Taxi data from a Public Azure Storage Account.

```
1 # Azure storage access info for open datasets yellow cab
2 storage_account = "azuresynapsestorage"
3 container = "sampledata"
4
5 sas_token = r"" # Blank since container is Anonymous access
6
7 # Set Spark config to access blob storage
8 spark.conf.set(f"fs.azure.sas.{container}.{storage_account}.blob.core.windows.net", sas_token)
9
10 # NOTE TO FELLOW AUTHORS:
11 # We can now easily read all of the parquet files in the directory just by pointing Spark at the directory
12 # But if we need to restrict the years that we read (maybe to match the weather data), then we
13 # can use the "pathGlobFilter" parameter. So line 18 (below) would be something like this:
14 # .parquet(f"wasbs://{container}@{storage_account}.blob.core.windows.net/Fabric/NYCTaxiDataset/yellow/", pathGlobFilter="--2018,19,21").parquet"
15
```

01 Data Ingestion... ConfidentialMicrosoft Extended · Saved

Search

Trial: 59 days left

Home Edit Run Data View

Run all Stop session Language PySpark (Python) Open in VS Code

Lakehouses Resources

Lakehouses

Add lakehouse

Add

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read Open Datasets manipulate the data to prepare for further analysis, including column projection, filtering, grouping and joins etc. create a Spark table to be used in other notebooks for modeling training create a Lakehouse table and build PBI reports on top of it

Importing the Data

```
1 from pys
2 import p
```

Data Ingestion New York Yellow Taxi Data Set

We are going to load the New York Yellow Taxi data from a Public Azure Storage Account.

```
1 # Azure storage access info for open datasets yellow cab
2 storage_account = "azuresynapsestorage"
3 container = "sampledata"
4
5 sas_token = r"" # Blank since container is Anonymous access
6
7 # Set Spark config to access blob storage
8 spark.conf.set(f"fs.azure.sas.{container}.{storage_account}.blob.core.windows.net", sas_token)
9
10 # NOTE TO FELLOW AUTHORS:
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12 # But if we need to restrict the years that we read (maybe to match the weather data), then we
13 # can use the "pathGlobFilter" parameter. So line 18 (below) would be something like this:
```

Add lakehouse

☐ New lakehouse

☒ Existing lakehouse

Add

Cancel

Choose the 3 lakehouses you created earlier and click on Add:

OneLake data hub

Choose the data you want to connect

All My data Endorsed in your org Favorites Filter by keyword

	Name	Owner	Refreshed	Location	Endorsement	Sensitivity
☒	BronzeLakehouse	Christina Pardali	-	FabricVBDFinal	-	Confidential\Microsoft E... @
☒	SilverLakehouse	Christina Pardali	-	FabricVBDFinal	-	Confidential\Microsoft E... @
☒	GoldLakehouse	Christina Pardali	-	FabricVBDFinal	-	Confidential\Microsoft E... @
☐	Bronze	Alex Niculai	-	Fabric VBD Final	-	Confidential\Microsoft E... @
☐	Silver	Christina Pardali	-	Final VBD Test	-	Confidential\Microsoft E... @
☐	Gold	Christina Pardali	-	Final VBD Test	-	Confidential\Microsoft E... @
☐	Bronze	Christina Pardali	-	Final VBD Test	-	Confidential\Microsoft E... @
☐	BronzeLakehouse	Christina Pardali	-	My Workspace	-	Confidential\Microsoft E... @
☐	Silver	Christina Pardali	-	Fabric Lab	-	Confidential\Microsoft E... @
☐	Bronze	Christina Pardali	-	Fabric Lab	-	Confidential\Microsoft E... @
☐	Gold	Christina Pardali	-	Fabric Lab	-	Confidential\Microsoft E... @
☐	DataflowsStagingLakehouse ...	Christina Pardali	-	Mv Workspace	-	Confidential\Microsoft E... @

Add Cancel

After clicking on Add, you will be able to see your lakehouses:

01: Data Ingestion... Confidential/Microsoft Extended · Saved

Home Edit Run Data View

Language: PySpark (Python) Open in VS Code

Home Create Browse OneLake data hub Monitoring hub Workspace FabricVBDFinal BronzeLakehouse SilverLakehouse GoldLakehouse Add Lakehouse Remove all Lakehouses

Enrich NYC Yellow Taxi Data with Holiday and Weather

We are going to enrich NYC Yellow Taxi Data as a process of multiple notebooks. This notebook provides examples of how to load and clean the NYC Yellow Taxi Data. On the other Notebooks the NYC Yellow Taxi Data will be enriched with Holiday and Weather with focusing on:

- read Open Datasets
- manipulate the data to prepare for further analysis, including column projection, filtering, grouping and joins etc.
- create a Spark table to be used in other notebooks for modeling training
- create a Lakehouse table and build PBI reports on top of it

Importing the Data types we need for this Notebook

```
1 from pyspark.sql.types import StructType, StructField, IntegerType, DoubleType, StringType, TimestampType
2 import pyspark.sql.functions as f
```

Data Ingestion New York Yellow Taxi Data Set

We are going to load the New York Yellow Taxi data from a Public Azure Storage Account.

```
1 # Azure storage access info for open datasets yellow cab
2 storage_account = "azuresynapsestorage"
3 container = "sampledata"
4
5 sas_token = "" # Blank since container is Anonymous access
6
7 # Set Spark config to access blob storage
8 spark.conf.set(f"fs.azure.sas.{container}.{storage_account}.blob.core.windows.net", sas_token)
9
10 # NOTE TO FELLOW AUTHORS:
11 # We can now easily read all of the parquet files in the directory just by pointing Spark at the directory
12 # But if we need to restrict the years that we read (maybe to match the weather data), then we
13 # can use the "pathGlobFilter" parameter. So line 18 (below) would be something like this:
14 # .parquet(f"wasbs://{container}@{storage_account}.blob.core.windows.net/Fabric/NYTaxiDataset/yellow/", pathGlobFilter="20[18,19,21].parquet"
15
```

Not connected Save option: Automatic Selected Cell 1 of 43 cells

In the next steps, you will need to pin a specific lakehouse for each notebook. If you want to set a lakehouse as the default lakehouse for a notebook, choose the Pin symbol next to the lakehouse:

The screenshot shows a Databricks workspace with a notebook titled "Enrich the Previously Ingested Dataset with Holidays Dataset". On the left, a sidebar titled "Lakehouses" lists several lakehouses: "BronzeLakehouse", "GoldLakehouse", and "SilverLakehouse". The "SilverLakehouse" is highlighted with a red box and a pin icon. The notebook content includes a title, a description of the task, and two code blocks. The first code block imports datetime and pyspark.sql.functions. The second code block loads holiday data from a Bronze Lakehouse. The notebook is running on a PySpark (Python) kernel.

Notebook 01 – Data Ingestion Bronze

This notebook provides examples of how to load and clean the NYC Yellow Taxi Data. The main steps you will take are:

- Load the taxi dataset from the public Azure storage account and save them in a dataframe
- Display the dataframe and explore the data
- Create new columns for the month number, day of month, day of week, and hour of day. This info is going to be used in the training model to factor in time-based seasonality
- Add a static feature for the country code to be able to join this dataframe with holiday data in the next notebook
- Extract the year, month, day of the month and day of the week that will be further used in the joins with Holiday and Weather datasets.
- Remove unneeded columns from the dataframe
- Print the schema of the DataFrame and gain a deeper understanding of the data types and the specific columns that have been selected. This allows us to more effectively analyze and manipulate the data to extract meaningful insights.

- Check the existing tables in the SilverLakehouse to ensure that you can save your dataframe in the next step in the Silver Lakehouse
- As the final step in our data processing pipeline, we can save the cleaned and filtered DataFrame, nyc_tlc_df_clean, as a table in the Silver layer. However, since we are saving the table in a different Lakehouse, namely Silver, we need to provide the ABFS path. Please note that the ABFS path will be unique to your specific lakehouse according to the steps you followed [in this section](#)
- Load the holiday dataset and save the dataframe as parquet file in the Bronze Lakehouse.
- Load the weather dataset and save the dataframe as parquet file in the Bronze Lakehouse

To navigate to the Data Ingestion notebook, first click on the workspace on the left pane and now search for your workspace name and open that workspace.

The screenshot shows the Microsoft Synapse Data Engineering interface. The top bar includes the Microsoft logo, 'Synapse Data Engineering', the workspace name 'Fabric VBD Final', a search bar, and a trial status 'Trial: 59 days left'. The left sidebar contains navigation icons for Home, Create, Browse, OnLake data hub, Monitoring hub, Workspaces, and Data Engineering. The 'Workspaces' pane is open, showing a search for 'fabric' and a list of pinned workspaces: 'Fabric VBD Final' (highlighted with a red box), 'FabricSession18September', 'Fabric Lab', 'Fabric_Session_Live', and 'FabricVBDFinal'. At the bottom of the sidebar, there are links for 'Deployment pipelines' and a '+ New workspace' button. The main area displays a table of resources with columns: Type, Owner, Refreshed, and Next refresh.

Type	Owner	Refreshed	Next refresh
Notebook		—	—
Notebook		—	—
Notebook		—	—
Notebook		—	—
Notebook		—	—
Dataset (default)		9/19/23, 12:42:59 PM	N/A
SQL endpoint		—	N/A
Lakehouse		—	—
Dataset (default)		9/19/23, 10:35:10 PM	N/A
SQL endpoint		—	N/A
Lakehouse		—	—
Dataset (default)		9/19/23, 12:57:19 PM	N/A
SQL endpoint		9/21/23, 7:47:53 PM	N/A

From the list of existing notebooks, select the “**01 – Data ingestion Bronze**”

The screenshot shows the Microsoft Synapse Data Engineering interface. The top bar includes the Microsoft logo, 'Synapse Data Engineering', 'Fabric VBD Final', a search bar, and a trial status 'Trial: 59 days left'. The left sidebar contains navigation icons for Home, Create, Browse, OneLake data hub, Monitoring hub, Workspaces, Fabric VBD Final, and Data Engineering. The main area displays a table of notebooks:

Name	Type	Owner	Refreshed	Next refresh
01 - Data Ingestion Bronze	Notebook		—	—
02 - Data Transformation Silver	Notebook		—	—
03 - Table Creation Gold	Notebook		—	—
04 - Lakehouse Optimization	Notebook		—	—

The first row, '01 - Data Ingestion Bronze', is highlighted with a red box. Below the table, there is a red square icon.

Once you have the notebook open, you can begin working. Make sure that you can see the different lakehouses on your left as described before:

The screenshot shows the Microsoft Synapse Data Engineering interface with the '01 Data Ingestion Bronze' notebook open. The top bar includes the Microsoft logo, 'Synapse Data Engineering', 'Fabric VBD Final', a search bar, and a trial status 'Trial: 59 days left'. The left sidebar contains navigation icons for Home, Create, Browse, OneLake data hub, Monitoring hub, Workspaces, Fabric VBD Final, and Data Engineering. The main area displays the notebook content:

Enrich NYC Yellow Taxi Data with Holiday and Weather

We are going to enrich NYC Yellow Taxi Data as a process of multiple notebooks This notebook provides examples of how to load and clean the NYC Yellow Taxi Data

On the other Notebooks the NYC Yellow Taxi Data will be enriched with Holiday and Weather with focusing on :

- read Open Datasets manipulate the data to prepare for further analysis, including column projection, filtering, grouping and joins etc. create a Spark table to be used in other notebooks for modeling training create a Lakehouse table and build PBI reports on top of it

Importing the Data types we need for this Notebook

```
1 from pyspark.sql.types import StructType, StructField, IntegerType, DoubleType, StringType, TimestampType
2 import pyspark.sql.functions as f
```

Data Ingestion New York Yellow Taxi Data Set

We are going to load the New York Yellow Taxi data from a Public Azure Storage Account.

```
1 # Azure storage access info for open datasets yellow cab
2 storage_account = "azuresynapsestorage"
3 container = "sampledata"
4
5 sas_token = "" # Blank since container is Anonymous access
6
7 # Set Spark config to access blob storage
8 spark.conf.set(f"fs.azure.sas.{container}.{storage_account}.blob.core.windows.net", sas_token)
9
10 # NOTE TO FELLOW AUTHORS:
11 # We can now easily read all of the parquet files in the directory just by pointing Spark at the directory
12 # But if we need to restrict the years that we read (maybe to match the weather data), then we
13 # can use the "pathGlobFilter" parameter. So line 18 (below) would be something like this:
14 # .parquet(f"wasbs://{container}@{storage_account}.blob.core.windows.net/Fabric/NYCTaxiDataset/yellow/", pathGlobFilter="2018,19,21".parquet"
15
```

Make sure that the BronzeLakehouse is pinned for this notebook.

The screenshot shows the Databricks workspace interface. On the left, the 'Lakehouses' section is expanded, and 'BronzeLakehouse' is pinned. The main notebook area displays the title 'Enrich NYC Yellow Taxi Data with Holiday and Weather'. The notebook content includes introductory text and two code cells. The first code cell imports pyspark.sql types and functions. The second code cell sets up Azure storage access for yellow cab data.

```
1 from pyspark.sql.types import StructType, StructField, IntegerType, DoubleType, StringType, TimestampType
2 import pyspark.sql.functions as f
```

```
1 # Azure storage access info for open datasets yellow cab
2 storage_account = "azurestorage"
3 container = "sampledata"
4
5 sas_token = "" # Blank since container is Anonymous access
6
7 # Set Spark config to access blob storage
8 spark.conf.set(f"fs.azure.sas.{container}.{storage_account}.blob.core.windows.net", sas_token)
```

Read through the notebook and execute the code cells as you go along. This will populate the bronze layer of your lakehouse with the data you need for the remainder of the workshop.

Notebook 02 – Data Transformation Silver

Now that we have taxi data downloaded, prepared, and saved in the Silver Layer as a table, let's add additional features to it by enriching it with information from the holiday dataset. Holiday-specific features will assist model accuracy, as major holidays are times where taxi demand increases dramatically, and supply becomes limited.

Let's see now the steps we take in this notebook to transform our data.

- Load the Holiday data from the bronze lakehouse and create a dataframe.
- Rename the *countryRegionCode* and date columns to match the respective field names from the taxi data and normalize the time so it can be used as a key.
- Check if the table exists before creating it in the next step.

- Save the clean file as a table in the Silver Layer by providing the correct ABFS path as described above.
- Create a new dataframe, namely *nyc_tlc_df_clean* by reading the *NYCTaxi_Clean* table Since we saved the *NYCTaxi_Clean* as a table in the Silver layer.
- Join the holiday data with the taxi data by performing a left-join. This will preserve all records from taxi data, but add in holiday data where it exists for the corresponding datetime and *country_code*, which in this case is always "US".
- Preview the data to verify that they were merged correctly.
- Store the *nyc_taxi_holiday_df* as a table named *NYCTaxi_Holiday* on the Silver layer. By specifying the ABFS path, we'll be able to easily create dataFrames from this table and use them for any future data transformations.

To navigate to the Data transformation Silver notebook, first click on the workspace on the left pane and now search for the workspace name you created and open that workspace.

The screenshot shows the Microsoft Synapse Data Engineering interface. On the left, the 'Workspaces' pane is open, displaying a search bar with 'fabric' and a list of workspaces. The workspace 'Fabric VBD Final' is highlighted with a red box. Below the search bar, there are sections for 'Pinned' (showing 'Fabric VBD Final' and 'FabricSession18September') and 'All' (showing 'Fabric Lab', 'Fabric_Session_Live', and 'FabricVBDFinal'). At the bottom of the pane, there is a 'Deployment pipelines' section and a '+ New workspace' button.

The main area on the right displays a table of workspace details. The table has columns for 'Type', 'Owner', 'Refreshed', and 'Next refresh'. The data is as follows:

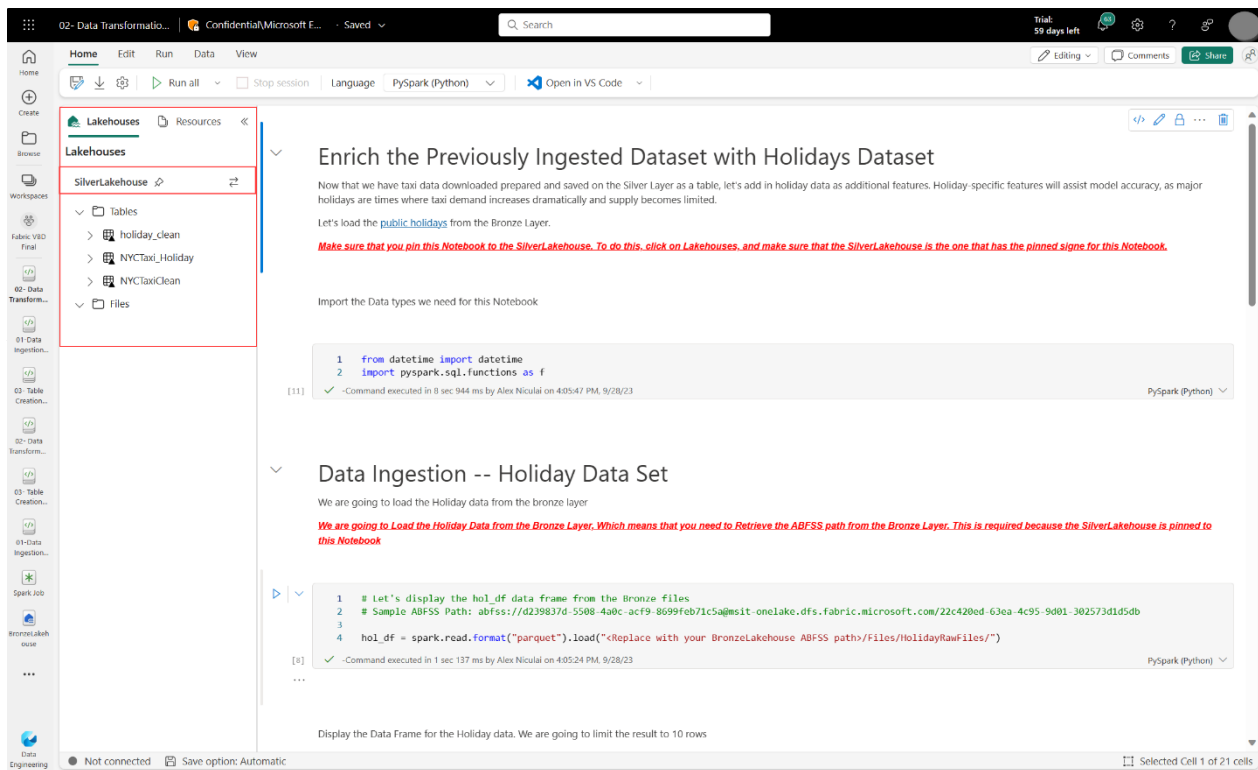
Type	Owner	Refreshed	Next refresh
Notebook		—	—
Notebook		—	—
Notebook		—	—
Notebook		—	—
Notebook		—	—
Dataset (default)		9/19/23, 12:42:59 PM	N/A
SQL endpoint		—	N/A
Lakehouse		—	—
Dataset (default)		9/19/23, 10:35:10 PM	N/A
SQL endpoint		—	N/A
Lakehouse		—	—
Dataset (default)		9/19/23, 12:57:19 PM	N/A
SQL endpoint		9/21/23, 7:47:53 PM	N/A

Now, from the list of existing notebooks, select the **02- Data Transformation Silver**

The screenshot shows the Microsoft Synapse Data Engineering interface. The top navigation bar includes the Microsoft logo, 'Synapse Data Engineering', and 'Fabric VBD Final'. A search bar is present, and a trial status indicates '59 days left'. The left sidebar contains navigation icons for Home, Create, Browse, OneLake data hub, Monitoring hub, Workspaces, and Fabric VBD Final. The main content area displays a table of notebooks. The notebook '02-Data Transformation Silver' is highlighted with a red box. Below the table, there is a red square icon.

Name	Type	Owner	Refreshed	Next refresh
01-Data Ingestion Bronze	Notebook		—	—
02-Data Transformation Silver	Notebook		—	—
03-Table Creation Gold	Notebook		—	—
04-Lakehouse Optimization	Notebook		—	—
			—	—
			—	—

Once you have the notebook open, you can begin working. Make sure that you can see the different lakehouses on your left as described before and that the SilverLakehouse is pinned for this notebook:



You can read through the notebook and execute the code cells to build the silver layer of your lakehouse.

Notebook 03 – Table Creation Gold

Now that we have taxi and holiday data downloaded, prepared, and saved on the Silver Layer as tables, let's add in weather data as additional features.

Let's see now the steps we take in this notebook to transform our data:

To enrich the *NYCTaxi_Holiday* table we created in the Silver Layer, we first need to filter out any non-empty holiday rows. We'll accomplish this by creating a temporary view and then building the *nyc_taxi_holiday_df* DataFrame on top of the *NYCTaxi_Holiday* table from the previous notebook.

If the Silver Lakehouse is already attached to the notebook and the table is visible, we can simply use the syntax `Silver.TableName`. However, if the table isn't listed, we'll need to use the ABFS path to access it.

- Load the Weather data from the public Azure Storage Account.
- Filter out the weather for New York City by latitude and longitude and remove the null temperature.
- Remove any unused columns to simplify our DataFrame.
- Enrich our dataset with aggregation statistics to gain further insights from our weather data. We group the cleaned weather DataFrame by datetime and calculate the mean snow depth, maximum precipitation time, mean temperature, and maximum precipitation depth using the `agg()` function. The resulting DataFrame is stored in *weather_df_grouped* and will be used for further analysis.
- , it may be helpful to Rename the columns to more descriptive names and ensure that the data format is consistent throughout the dataset to improve our ability to work with the *weather_df_grouped* dataframe. This can make it easier to understand and manipulate the data.
- Enhance the *NYCTaxi_Holiday* dataset by incorporating data from the *weather_df_grouped* dataframe by performing a left-join.
- Save the dataframe *nyc_taxi_holiday_weather_df* as a table in the Gold Layer by specifying the ABFS path of the Gold Lakehouse.
- Build a dataframe on the curated table from Gold Layer *NYCTaxi_Holiday_Weather*.
- Remove invalid rows with less than 0 taxi fare or tip.
- Explore the gold table.

To navigate to the Table Creation Gold notebook, first click on the workspace on the left pane and now search for your workspace. Open it from the list of search results.

Microsoft

Synapse Data Engineering

Fabric VBD Final

Search

Trial: 59 days left

29

?

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

Workspaces

fabric

Pinned

Fabric VBD Final

FabricSession18September

All

Fabric Lab

Fabric_Session_Live

FabricVBDFinal

Deployment pipelines

New workspace

te app

Filter by keyword

Filter

Type	Owner	Refreshed	Next refresh
Notebook		—	—
Notebook		—	—
Notebook		—	—
Notebook		—	—
Notebook		—	—
Dataset (default)		9/19/23, 12:42:59 PM	N/A
SQL endpoint		—	N/A
Lakehouse		—	—
Dataset (default)		9/19/23, 10:35:10 PM	N/A
SQL endpoint		—	N/A
Lakehouse		—	—
Dataset (default)		9/19/23, 12:57:19 PM	N/A
SQL endpoint		9/21/23, 7:47:53 PM	N/A

Now, from the list of existing notebooks, select the Table Creation Gold notebook

Microsoft | Synapse Data Engineering | Fabric VBD Final

Search

Trial: 59 days left

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

Fabric VBD Final

+ New

Upload

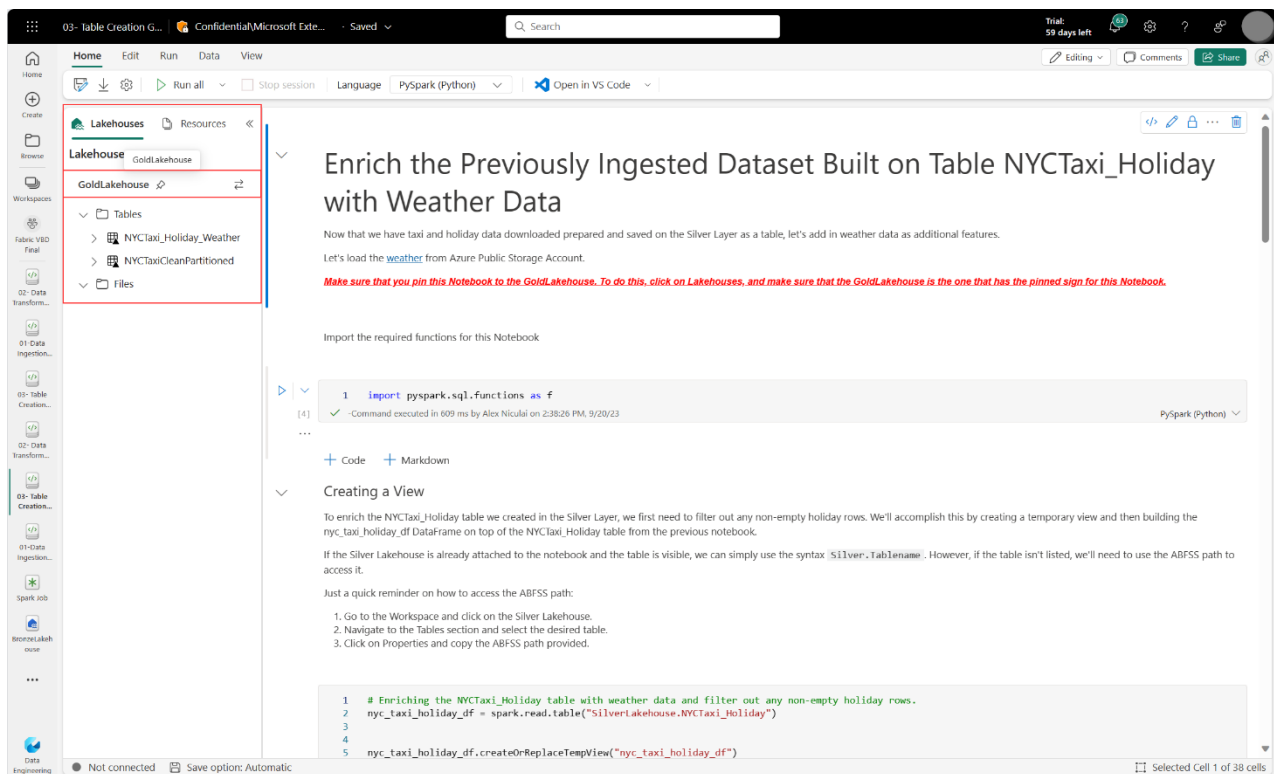
Create app

Filter by keyword

Filter (1)

	Name	Type	Owner	Refreshed	Next refresh
				—	—
	01-Data Ingestion Bronze	Notebook		—	—
	02-Data Transformation Silver	Notebook		—	—
	03-Table Creation Gold	Notebook		—	—
	04-Lakehouse Optimization	Notebook		—	—
				—	—
				—	—

Once you have the notebook open, you can begin working. Make sure that you can see the different lakehouses on your left as described before and that the GoldLakehouse is pinned for this notebook:



Read through this notebook and execute the code cells as you go along.

Direct Insights using PBI

Power BI is natively integrated in the whole Fabric experience and this native integration brings unique mode of accessing the data, called DirectLake, from the lakehouse to provide the most performant query and reporting experience. DirectLake mode is a groundbreaking new engine capability to analyze very large datasets in Power BI. The technology is based on the idea of loading parquet-formatted files directly from a data lake without having to query a data warehouse or lakehouse endpoint and without having to import or duplicate data into a Power BI dataset.

DirectLake is a fast path to load the data from the data lake straight into the Power BI engine, ready for analysis.

In traditional DirectQuery mode, the Power BI engine queries the data directly from the data source every time it is queried and hence query performance depends on the speed data can be retrieved from the data source. It avoids having to copy the data i.e., any

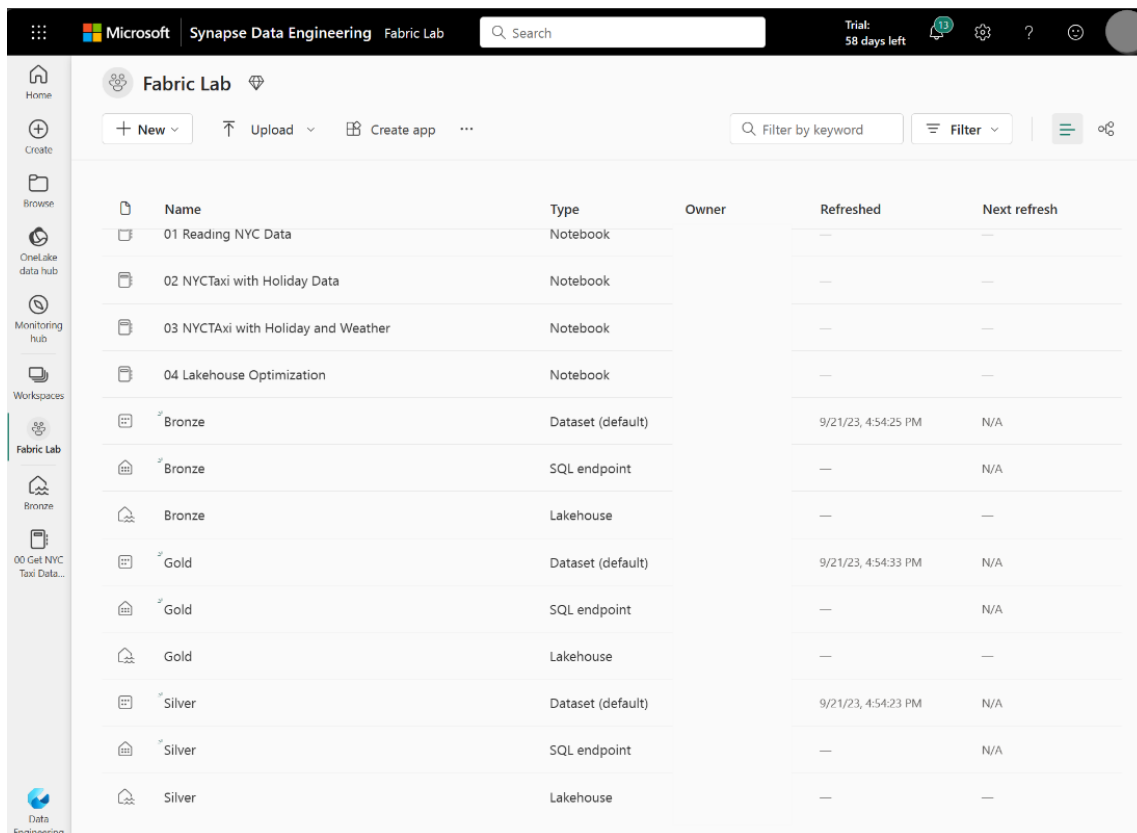
changes at the source are immediately reflected in the query results while in the import mode, on the other hand, performance is much better because the data is readily available in memory without having to query the data source each time, but the Power BI engine must first copy the data into the dataset at refresh time. Any changes at the source are only picked up during the next data refresh.

DirectLake mode now eliminates this import requirement by loading the data files directly into memory. Because there is no explicit import process, it is possible to pick up any changes at the source as they occur, thus combining the advantages of DirectQuery and import mode while avoiding their disadvantages. DirectLake mode is therefore the ideal choice for analyzing very large datasets as well as datasets with frequent updates at the source.

In this workshop, we will use the option auto create report to generate our report. It is necessary to have completed all the previous steps to be able to generate the report.

The steps you should follow are:

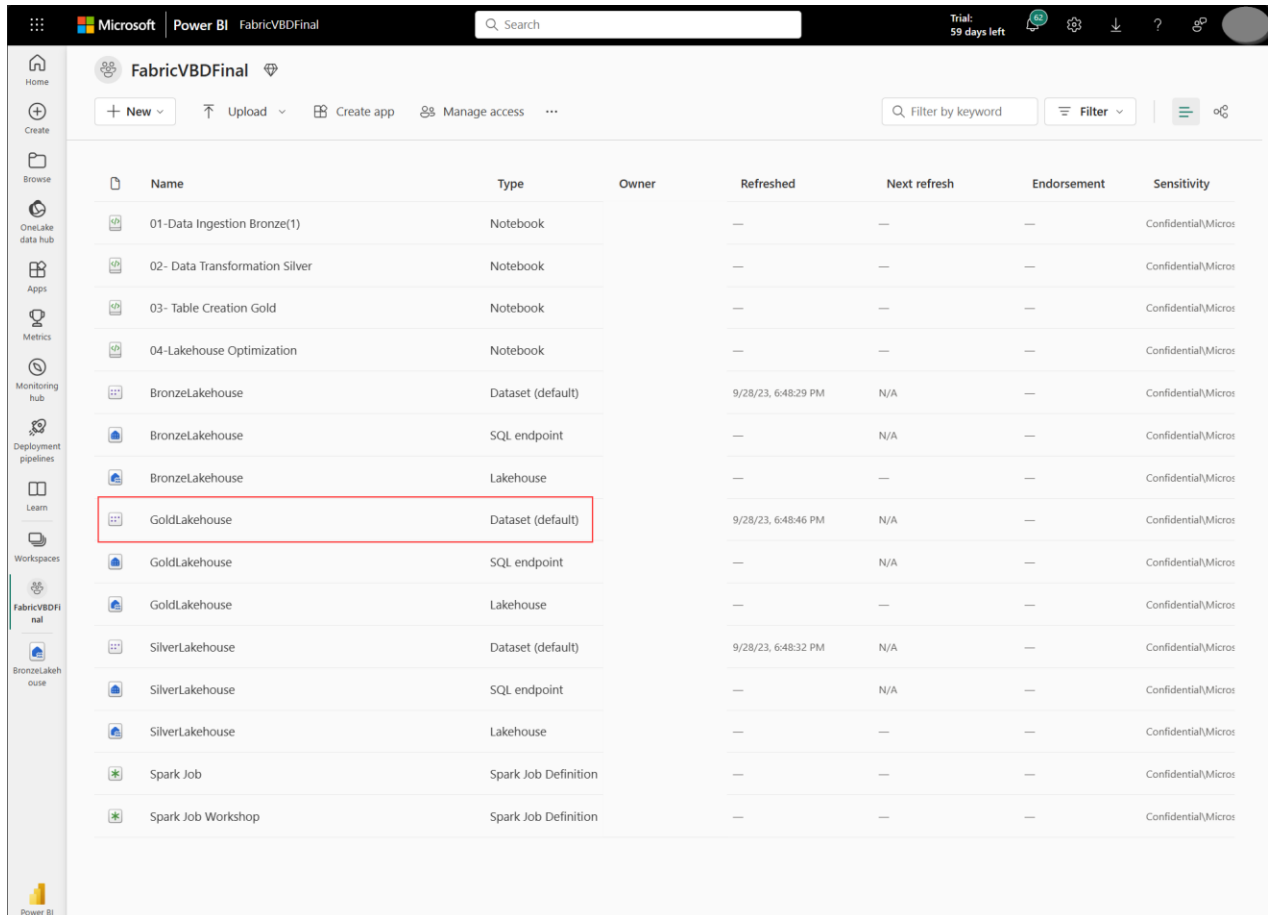
Go to your workspace where all items are shown:



The screenshot displays the Microsoft Synapse Data Engineering Fabric Lab interface. The top navigation bar includes the Microsoft logo, 'Synapse Data Engineering', 'Fabric Lab', a search bar, and a trial status indicator ('Trial: 58 days left'). The left sidebar shows navigation options: Home, Create, Browse, OneLake data hub, Monitoring hub, Workspaces, Fabric Lab (selected), Bronze, and Data Engineering. The main content area shows a list of items in a workspace, with columns for Name, Type, Owner, Refreshed, and Next refresh. The items are organized into a hierarchy: 01 Reading NYC Data (Notebook), 02 NYCTaxi with Holiday Data (Notebook), 03 NYCTaxi with Holiday and Weather (Notebook), 04 Lakehouse Optimization (Notebook), and a group of items under '00 Get NYC Taxi Data...'. The items include Bronze, Gold, and Silver datasets (default), SQL endpoints, and Lakehouses.

Name	Type	Owner	Refreshed	Next refresh
01 Reading NYC Data	Notebook		—	—
02 NYCTaxi with Holiday Data	Notebook		—	—
03 NYCTaxi with Holiday and Weather	Notebook		—	—
04 Lakehouse Optimization	Notebook		—	—
00 Get NYC Taxi Data... Bronze	Dataset (default)		9/21/23, 4:54:25 PM	N/A
Bronze	SQL endpoint		—	N/A
Bronze	Lakehouse		—	—
Gold	Dataset (default)		9/21/23, 4:54:33 PM	N/A
Gold	SQL endpoint		—	N/A
Gold	Lakehouse		—	—
Silver	Dataset (default)		9/21/23, 4:54:23 PM	N/A
Silver	SQL endpoint		—	N/A
Silver	Lakehouse		—	—

Find the dataset that corresponds to the Gold Lakehouse named **GoldLakehouse** and click the three-dot context menu link that appears when you hover over it in the list:



The screenshot shows the Microsoft Fabric VBD Final workspace. The left sidebar contains navigation icons for Home, Create, Browse, OneLake data hub, Apps, Metrics, Monitoring hub, Deployment pipelines, Learn, Workspaces, and Power BI. The main area displays a table of resources. The 'GoldLakehouse' dataset is highlighted with a red box.

Name	Type	Owner	Refreshed	Next refresh	Endorsement	Sensitivity
01-Data Ingestion Bronze(1)	Notebook		—	—	—	Confidential\Micros
02- Data Transformation Silver	Notebook		—	—	—	Confidential\Micros
03- Table Creation Gold	Notebook		—	—	—	Confidential\Micros
04-Lakehouse Optimization	Notebook		—	—	—	Confidential\Micros
BronzeLakehouse	Dataset (default)		9/28/23, 6:48:29 PM	N/A	—	Confidential\Micros
BronzeLakehouse	SQL endpoint		—	N/A	—	Confidential\Micros
BronzeLakehouse	Lakehouse		—	—	—	Confidential\Micros
GoldLakehouse	Dataset (default)		9/28/23, 6:48:46 PM	N/A	—	Confidential\Micros
GoldLakehouse	SQL endpoint		—	N/A	—	Confidential\Micros
GoldLakehouse	Lakehouse		—	—	—	Confidential\Micros
SilverLakehouse	Dataset (default)		9/28/23, 6:48:32 PM	N/A	—	Confidential\Micros
SilverLakehouse	SQL endpoint		—	N/A	—	Confidential\Micros
SilverLakehouse	Lakehouse		—	—	—	Confidential\Micros
Spark Job	Spark Job Definition		—	—	—	Confidential\Micros
Spark Job Workshop	Spark Job Definition		—	—	—	Confidential\Micros

Choose the Auto-create report option:

Microsoft | Synapse Data Engineering Fabric VBD Final

Search

Trial: 59 days left

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

Fabric VBD Final

+ New Upload Create app

Filter by keyword Filter

Name	Type	Owner	Refreshed	Next refresh
FilesTemp	Dataset (default)		9/19/23, 10:35:10 PM	N/A
FilesTemp	SQL endpoint		—	N/A
FilesTemp	Lakehouse		—	—
Gold	Dataset (default)		9/19/23, 12:57:19 PM	N/A
Gold			9/21/23, 7:47:53 PM	N/A
Gold			—	—
ij			—	—
Notebook 1			—	—
Silver			9/19/23, 12:56:47 PM	N/A
Silver			—	N/A
Silver			—	—
Spark Job CSV to Delta file	Spark Job Definition		—	—
Spark Job Definition	Notebook		—	—

...

Analyze in Excel

Create report

Auto-create report

Create paginated report

Security

Open data model

Settings

Manage permissions

View lineage

Microsoft

Synapse Data Engineering

Fabric VBD Final

Search

Trial:
59 days left

File

Export

Save

Edit

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Data Engineering

Generating your report...

Hint: To see the best results, try pre-selecting the data you're most interested in.

Pre-select data

-

+

85%



By now we have created PBI auto report using Direct Lake capability, you can add, or remove columns of choice from the right side of the screen to further customize the PBI report.

Lakehouse Performance Tuning

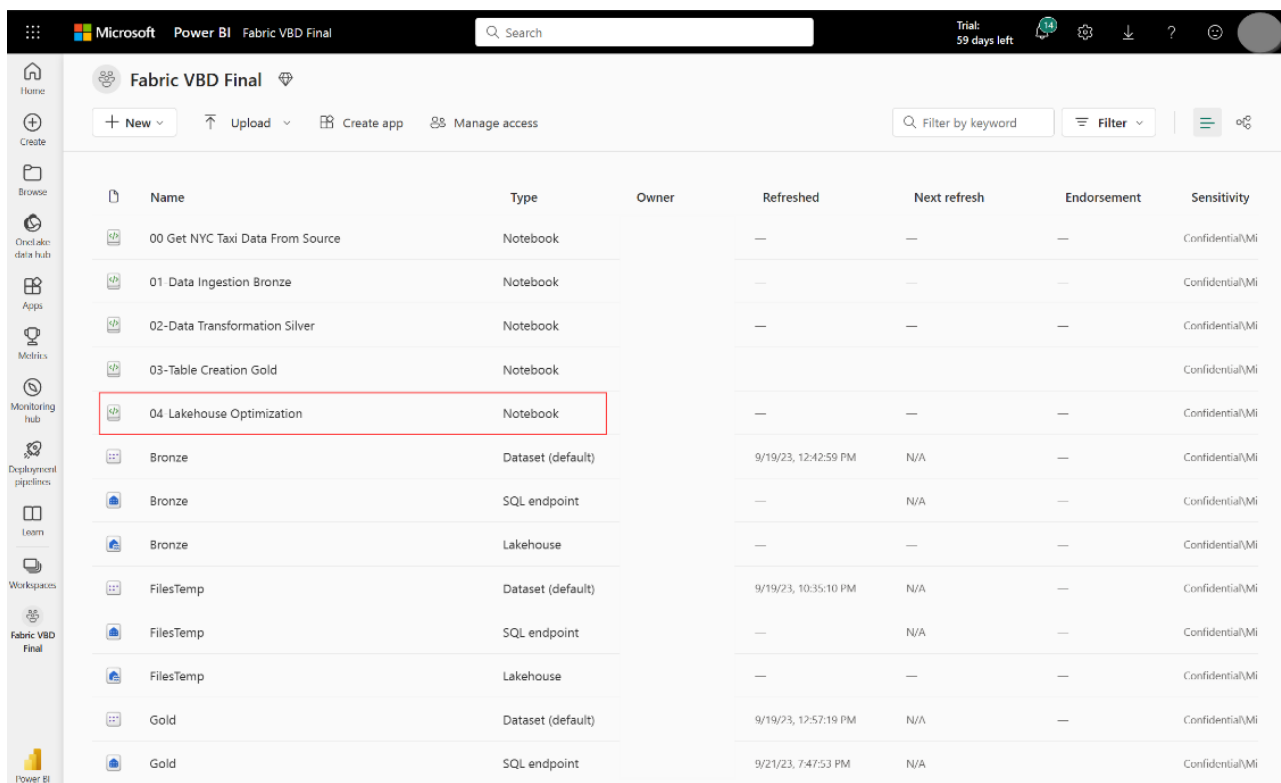
The Lakehouse and the Delta Lake table format are central to Microsoft Fabric, assuring that tables are optimized for analytics is a key requirement. This notebook covers Delta Lake table optimization concepts, configurations and how to apply it to most common Big Data usage patterns:

1. **V-order:** V-Order is a write time optimization to the parquet file format that enables lightning-fast reads under the Microsoft Fabric compute engines, such as Power BI, SQL, Spark and others. V-Order is enabled by default in Microsoft Fabric and in Apache Spark
2. **Optimize write:** Optimize Write is a Delta Lake on Microsoft Fabric and Azure Synapse Analytics feature in the Apache Spark engine that reduces the number of files written and aims to increase individual file size of the written data. It dynamically optimizes partitions while generating files with a default 128 MB size. The target file size may be changed per workload requirements using configurations.
3. **Partitioning:** Partitioning Delta table helps when data skipping if only a specific set of data is queried on the partitioned column . You should always partition the table with columns of low cardinality . Having too many partitions can lead to smaller files and performance degradation. Partitioning also helps with data deletion
4. **Z-order:** Partitioning Delta table helps when data skipping if only a specific set of data is queried on the partitioned column . You should always partition the table with columns of low cardinality . Having too many partitions can lead to smaller files and performance degradation. Partitioning also helps with data deletion.
5. **Merge optimization:** Delta Lake MERGE command allows users to update a delta table with advanced conditions. It can update data from a source table, view or DataFrame into a target table by using MERGE command. However, the current algorithm in the open source distribution of Delta Lake isn't fully optimized for handling unmodified rows. The Microsoft Spark Delta team implemented a custom Low Shuffle Merge optimization, unmodified rows are excluded from an expensive shuffling operation that is needed for updating matched rows. The implementation is controlled by the `spark.microsoft.delta.merge.lowShuffle.enabled` configuration, enabled by default in the runtime. It requires no code changes and is fully compatible with the open-source distribution of Delta Lake.
6. **Vaccum :** You can remove files no longer referenced by a Delta table and are older than the retention threshold by running the vacuum command on the table. vacuum

is not triggered automatically. The default retention threshold for the files is 7 days. You will lose the ability to time travel beyond the retention threshold after running Vacuum on a table.

Notebook 04 – Lakehouse Optimization

Open the **04-Lakehouse optimization** notebook that you imported in section 1. Proceed with each execution of notebook. Each performance technique is further illustrated in the notebook.



The screenshot shows the Microsoft Fabric VBD Final workspace. The top navigation bar includes the Microsoft logo, 'Power BI', 'Fabric VBD Final', a search bar, and a trial status 'Trial: 59 days left'. The left sidebar contains navigation icons for Home, Create, Browse, On-look data hub, Apps, Metrics, Monitoring hub, Deployed pipelines, Learn, Workspaces, and Fabric VBD Final. The main content area displays a table of items with columns: Name, Type, Owner, Refreshed, Next refresh, Endorsement, and Sensitivity. The notebook '04 Lakehouse Optimization' is highlighted with a red box.

Name	Type	Owner	Refreshed	Next refresh	Endorsement	Sensitivity
00 Get NYC Taxi Data From Source	Notebook		—	—	—	Confidential\Mi
01 Data Ingestion Bronze	Notebook		—	—	—	Confidential\Mi
02-Data Transformation Silver	Notebook		—	—	—	Confidential\Mi
03-Table Creation Gold	Notebook		—	—	—	Confidential\Mi
04 Lakehouse Optimization	Notebook		—	—	—	Confidential\Mi
Bronze	Dataset (default)		9/19/23, 12:42:59 PM	N/A	—	Confidential\Mi
Bronze	SQL endpoint		—	N/A	—	Confidential\Mi
Bronze	Lakehouse		—	—	—	Confidential\Mi
FilesTemp	Dataset (default)		9/19/23, 10:35:10 PM	N/A	—	Confidential\Mi
FilesTemp	SQL endpoint		—	N/A	—	Confidential\Mi
FilesTemp	Lakehouse		—	—	—	Confidential\Mi
Gold	Dataset (default)		9/19/23, 12:57:19 PM	N/A	—	Confidential\Mi
Gold	SQL endpoint		9/21/23, 7:47:53 PM	N/A	—	Confidential\Mi

Spark Job Definition

An Apache Spark Job Definition is a Microsoft Fabric code item that allows you to submit batch/streaming job to Spark cluster. By uploading the binary files from compilation output of different languages, .jar from Java for example, you can apply different transformation logic to the data hosted on lakehouse. Besides the binary file, you can further customize the behavior of the job by uploading additional libraries and command line arguments.

To run a Spark job definition, you must have at least one lakehouse associated with it. This default lakehouse context serves as the default file system for Spark runtime. For any Spark code using relative path to read/write data, the data is served from the default lakehouse.

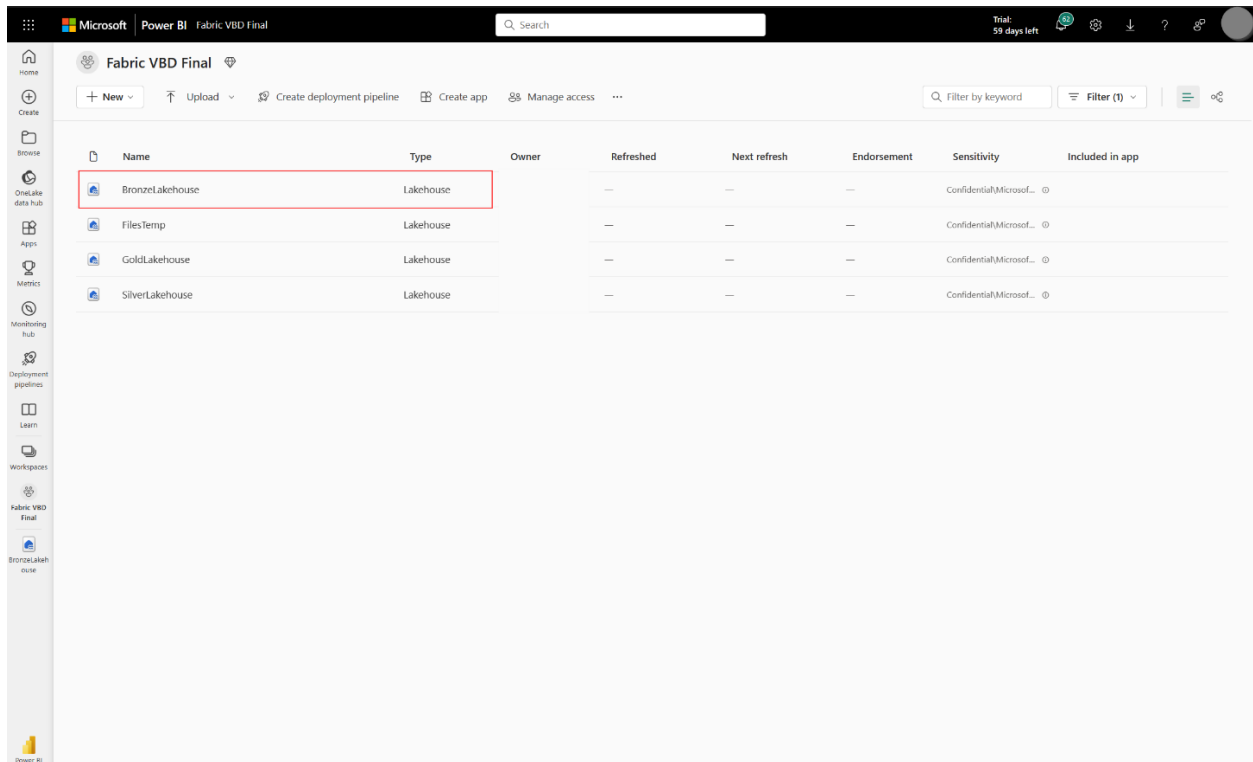
The goal of our Spark job is to convert a parquet file to delta table. In this exercise we will write the Holiday parquet file as a delta table into our lakehouse.

In the next steps described below, you will upload a Python file. First, you must download this Python file and make some changes:

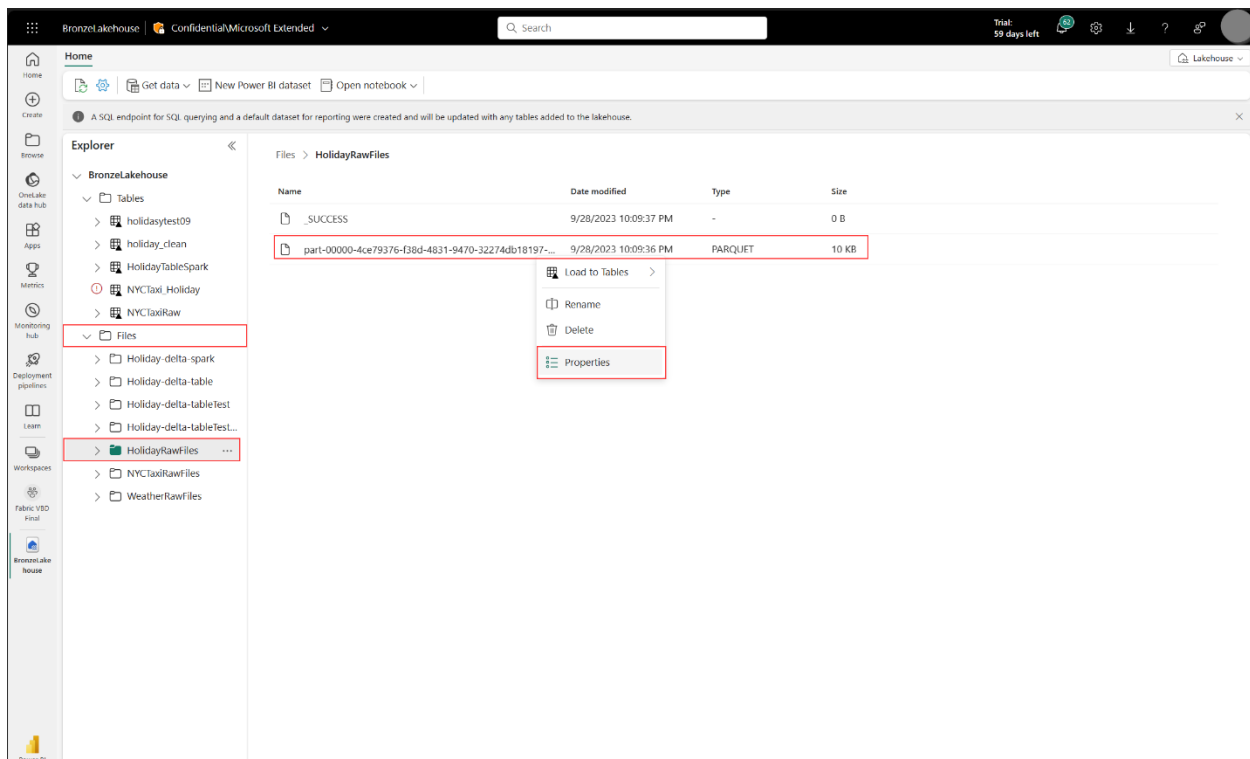
1. Download the Main job definition file from [Here](#) and open it
2. The *Spark Job.py* file includes comments to understand better what the code is doing but here is a summary of the important steps:
 - Import necessary libraries and constants
 - Create a spark session to run the code
 - Provide the parquet source file path
 - Provide the delta Table destination path
 - Read the parquet file from the path provided
 - Write the parquet file as delta table in the provided path
3. Before we proceed with job creation we need to edit the file *Spark Job.py* downloaded in previous step and update the location for your data files from the

BronzeLakehouse which we are going to use as part of this Spark job. To find the file location path in your lakehouse, follow these steps:

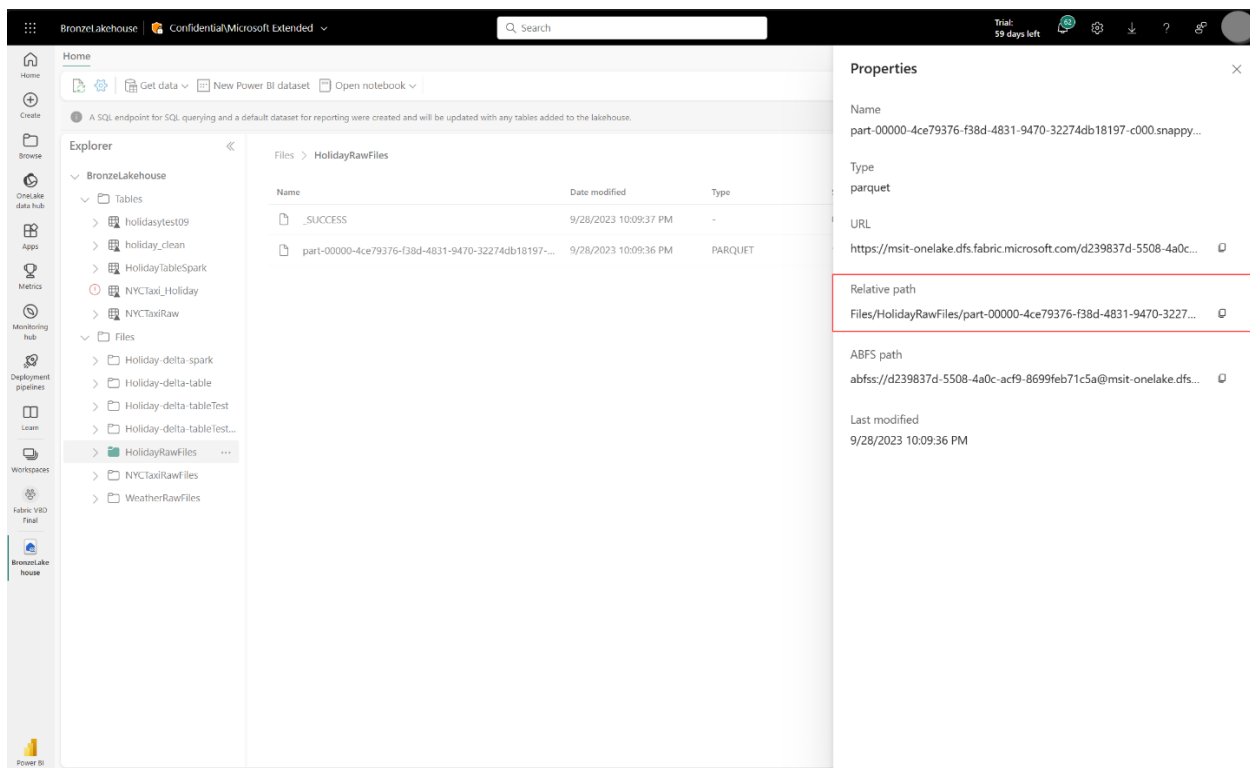
- Go to the BronzeLakehouse from your workspace overview:



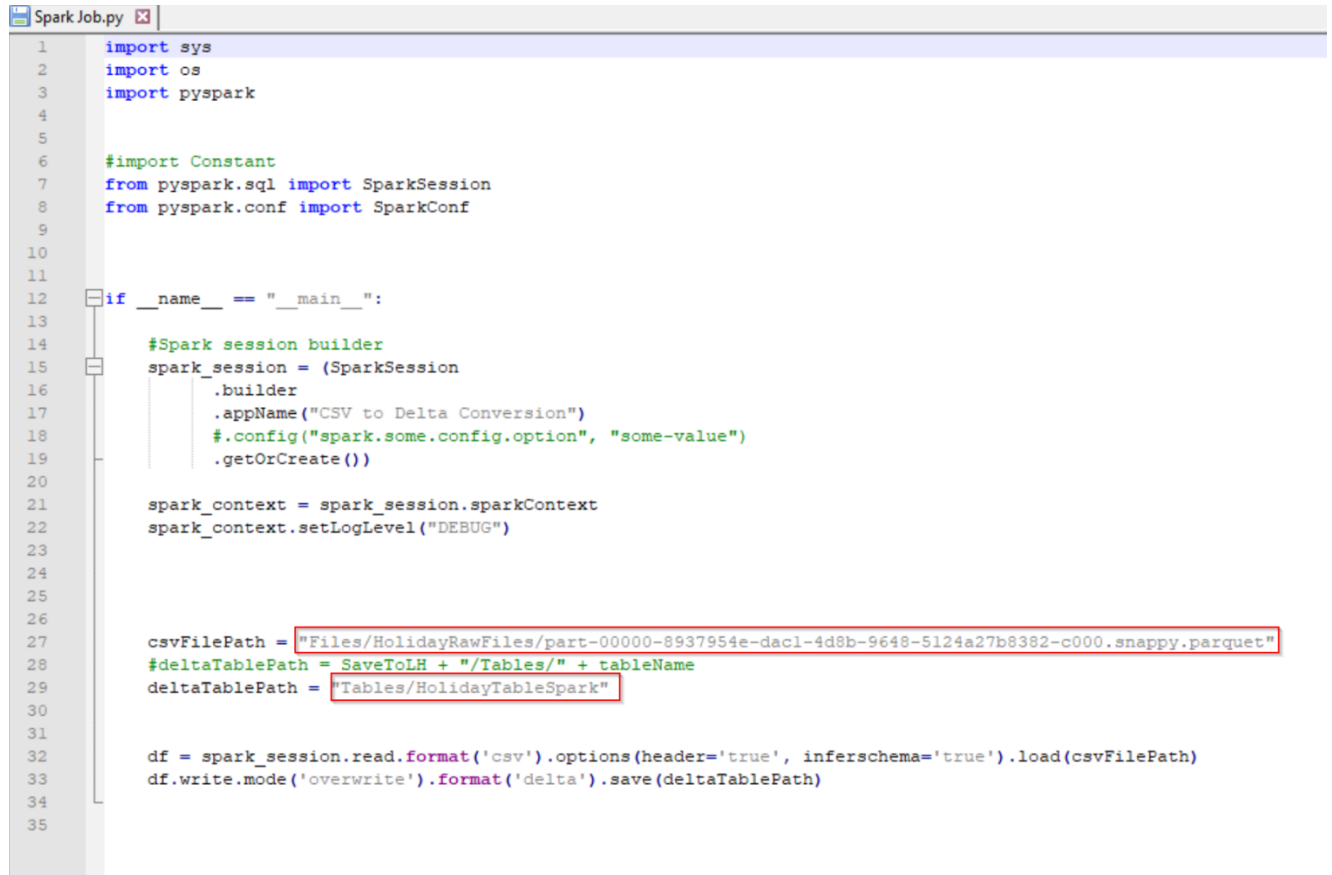
- Go to the Files section and click on the *HolidayRawFiles*. On the right side, you will see the parquet file. Click on the three dots of the parquet file as shown in the picture below and then click on Properties:



- Copy the relative path:



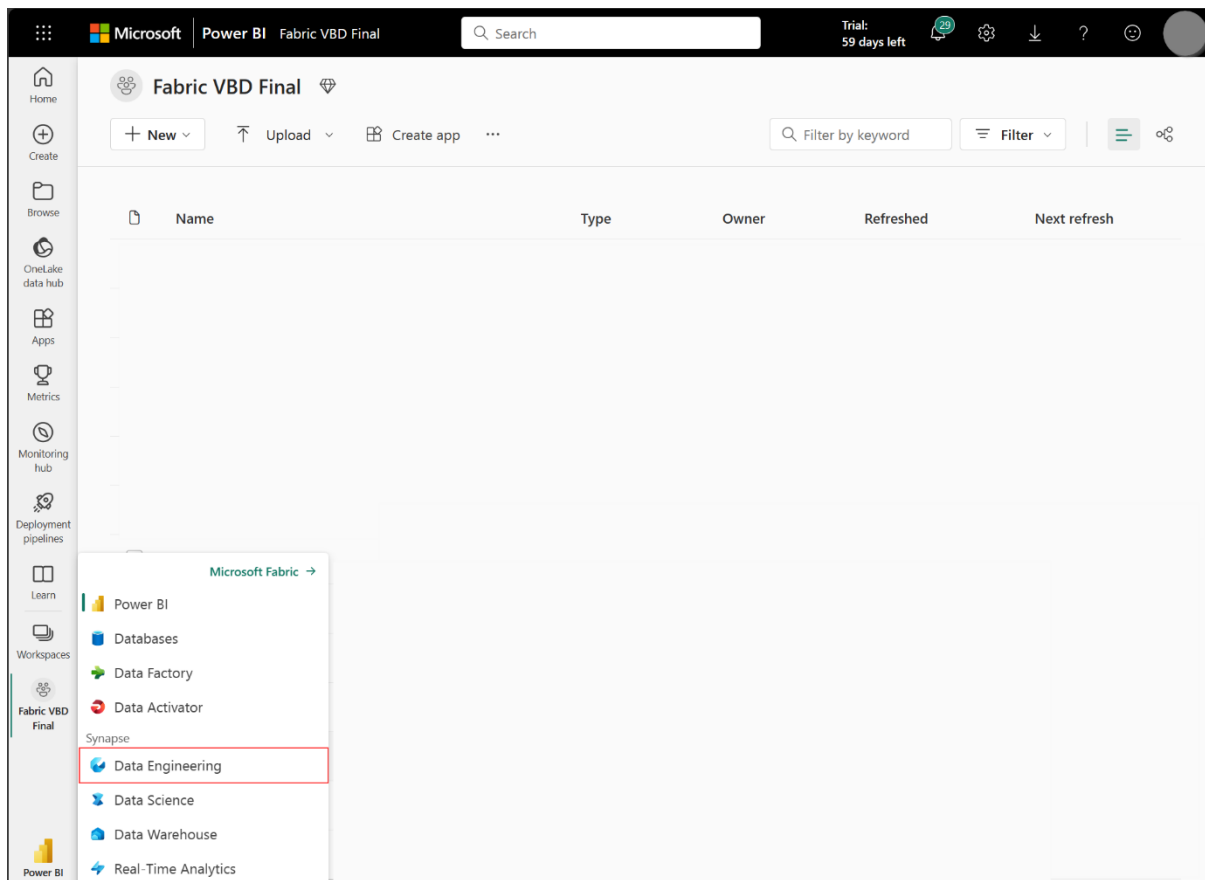
- a. Once you have copied the file path from *BronzeLakehouse* we need to update this path in *Spark Job.py* file. We need to modify line 27 and update the relative path you just copied in the previous step.
- b. The line 29 will contain the table name, in our case: *HolidayTableSpark*



```
1  import sys
2  import os
3  import pyspark
4
5
6  #import Constant
7  from pyspark.sql import SparkSession
8  from pyspark.conf import SparkConf
9
10
11
12  if __name__ == "__main__":
13
14      #Spark session builder
15      spark_session = (SparkSession
16                      .builder
17                      .appName("CSV to Delta Conversion")
18                      .config("spark.some.config.option", "some-value")
19                      .getOrCreate())
20
21      spark_context = spark_session.sparkContext
22      spark_context.setLogLevel("DEBUG")
23
24
25
26
27      csvFilePath = "Files/HolidayRawFiles/part-00000-8937954e-dacl-4d8b-9648-5124a27b8382-c000.snappy.parquet"
28      #deltaTablePath = SaveToLH + "/Tables/" + tableName
29      deltaTablePath = "Tables/HolidayTableSpark"
30
31
32      df = spark_session.read.format('csv').options(header='true', inferschema='true').load(csvFilePath)
33      df.write.mode('overwrite').format('delta').save(deltaTablePath)
34
35
```

- c. After you modify the *Spark Job.py* file, save it

After saving the Python file it's time to create your first spark job definition. You have to go to the Data Engineering part as shown here:



After you land on the Data Engineering homepage, click on Spark job definition:

Microsoft

Synapse Data Engineering

Home

Search

Trial

59 days left

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

FabricVBDFinal

Data Engineering

New

Current workspace: FabricVBDFinal

Items will be saved to this workspace.

Lakehouse (Preview)

Notebook (Preview)

Spark Job Definition ...

Data pipeline (Preview)

Import notebook

Use a sample

Recommended

You frequently open this

My workspace

You favorited this

01-Data Ingestion Bronze

You frequently open this

03-Table Creation Gold

You favorited this

02-Data Transformation Silver

You frequently o

00 Get NYC Taxi

Quick access

Recent

Favorites

Filter by keyword

Filter

Name	Type	Opened	Owner	Endorsement	Sensitivity
FabricVBDFinal	Workspace	now	—	—	—
My workspace	Workspace	a minute ago	—	—	—

Choose a name:

Microsoft

Synapse Data Engineering

Home

Search

Trial: 59 days left

52

?

Home

Create

Browse

OnLake data hub

Monitoring hub

Workspaces

FabricVBDFI nal

Data Engineering

New

Current workspace: FabricVBDFinal

Items will be saved to this workspace.

Lakehouse (Preview)

Notebook (Preview)

Spark Job Definition ...

Data pipeline (Preview)

Import notebook

Use a sample

Recommended

You frequently open this

You frequently open this

You favoredited this

My workspace

01-Data Ingestion Bronze

03-Table Creation Gold

02-Data Transformation Si

Quick access

Recent

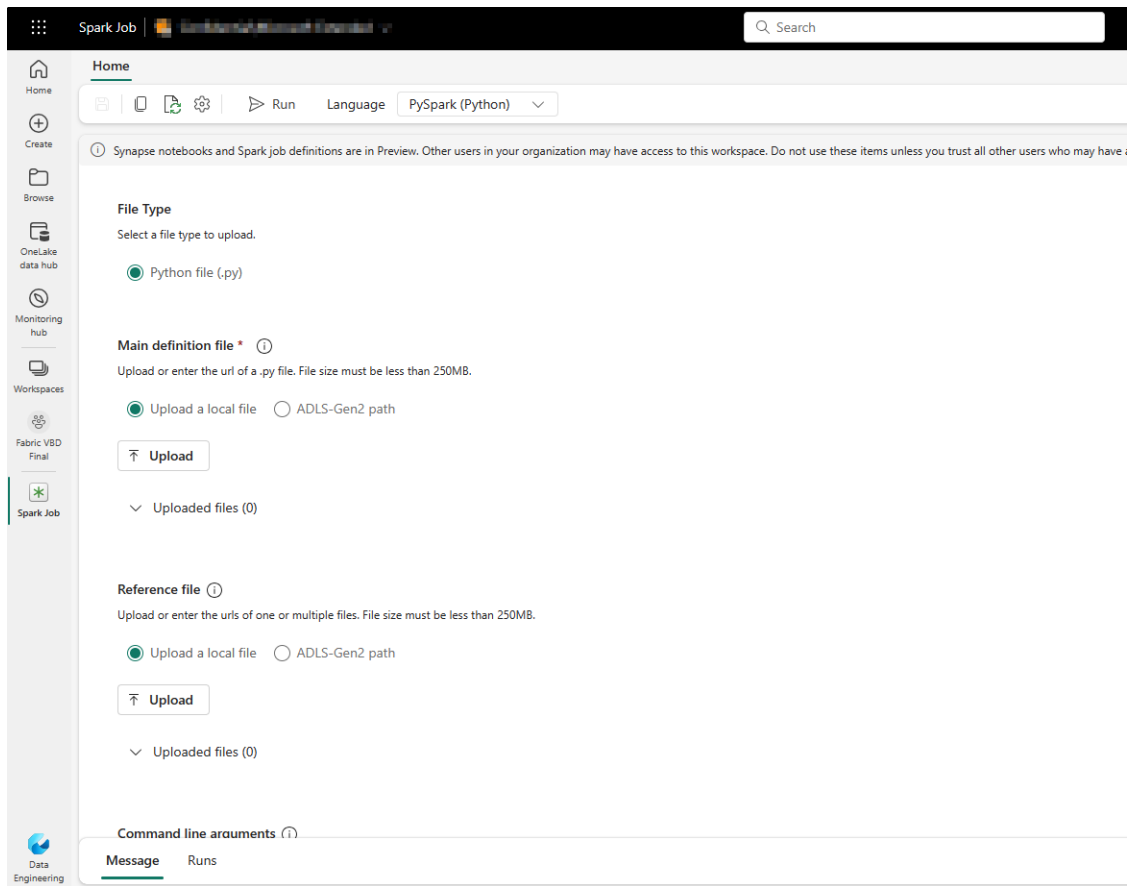
Favorites

Filter by keyword

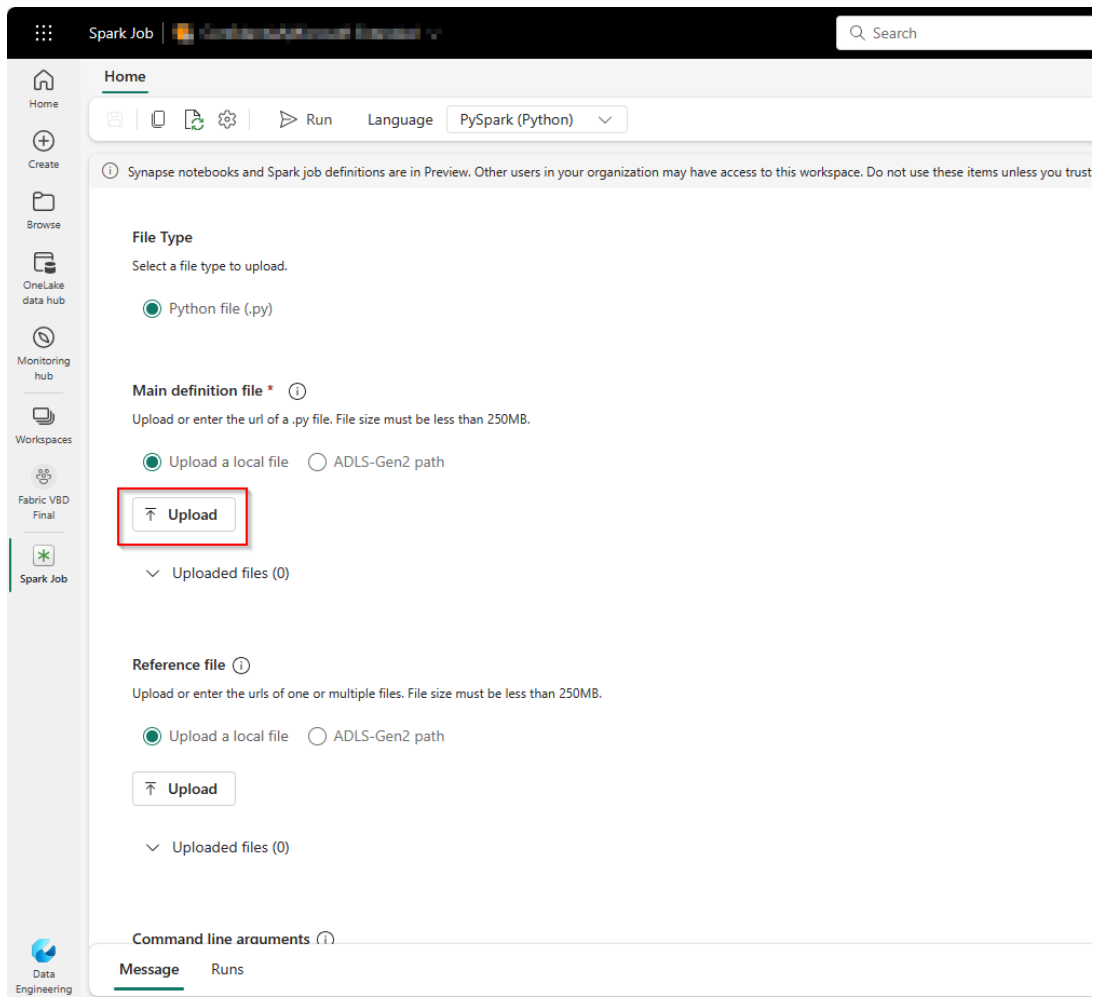
Filter

Name	Type	Opened	Owner
FabricVBDFinal	Workspace	now	—
My workspace	Workspace	a minute ago	—

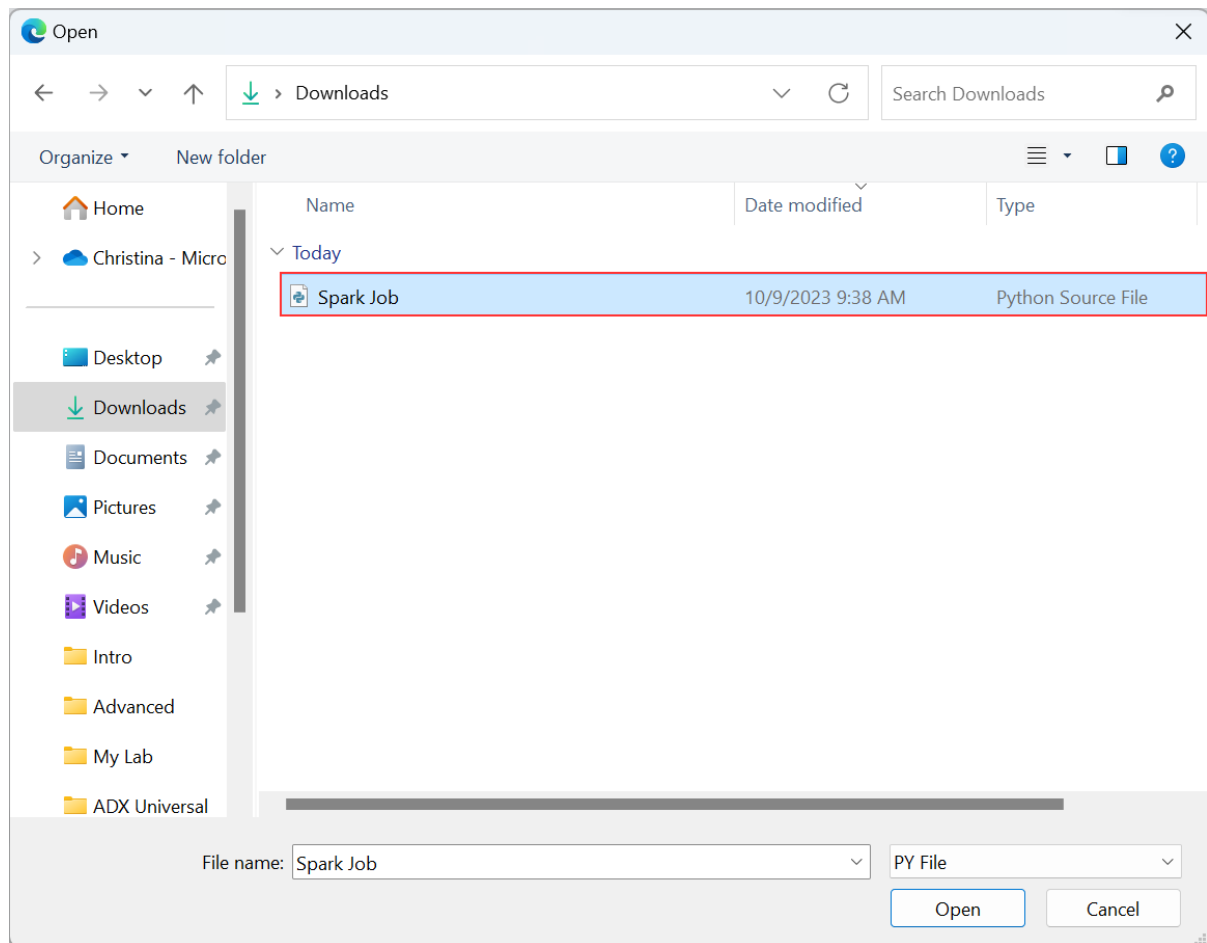
Your Spark Job Definition will be created:



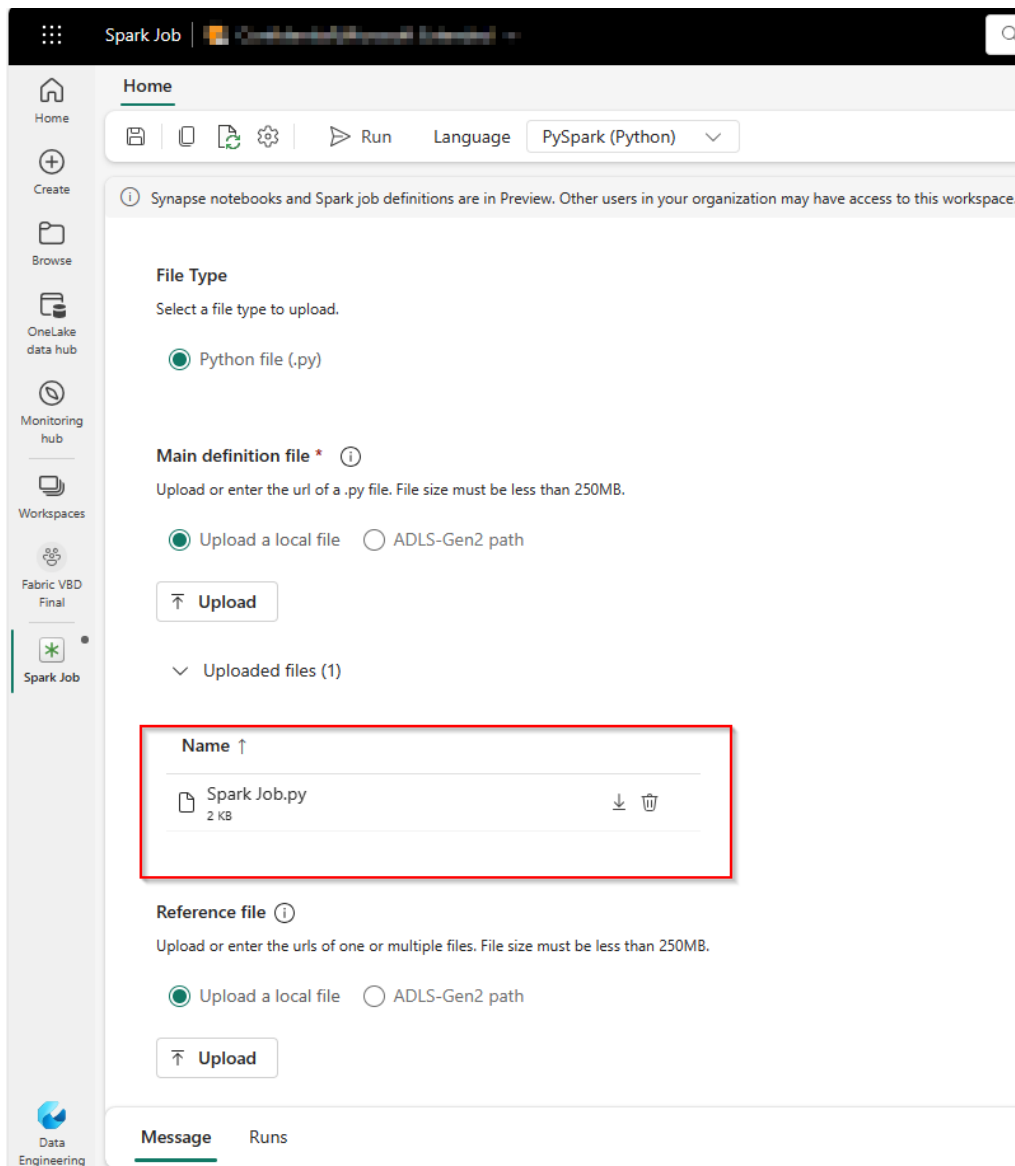
1. Leave the File type as it is
2. For the main definition file, choose the option Upload a local file and then click on Upload.



4. Choose the *Spark Job.py* file you downloaded and edited earlier.



5. After the *Spark Job.py* file is uploaded you will see it in the uploaded files section.



6. We are not uploading any reference file and command line arguments in this exercise. Leave them as it-is.

The next step is to choose a Lakehouse reference. As mentioned above, to run a spark job definition you must have at least one lakehouse associated with it. This default lakehouse context serves as the default file system for Spark runtime.

Click on Add on the Lakehouse reference section:

⋮

Spark Job

🔍

Search

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Spark Job

Data Engineering

Home

📁 📄 ⚙️ ▶️ Run Language PySpark (Python) ▼

ⓘ

Synapse notebooks and Spark job definitions are in Preview. Other users in your organization may have access to this workspace. Do not use these

⬆️ Upload

▼ Uploaded files (1)

Name ↑

📄 Spark Job.py

2 KB

⬇️ 🗑️

Reference file ⓘ

Upload or enter the urls of one or multiple files. File size must be less than 250MB.

☒ Upload a local file

☐ ADLS-Gen2 path

⬆️ Upload

▼ Uploaded files (0)

Command line arguments ⓘ

e.g. "abfss://.../path/to/shakespeare.txt abfss://.../path/to/result"

Lakehouse Reference * ⓘ

+ Add

Message

Runs

Choose the data you want to connect

All My data Endorsed in your org Favorites

Search: bronzelakehouse X

Explorer >							
		Name	Owner	Refreshed	Location	Endorsement	Sensitivity
	☐	bronzelakehouse	Michael (Michael)	-	Fabric Workshop VBD	-	Confidential (Microsoft) L1
	☐	bronzelakehouse	James (James)	-	Fabric in a Day Workshop	-	Confidential (Microsoft) L1
☒		BronzeLakehouse	Mike (Michael)	-	Fabric VBD Final	-	Confidential (Microsoft) L1

Add

Cancel

Choose your *BronzeLakehouse*:

After the Lakehouse is selected, you will see it in the All Lakehouses section:

Fabric VBD
Final

Spark Job

Data
Engineering

✓ Uploaded files (0)

Command line arguments ⓘ
e.g. "abfss://.../path/to/shakespeare.txt abfss://.../path/to/result"

Lakehouse Reference * ⓘ
+ Add
✓ All lakehouses (1)

Name ↑		Workspace
📄 BronzeLakehouse	📌	Fabric VBD Final

Message Runs

Click Save on the top left corner to Save the Job

Spark Job

Configuration | Resources | Help

Search

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Spark Job

Data Engineering

File Type

Select a file type to upload.

☒ Python file (.py)

Main definition file *

Upload or enter the url of a .py file. File size must be less than 250MB.

☒ Upload a local file ☐ ADLS-Gen2 path

Upload

Uploaded files (1)

Name ↑

Spark Job.py

2 KB

Download

Delete

Reference file

Upload or enter the urls of one or multiple files. File size must be less than 250MB.

☒ Upload a local file ☐ ADLS-Gen2 path

Upload

Message

Runs

The last step is to click Run on the top of the page to run your Spark Job:

Spark Job

Search

Home

Create

Browse

OneLake data hub

Monitoring hub

Workspaces

Fabric VBD Final

Spark Job

Data Engineering

Home

File

Code

Settings

Run

Language PySpark (Python)

Synapse notebooks and Spark job definitions are in Preview. Other users in your organization may have access to this workspace. Do not use these items unless you trust all other users who may have access to this workspace.

File Type

Select a file type to upload.

☒ Python file (.py)

Main definition file *

Upload or enter the url of a .py file. File size must be less than 250MB.

☒ Upload a local file ☐ ADLS-Gen2 path

Upload

Uploaded files (1)

Name ↑

Spark Job.py

2 KB

Download

Delete

Reference file

Upload or enter the urls of one or multiple files. File size must be less than 250MB.

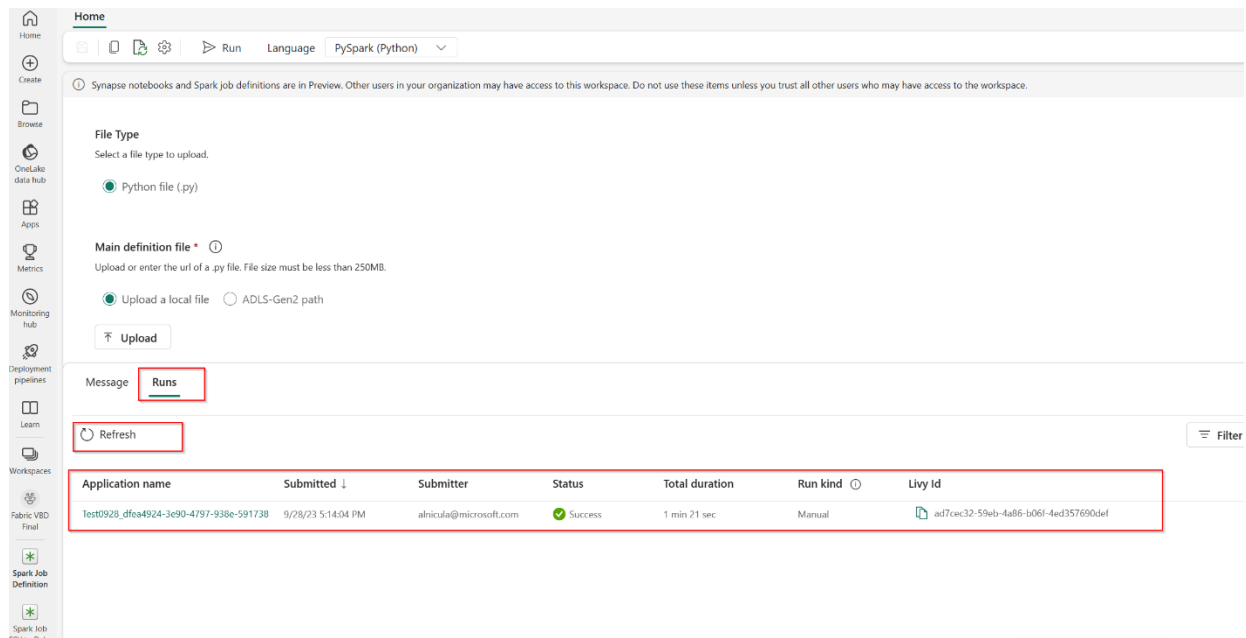
☒ Upload a local file ☐ ADLS-Gen2 path

Upload

Message

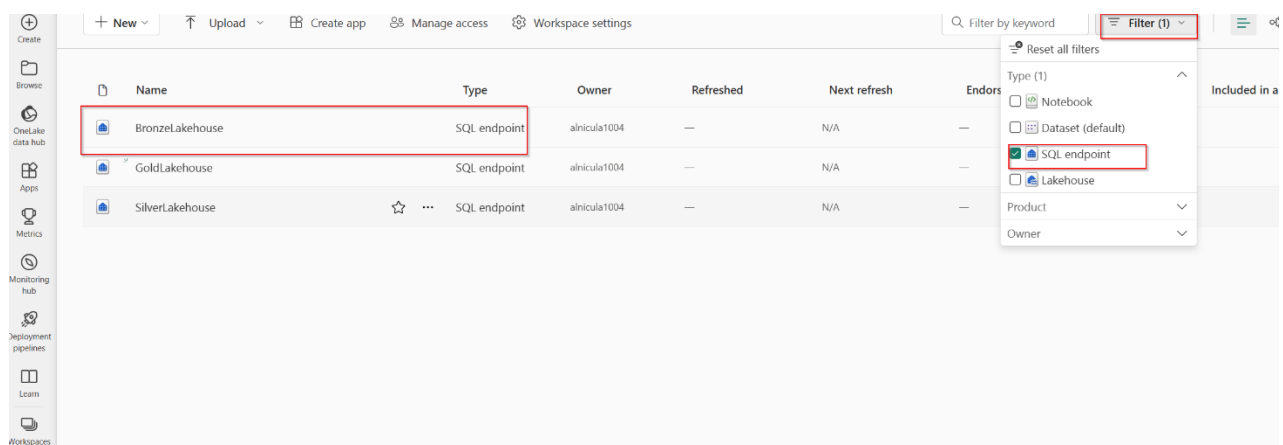
Runs

Once you started the Spark job, click on Run and hit for refresh:

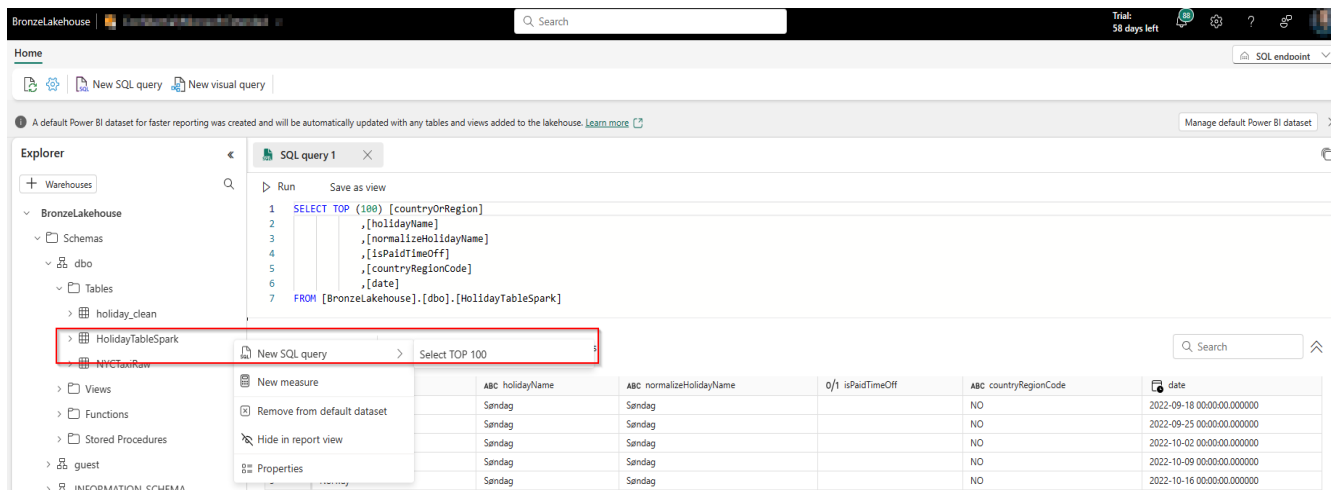


After the job has successfully run, navigate to your Workspace, Go to Filter, and check the SQL endpoint only.

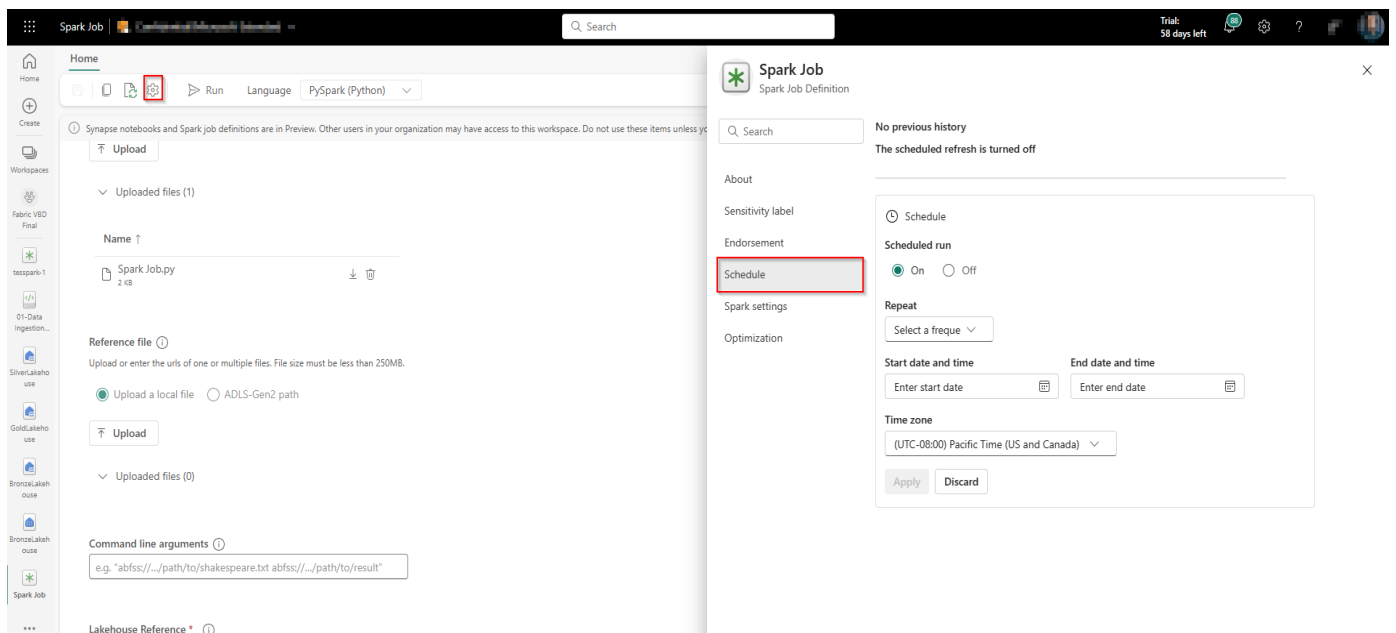
Select the BronzeLakehouse. You should see your table under the Tables section



Click on the newly created table, more options and then Select Top 100 to explore the dataset:



There is also an option to schedule your Spark Job, to do so click on the setting and then schedule to see the available scheduling options



At this point we have created a Delta table from the Parquet file using Spark job using this example, similarly, you can automate other lakehouse tasks/jobs using Spark job definitions.

Further Reading

[Create an Apache Spark job definition - Microsoft Fabric | Microsoft Learn](#)

Lakehouse Administration

In this workshop, you have learned how to accomplish typical data engineering tasks, like using notebooks for data ingestion and using Power BI to surface new insights. In this section, we will look at how you can administer and maintain your Fabric Lakehouse to help your users effectively access the data.

Prerequisites

Before continuing with the exercises in this section, you will need to prepare by completing some prerequisites.

1. Be sure you have access to an ADLS Gen2 storage account in your Azure subscription. If you do not already have an account, you can create a new one: [Create a storage account for Azure Data Lake Storage Gen2](#)
2. If you created a new storage account for the previous prerequisite, be sure that you also create a storage container in the account: [Create a container](#)
3. Create a SAS token for the storage container that has read, write, delete, list, add, create, update, and process permissions: [Create shared access signature \(SAS\) tokens for storage containers and blobs](#)

Managing Shortcuts

By default, new datasets you create in Microsoft Fabric will be stored in OneLake. However, modern enterprises have data stored in many other locations, including pre-existing Azure storage accounts and (for some reason) AWS S3 buckets. OneLake makes it easy to work with data stored in a variety of locations through a system called “shortcuts”. These allow an administrator to create a centrally managed pointer to another storage location. Fabric users simply reference the pointer in their queries without worrying about the underlying details of the storage connection.

A shortcut is a symbolic link which points from one data location to another. It is created to make data from a warehouse part of the lakehouse, to consolidate data across items or workspaces without changing the ownership of the data. Data can be reused multiple

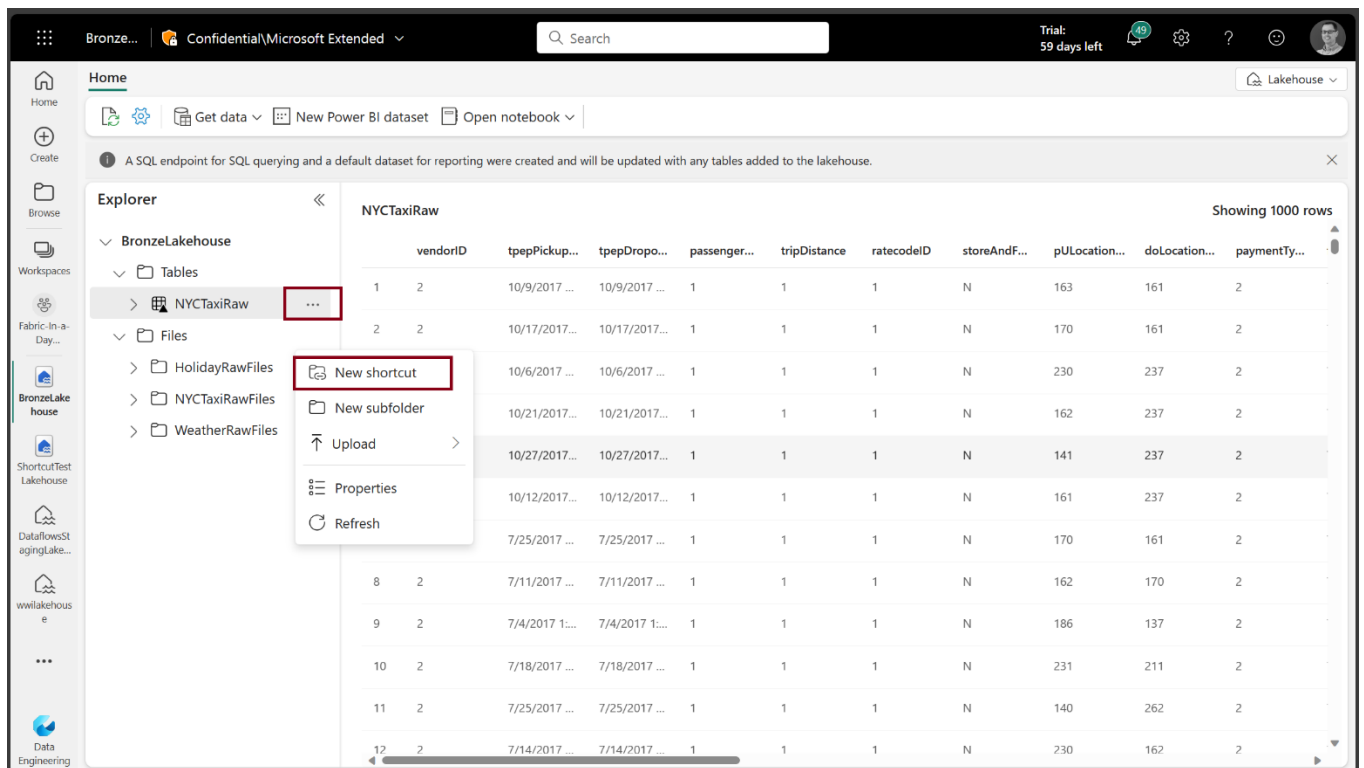
times without data duplication and existing Azure storage accounts and Amazon S3 buckets can be managed externally to Fabric and Microsoft while still being virtualized into OneLake with shortcuts. All data is mapped to a unified namespace and can be accessed using the same APIs, including the ADLS Gen2 DFS APIs.

Let's begin by creating a shortcut to an Azure storage account. We will create the new shortcut in the BronzeLakehouse that you created earlier in this workshop. For this exercise, we will use the Azure Storage account that you created as part of the prerequisites for this workshop.

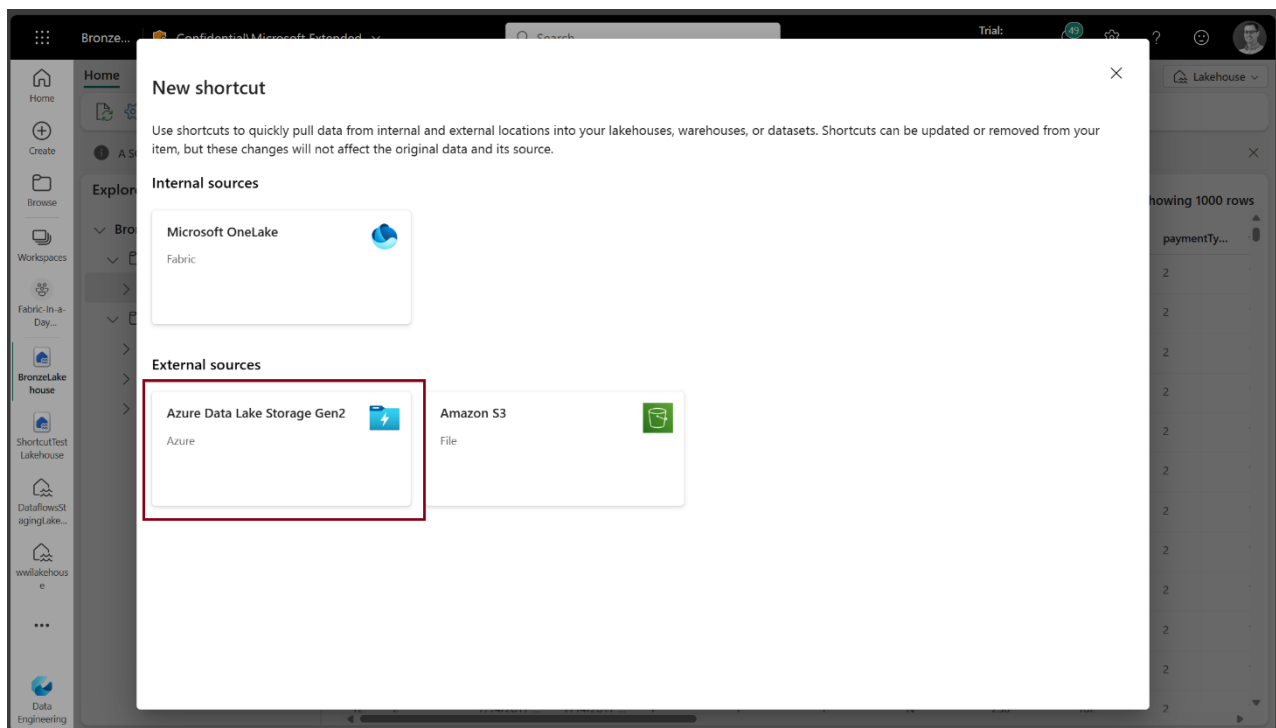
1. In the Fabric portal, click “Browse” in the toolbar on the left edge of the screen. In the list of workspace resources, click on the BronzeLakehouse.

Name	Type	Opened
01-Data Ingestion Bronze	Notebook	4 hours ago
03- Clone Table Creation Gold Copy	Notebook	5 hours ago
Fabric VBD Final	Workspace	5 hours ago
02- Clone Data Transformation Silver	Notebook	5 hours ago
Fabric-In-a-Day Workshop	Workspace	6 hours ago
01-Data Ingestion Bronze	Notebook	6 hours ago
SilverLakehouse	Lakehouse	6 hours ago
BronzeLakehouse	Lakehouse	7 hours ago
GoldLakehouse	Lakehouse	7 hours ago
ShortcutTestLakehouse	Lakehouse	a day ago
JP Data Science	Warehouse	a day ago
Gold	Lakehouse	a day ago
Bronze	Lakehouse	a day ago

2. In the Explorer pane, click the three dots next to the “Files” header to open the context menu. Select “New shortcut” from the menu. This will open the New Shortcut Wizard.



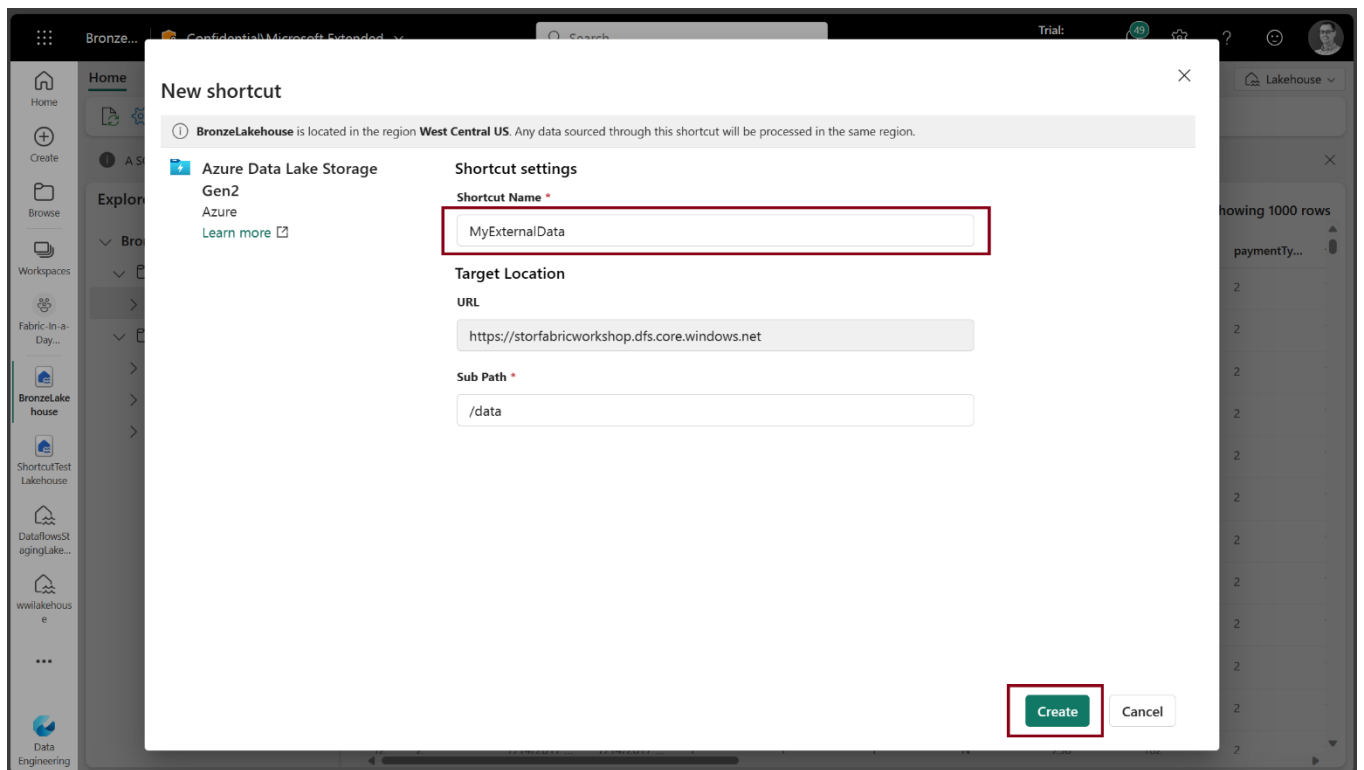
3. On the first page of the wizard, select “Azure Data Lake Storage Gen2”



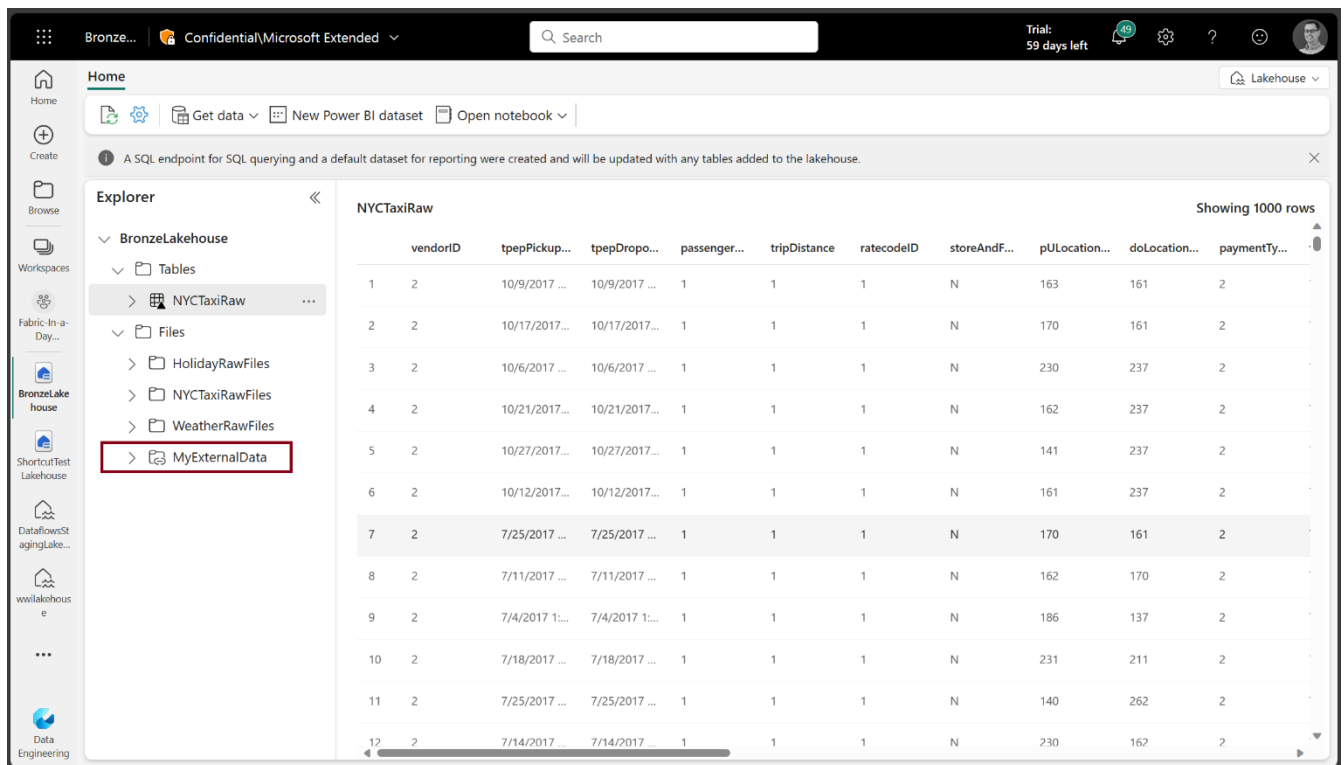
4. On the next page of the wizard, you will provide the details for connecting to your storage account.
 - a. In the “URL” field, enter the URL that points to your storage account. It will use the following format:
`https://<account name>.dfs.core.windows.net/<container name>`
 - b. For “Authentication kind”, select Shared Access Signature (SAS)
 - c. In the “SAS token” field, paste the SAS token that you generated as part of the prerequisites for this workshop.
 - d. Once you’ve entered the details, click the “Next” button at the bottom of the screen.

The screenshot shows the 'New shortcut' dialog box in the Microsoft Fabric interface. The dialog is titled 'New shortcut' and has a close button (X) in the top right corner. Below the title bar, there is a note: 'BronzeLakehouse is located in the region West Central US. Any data sourced through this shortcut will be processed in the same region.' The main content area is divided into two sections: 'Connection settings' and 'Connection credentials'. In the 'Connection settings' section, the 'URL' field is highlighted with a red arrow and contains the text 'https://storfabricworkshop.dfs.core.windows.net/data'. In the 'Connection credentials' section, the 'Authentication kind' dropdown is highlighted with a red arrow and shows 'Shared Access Signature (SAS)'. The 'SAS token' field is also highlighted with a red arrow and contains a series of asterisks. At the bottom right of the dialog, the 'Next' button is highlighted with a red arrow. The background shows the Microsoft Fabric interface with a sidebar on the left and a main area on the right.

5. On the last page of the wizard, enter “MyExternalData” as the name of the shortcut. Then click the “Create” button.



You will now see your shortcut listed in the lakehouse explorer pane.



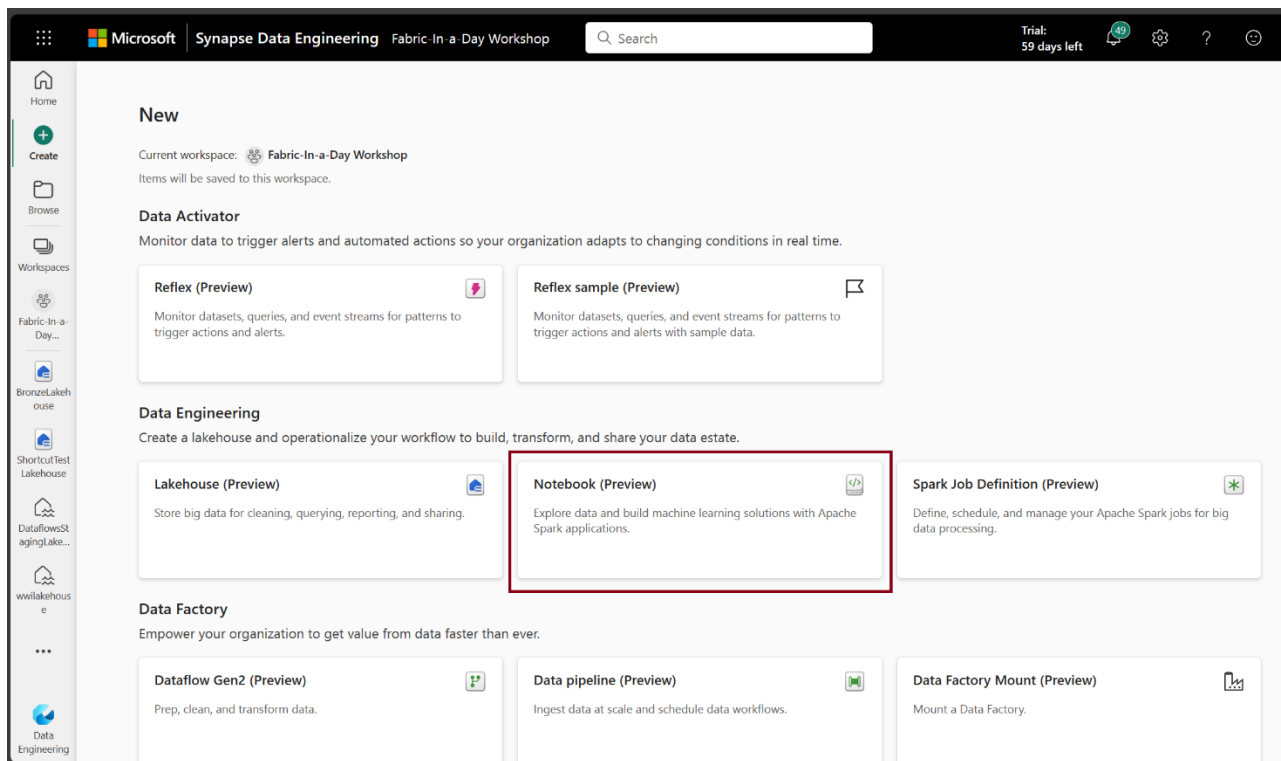
Now that you have created a shortcut, let's see how we can use that in our queries.

To begin with, we will need some data to query. Download the following CSV file:

<https://azuresynapsestorage.blob.core.windows.net/sampledData/SearchLog/SearchLog.csv>

Upload the CSV file to the root of the container to which you linked with the shortcut.

In the Fabric portal, click the “Create” button on the left side of the screen. Then select “Notebook” to create a new notebook.



In the new notebook, use the “Lakehouses” panel on the left of the screen to connect to the BronzeLakehouse.

In the first cell of the notebook, enter the following code:

```
df = spark.read.csv("Files/MyExternalData/SearchLog.csv")
display(df)
```

In the second cell of the notebook, enter the following code:

```
df = spark.sql("SELECT * FROM csv.`Files/MyExternalData/SearchLog.csv` LIMIT 1000")
display(df)
```

Notice that using both PySpark and Spark SQL, you were able to access the file in your external storage account simply by referencing the shortcut. You did not have to provide any connection details for the storage account in your code!

Lakehouse Sharing

In some cases, you may wish to share your lakehouse data with other users in your organization but you do not want to give them full access to your workspace or to the other items in the workspace. In this scenario, you can use Lakehouse Sharing to give access to a user or to a group of users.

When you share your lakehouse, you are also granting access to the associated SQL endpoint and to the endpoint's default dataset. By default, this will be read-only access. Once your lakehouse has been shared, users will see it listed under the “Shared With Me” section in the Fabric UI.

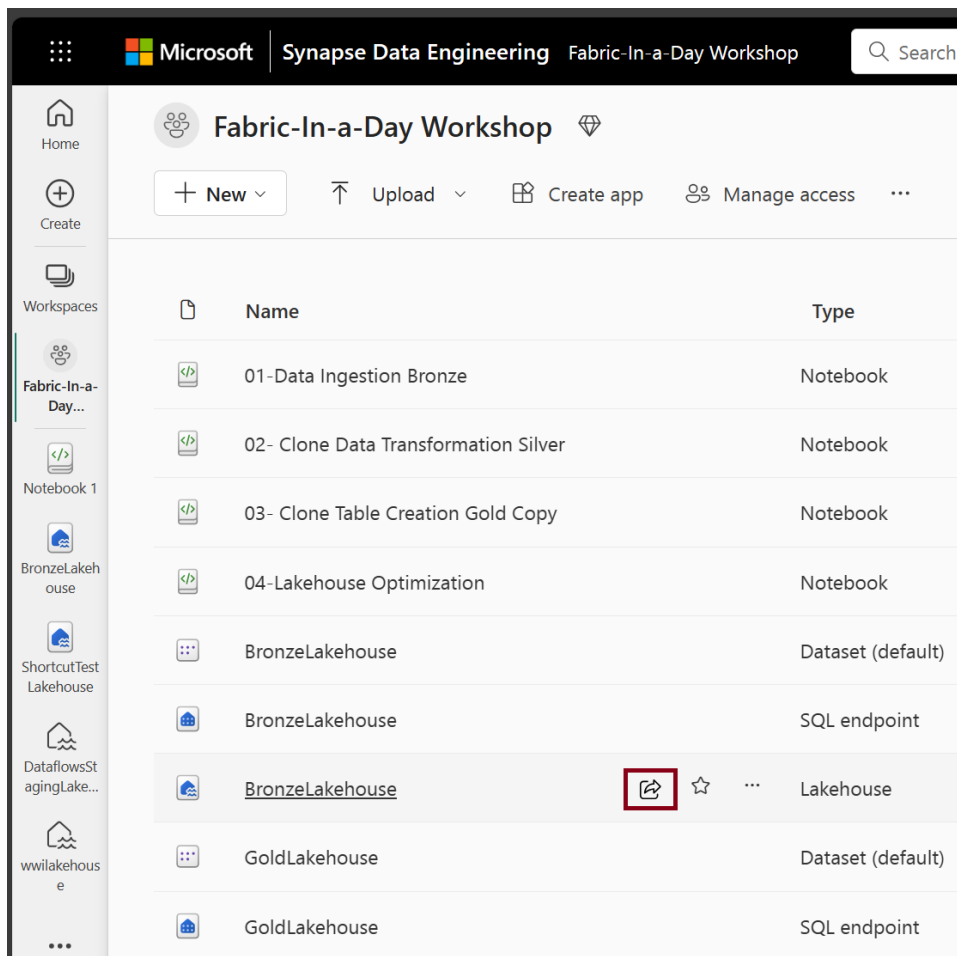
Let's walk through how to share your lakehouse with another user.

1. In Fabric, click the “Workspaces” button in the toolbar on the left side of the screen.
2. A list of all of your Fabric workspaces will be shown. Find the workspace containing the lakehouse that you would like to share. Click on the workspace to open it.

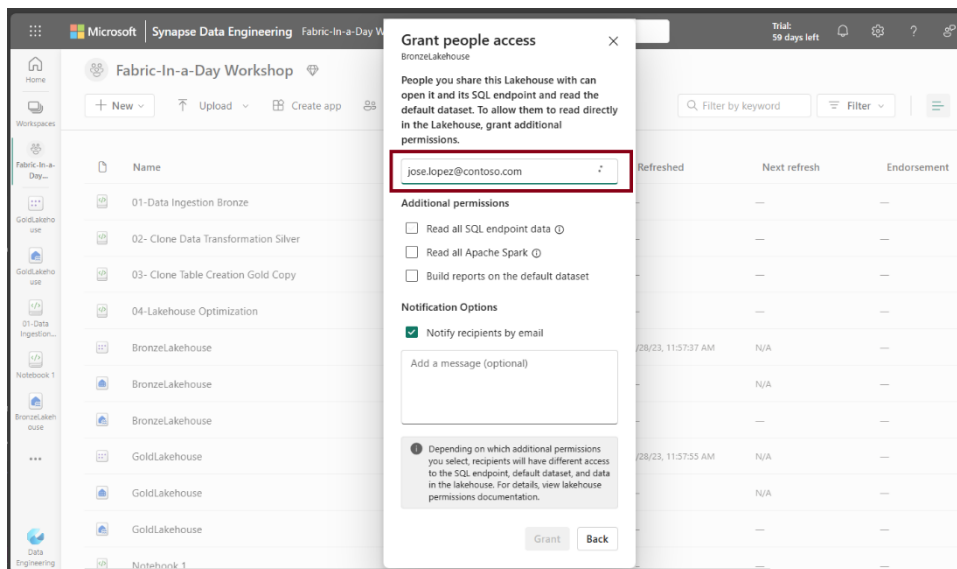
The screenshot shows the Microsoft Fabric user interface. On the left sidebar, the 'Workspaces' button is highlighted with a red box. The main workspace area shows a list of workspaces, with 'Fabric-In-a-Day Workshop' selected and highlighted with a red box. Below the workspace list, a PySpark code cell is shown with a successful execution result. At the bottom, a table of search logs is displayed.

Index	_c0	_c1	_c2	_c3
1	id	time	market	searchtext
2	399266	2019-10-15T11:53:04Z	en-us	how to make nachos
3	382045	2019-10-15T11:53:25Z	en-gb	best ski resorts
4	382045	2019-10-16T11:53:42Z	en-gb	broken leg
5	106479	2019-10-16T11:53:10Z	en-ca	south park episodes
6	906441	2019-10-16T11:54:18Z	en-us	cosmos
7	351530	2019-10-16T11:54:29Z	en-fr	microsoft
8	640806	2019-10-16T11:54:32Z	en-us	wireless headphones

3. In the Workspace view, you will see all of the resources in the workspace. Find the BronzeLakehouse. In this exercise, we will share that lakehouse.
4. When your cursor hovers over the lakehouse, a “Share” icon will appear. Click the icon to open the “Grant people access” dialog box.



5. In the dialog box, enter the names or email addresses of the users or groups with whom you would like to share the lakehouse.



6. If you would like to grant additional permissions, select them in the list.
7. When you are ready, click the “Grant” button to share the lakehouse.

If you would like to revoke access to the lakehouse, you can select “Manage Permissions” from the lakehouse pop-up menu. This will show you everyone who has access to the lakehouse. Use the pop-up menu for each user or group to select “Remove access”.

Securing Your Lakehouse

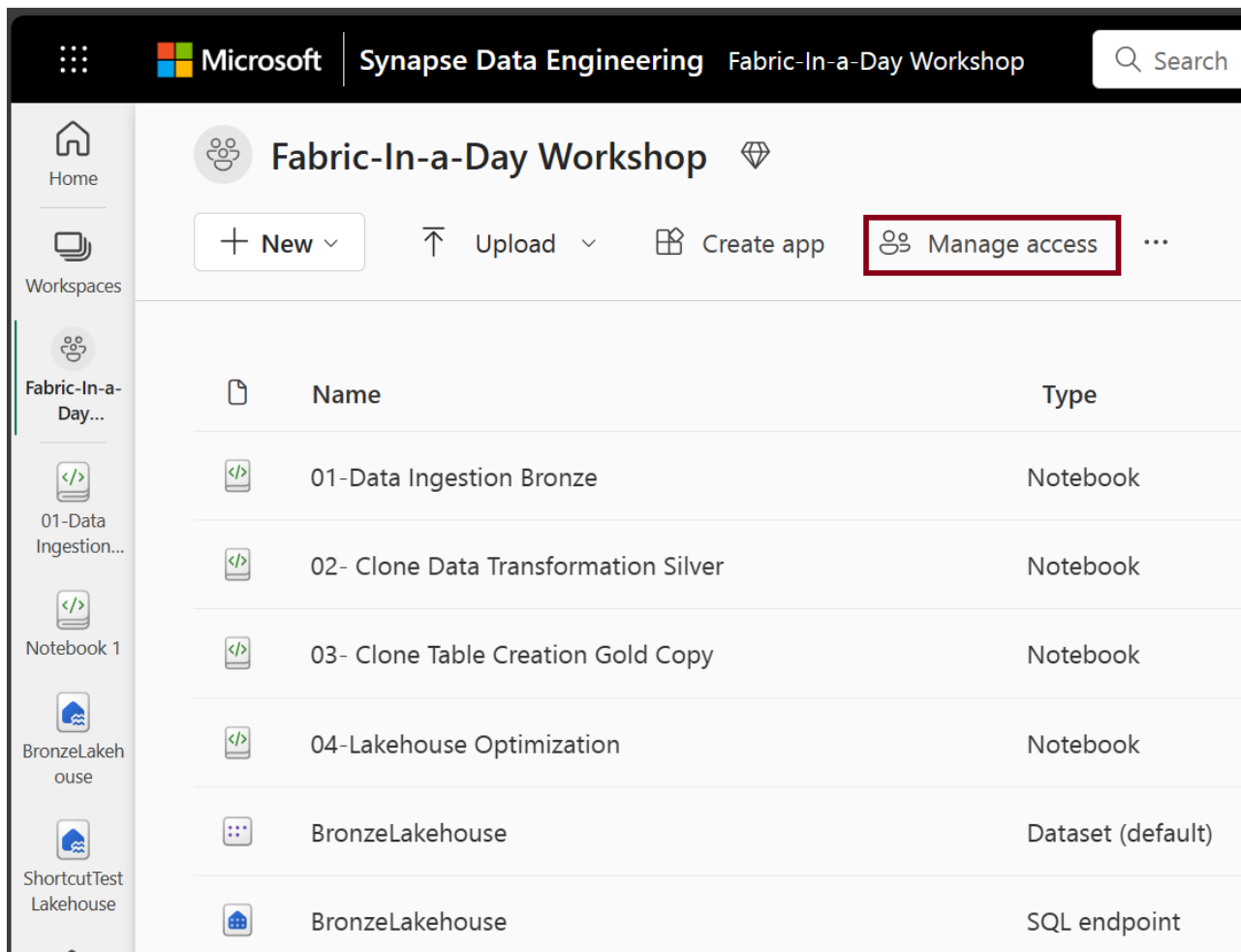
OneLake’s security model is built around your organization’s structure of experiences within Microsoft Fabric. Security is built on top of Azure Active Directory (Azure AD) authentication and is compatible with user identities, service principals, and managed identities. Using Azure AD together with Fabric, you can ensure that you keep your data safe without overly complex user schemes nor with redundant copies of data.

In Fabric, the primary boundary for security is the workspace. Each workspace is a single project area or domain where users can collaborate on shared data. Security in the workspace is managed through Fabric workspace roles. These roles allow you to manage who can do what within the workspace.

The basic roles in Fabric are: Admin, Member, Contributor, and Viewer. Roles can be assigned to individuals or to security groups, Microsoft 365 groups, and distribution lists. By default, the permissions granted by these roles will be applied to all objects within the workspace.

Let’s see how this works by granting access to your workspace to another user.

1. In the Fabric portal, navigate to your workspace.
2. At the top of the screen, click the “Manage access” button.



3. The “Manage access” panel will open on the right side of the screen. Click the “Add people or groups” button.
4. Enter the name or email address of the user or group that you want to access your workspace. You can add multiple people and groups here.
5. Next to the “Add” button, select the role that you want to assign. The default is “Viewer” (which grants read-only access), but you can select other roles with elevated permissions.
6. Click the “Add” button.

The selected users or groups will be granted the role you selected. They can now access the objects in your Fabric workspace.

Further Reading:

[OneLake security](#)

[Roles in workspaces in Microsoft Fabric](#)

Connecting With External Tools

So far in this workshop, we have used tools built into the Fabric portal to work with data in our lakehouse. However, many users prefer to use familiar tools that run locally on their workstations to work with their data. Fabric supports a wide range of external tools, allowing users to choose their experience for connecting to the lakehouse.

[NOTE: You are encouraged to complete the “Azure Storage Explorer” section as part of this workshop. The other sections are optional since you may not have these tools installed on your workstation. The instructions for these tools are provided for your reference.]

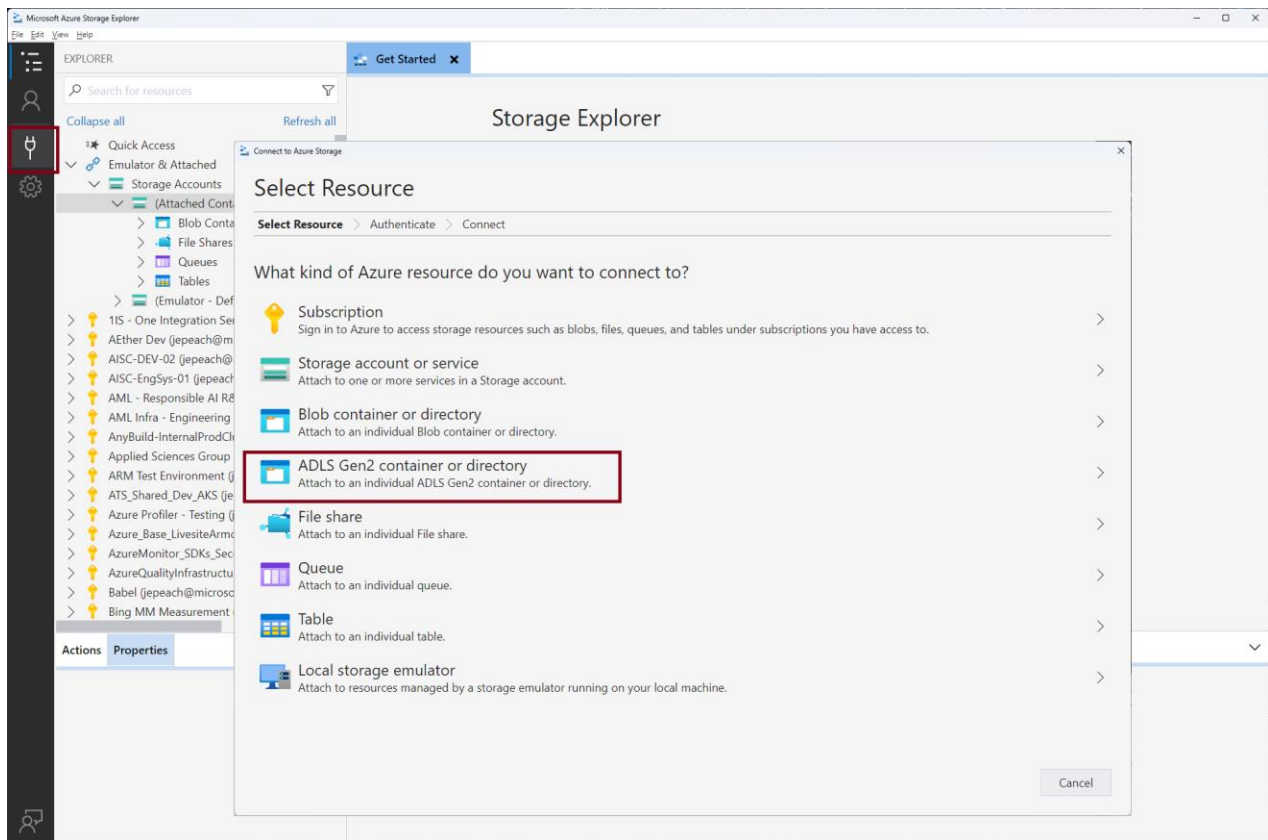
Azure Storage Explorer

Azure Storage Explorer is a local application that allows you to browse and interact with various storage systems in Microsoft Azure, including Azure Storage accounts and Azure CosmosDB databases. Since OneLake is underpinned by an Azure Storage account, we can use Azure Storage Explorer as a convenient way to interact with the files we have stored in OneLake.

You can download Azure Storage Explorer here: [Get started with Storage Explorer](#)

To connect to OneLake with Azure Storage Explorer, follow these steps:

1. Launch Azure Storage Explorer on your computer. Select the “Connect” icon from the toolbar along the left edge of the screen. This will open the “Connect to Azure Storage” dialog box, which guides you through the process of establishing a connection.
2. In the “Connect to Azure Storage” dialog box, click “ADLS Gen2 container or directory”



3. On the next page, select “Sign in using OAuth” as your method for connecting. Then click “Next”. On the subsequent page, select the account with which you want to authenticate. If your authentication is not current, you may be prompted to enter your Azure Active Directory credentials.
4. Finally, you will be able to enter the connection information. For “Display name”, you can enter “OneLake” (or any other name that will help you remember what this connects to). In the “Blob container or directory URL”, you will enter the connection

information for your OneLake storage account.

Connect to Azure Storage

Enter Connection Info

Select Resource > Select Connection Method > Select Account & Tenant > **Enter Connection Info** > Summary

Display name:
OneLake

Blob container or directory URL:
`https://msit-onelake.dfs.fabric.microsoft.com/af9e58b3-826d-4381-83d5-792069387ce6/2f4fdea4-4913-43ce-85df-9e5dfcdcc394`

⚠ The endpoint suffix is not a known blob or DFS suffix. Make sure the URL is correct before continuing.

Back Next Cancel

This URL will use the following pattern:

`https://onelake.dfs.fabric.microsoft.com/{workspace-id}/{lakehouse-id}/`

The workspace ID and lakehouse ID will be GUIDs. The easiest way to find this information is to open your lakehouse in the Fabric portal. In the Explorer, hover over the “Tables” folder and click the three dots to open the context menu. From the menu select “Properties”.

On the “Properties” pane, you will see a URL field. This URL points specifically to the “Tables” folder. If you remove “Tables” from the end of the URL, you can use it to access all of the resources in the lakehouse with Azure Data Explorer.

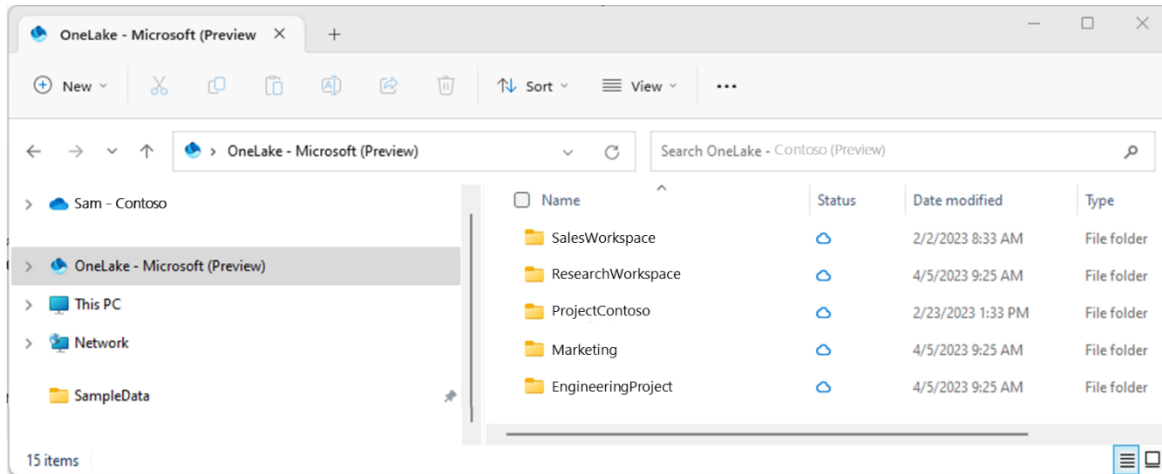
Note: Azure Storage Explorer may display a warning that the URL does not refer to a “known blob or DFS suffix.” Microsoft Fabric is still in public preview, and Azure Storage Explorer has not yet been updated to recognize the OneLake URL scheme. Expect Storage Explorer to be updated as Microsoft Fabric nears General Availability.

OneLake File Explorer

OneLake File Explorer is an innovative new feature of Fabric that allows access to data in OneLake to be seamlessly integrated into your local Windows File Explorer. We like to refer to this feature as “OneDrive for your data” because it allows you access your lakehouse data just as easily as you can access documents stored in OneDrive!

You can download OneLake File Explorer here: [Install OneLake File Explorer](#)

Once you complete the installation process, the tool will guide you through the configuration. You will then be able to access your data through Windows File Explorer. This allows you to easily upload and download data to OneLake.



SQL Server Management Studio (SSMS)

SSMS is the most popular tool for managing and working with Microsoft SQL Server databases. You can also use it to connect to a warehouse in Microsoft Fabric.

You can download SSMS here: [Download SQL Server Management Studio \(SSMS\)](#)

To connect to your Fabric warehouse, follow these steps:

1. In Microsoft Fabric, click the “Browse” button in the toolbar on the left side of the screen.
2. In your list of resources, find the warehouse you want to access. Click on it to open it.
3. When your warehouse opens, click the “Settings” icon in the toolbar at the top of the screen.
4. In the “About” panel, you will see the “SQL connection string”. This provides all of the details for connecting to your warehouse. Copy the connection string.
5. In SSMS, click the “Connect” button in the Object Explorer and select “Database Engine...” from the drop-down list.
6. In the “Connect to Server” dialog box, paste your SQL connection string into the “Server” field. For authentication, only “Azure Active Directory – Universal with MFA” is supported.

Once you are connected to your Fabric warehouse, you can use SSMS to manage database objects (like tables, views, and store procedures). You can also use the query editor to write SQL code against your Fabric warehouse.

Azure Data Studio

Azure Data Studio is a modern open-source, cross-platform hybrid data analytics tool designed to simplify the data landscape. Similar to SQL Server Management Studio, it is a powerful tool allowing data engineers and analysts to connect to a variety of database sources.

You can download Azure Data Studio here: [Download and install Azure Data Studio](#)

Connecting to your Fabric warehouse with Azure Data Studio is very similar to the steps above for connecting with SSMS. You will need to retrieve your SQL connection string from the warehouse as explained above. Then, in Azure Data Studio, on the “Welcome” page, select “New Connection”. On the “Connection Details” dialog box, paste your SQL connection string in the “Server” field and select “Windows Authentication” for the “Authentication type” field.

Further Reading:

[Connectivity to data warehousing](#)

[Integrate OneLake with Azure Storage Explorer](#)

[Access Fabric data locally with OneLake file explorer](#)